

B.Tech. I year, II sem
Physics-PH-113

Module-2: Electromagnetic Theory

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Electromagnetism

Electromagnetism is a branch of physics that involves the study of electromagnetic forces, which is the combination of electric and magnetic forces. It is a type of interaction that occurs between electrically charged particles and responsible for all magnetic, electrical, and electromagnetic phenomena.

Before the invention of electromagnetism, people or scientists used to think electricity and magnetism are two different topics. But electricity and magnetism are inter related.

Static charge produces the electric field and moving charge produces magnetic field. Also time varying magnetic field produces electric field and time varying electric field is the source of magnetic field.

These observation laid a foundation for Electromagnetism.

Scalar and Vector Quantity

A **scalar** is a quantity that is fully described by a **magnitude only**.

A **vector** is a quantity that has both a **magnitude and a direction**.

Scalar Quantities

length, area, volume,
Speed, mass, density,
pressure, temperature,
energy



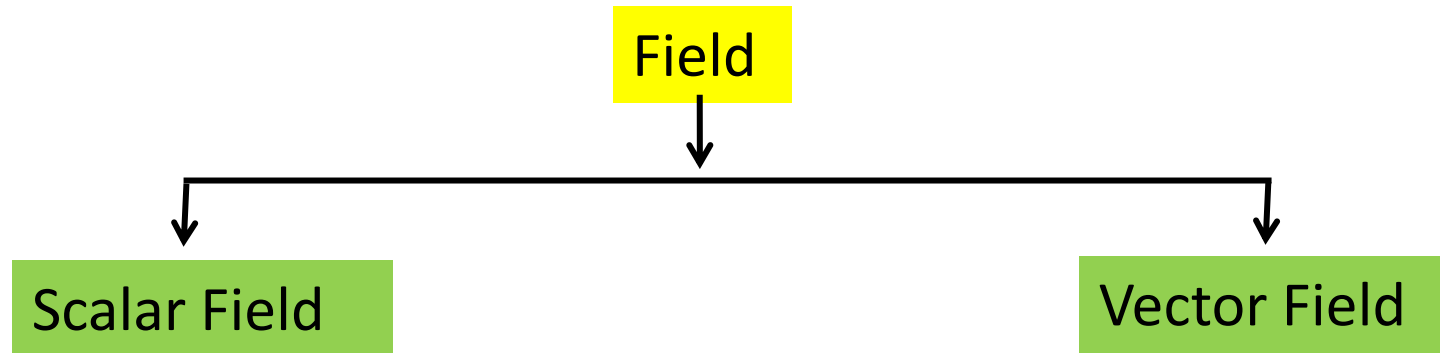
Vector Quantities

displacement,
velocity,
Acceleration,
Momentum, force



Scalar and Vector Fields

Field: The field or space in which the physical quantity depends upon the coordinates of the point of observation is called field of that physical quantity.



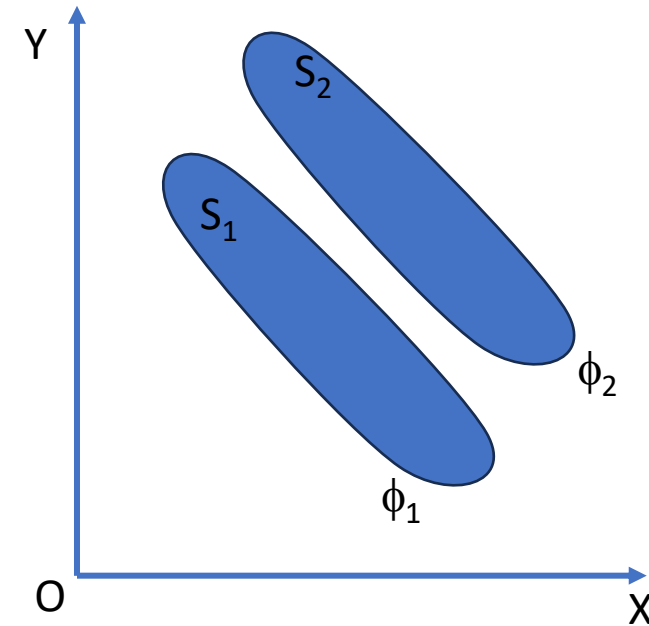
Scalar Field

- **Scalar field:** In space, when a scalar quantity is a function of coordinate of a point of observation is called scalar field.

A scalar field of a scalar quantity ϕ is defined at point $P(x,y,z)$

$$\phi = \phi(x, y, z)$$

- Scalar quantities are graphically represented by drawing surfaces at which the value of the scalar quantity remains a constant.
- Since one point in space corresponds to only one value of the scalar quantity i.e. scalar fields are single valued functions, hence these surfaces, called equiscalar surfaces, do not intersect with each other.



Equiscalar surfaces representing scalar quantity ϕ

Examples of Scalar Field:

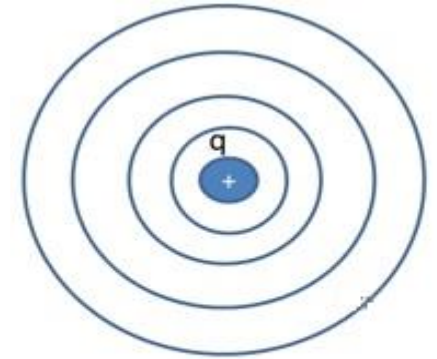
1. If we place a **point charge** at any place, then electric potential around it will depend on the position of the point.

$$V(r) = \frac{kq}{r} \quad \text{where} \quad r = \sqrt{x^2 + y^2 + z^2} \quad k = \frac{1}{4\pi\epsilon_0}$$

ϵ_0 is the permittivity of free space = 8.8542×10^{-12} F/m

Electric potential $V(r)$ is a scalar quantity

so field around the charge is called scalar potential field.



Equipotential surface for a single charge

2. **Gravitational Potential**

$$V(r) = \frac{GM}{r}$$

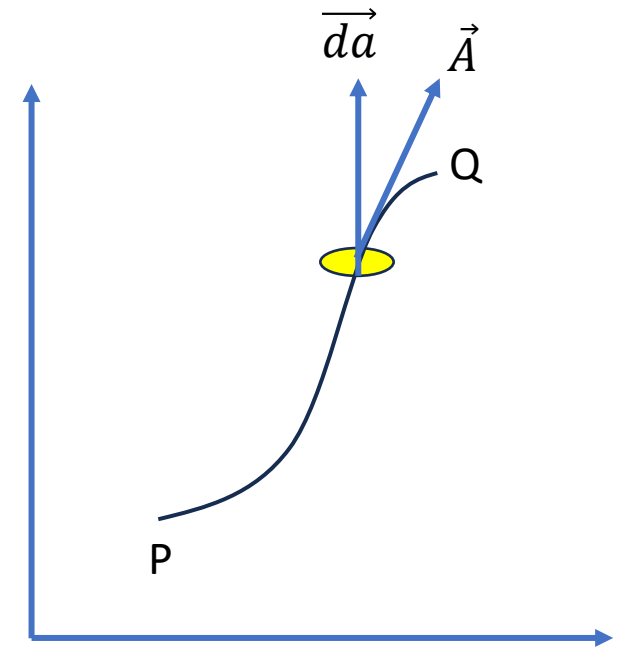
3. **Temperature $T(x, y, z)$**

Vector Field: In space, when a vector quantity is a function of coordinate of a point of observation is called a vector field.

The Vector field is expressed as

$$\vec{A} = A(x, y, z)$$

- Vector fields are graphically represented by lines. These lines are called field lines or flux lines.
- These lines are drawn in the field in such a way that the tangent at any point of the line gives the direction of the vector field at that point.
- The field lines drawn in electric and magnetic fields are also called lines of force.
- To express the magnitude of the vector field at any point of the field, an infinitesimal area element is drawn perpendicular to the field line. The number of field lines passing through the unit area, represents the magnitude of vector field.
- Hence, the magnitude of a vector field at any point of the field is equal to the number of field lines going out of the unit area taken normal to the direction of the vector at that point.



Representation of vector field

- Suppose the number of field lines or flux going out of an area element $\vec{da}$ perpendicular to the field lines at any point is $d\phi$. Then the magnitude of the intensity of the vector field will be.

$$|\vec{A}| = \frac{d\phi}{da} \quad (...1)$$

∴ Vector field

$$\vec{A} = \left(\frac{d\phi}{da}\right) \hat{n} \quad (...2)$$

Where $\hat{n}$ is the unit vector perpendicular to the area da

$$\hat{n} da = \vec{da} \quad (...3)$$

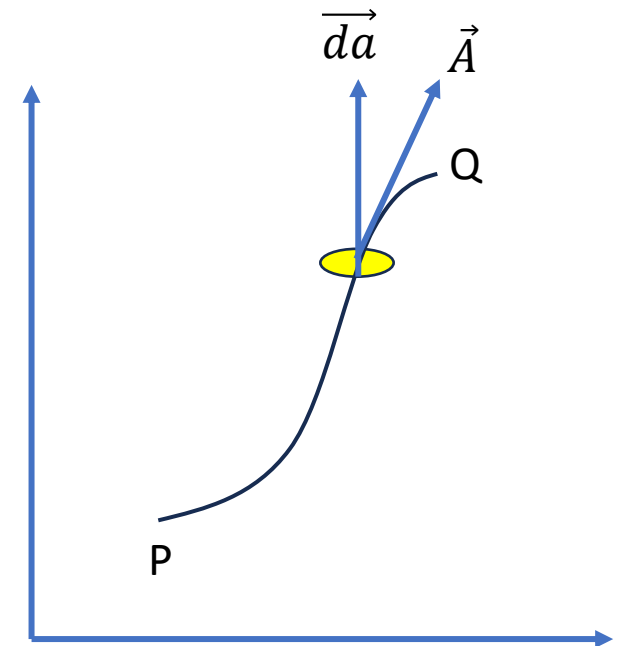
Taking the dot product of unit vector $\hat{n}$ with the vector field $\vec{A}$,

$$\vec{A} \cdot \hat{n} = \left(\frac{d\phi}{da}\right) \hat{n} \cdot \hat{n} = \frac{d\phi}{da}$$

$$\text{or } d\phi = (\vec{A} \cdot \hat{n}) da = \vec{A} \cdot \hat{n} da$$

$$\text{Flux } d\phi = \vec{A} \cdot \vec{da} \quad (...4)$$

Like scalar fields, positional function of a vector field is also a single valued function i.e. a vector field lines do not intersect with each other.



Representation of vector field

Examples:

1. Electric field

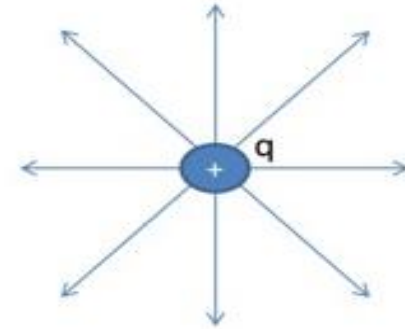
$$\vec{E} = \frac{kq}{r^2} \hat{r}$$

2. Magnetic field

$$d\vec{B} = \frac{\mu_o Idl \sin \theta}{4\pi r^2} \hat{r}$$

3. Gravitational field.

$$\vec{g} = -\frac{GM}{r^2} \hat{r}$$



Electric field lines for a single positive charge

DEL OPERATOR or GRADIENT OPERATOR (∇) :-

- The Del, or nabla, is an operator used in mathematics (particularly in vector calculus) as a vector differential operator, usually represented by the nabla symbol ∇ .
- In Cartesian coordinates Del is written by

$$\nabla = \frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z$$

- When del operator applied to a function defined on a one-dimensional domain, it denotes the standard derivative of the function as defined in calculus.
- This vector differential operator, otherwise known as the **gradient operator**, is not a vector itself, but when it operates on a scalar function.

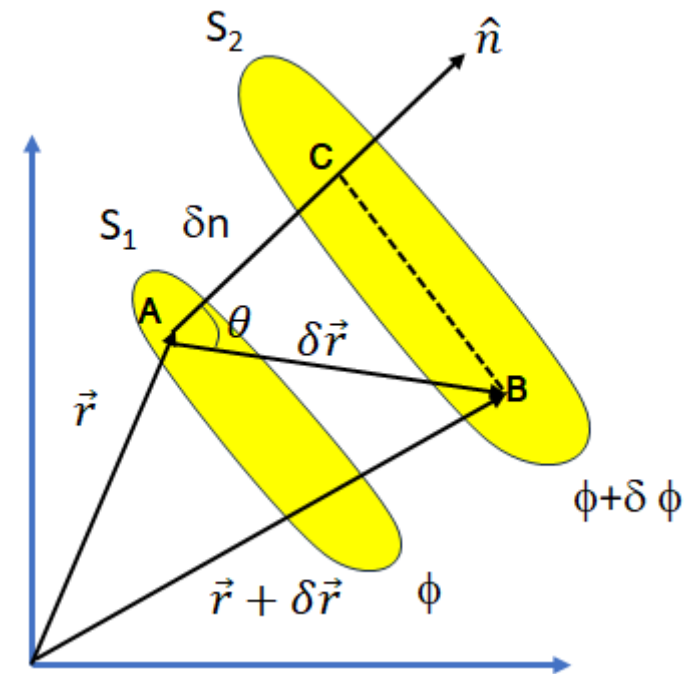
- **The operator is useful in defining**

1. The gradient of a scalar V , written as =>
2. The divergence of a vector A , written as =>
3. The curl of a vector A , written as =>
4. The Laplacian of a scalar V , written as =>

- | | |
|-------------------|------------------|
| ∇V | (grad V) |
| $\nabla \cdot A$ | (div A) |
| $\nabla \times A$ | (Curl A) |
| $\nabla^2 V$ | (Laplacian V) |

Gradient of Scalar Field

- Suppose in a scalar field two surfaces S_1 and S_2 represent surfaces with constant scalar quantities ϕ and $\phi + \delta\phi$ respectively. The position vector of any point A on S_1 relative to the origin O is $\vec{r}$ and the position vector of another point B on Surface two is $\vec{r} + \delta\vec{r}$.
- $\therefore$ displacement of B with respect to A is $AB = \delta\vec{r}$ and the rate of change of the scalar function in the direction of AB = $\frac{\delta\phi}{\delta r}$ (...1)
- Now we draw a perpendicular AC from A on surface S_2 . The minimum distance $AC = \delta n$ between the surface S_1 and S_2 will be in the direction of their normal.
- $\therefore$ The rate of change of scalar function ϕ in the direction of AC = $\frac{\delta\phi}{\delta n}$ (...2)



Gradient of a scalar quantity

Gradient of Scalar Field

If the distance δr is infinitesimal, then $\triangle ABC$ can be considered a right-angled triangle. Suppose the angle between $\delta \vec{r}$ and $\delta \vec{n}$ is θ , then

$$AC = AB \cos \theta$$

$$\text{or } \delta n = \delta r \cos \theta \quad (...3)$$

Substituting the value of δr in equation (1), we get

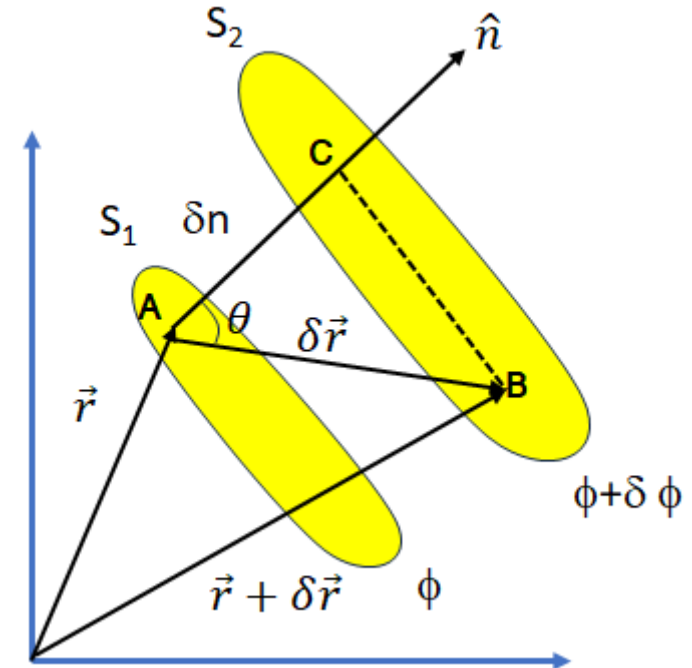
Rate of change of ϕ in the direction of AB

$$= \frac{\delta \phi}{\delta n} \cos \theta$$

As the maximum value of $\cos \theta = 1$

$\therefore$ The maximum rate of change of ϕ

$$= \left(\frac{\delta \phi}{\delta r} \right)_{\max} = \left(\frac{\delta \phi}{\delta n} \right)$$



Gradient of a scalar quantity

Gradient of Scalar Field

In the limit $\delta r \rightarrow 0$

$$\left(\frac{d\phi}{dr}\right)_{max} = \left(\frac{d\phi}{dn}\right)$$

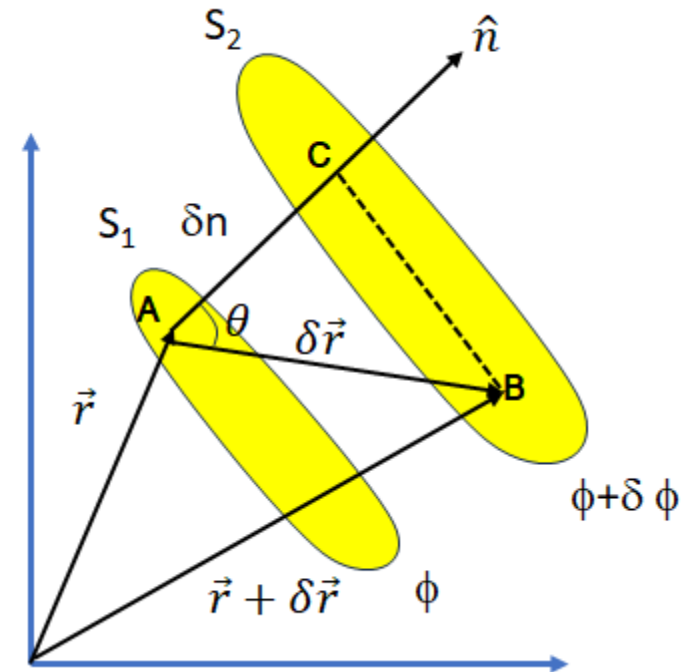
Hence, the rate of change of scalar function is maximum in a direction normal to the equi scalar surface.

In a scalar field, the maximum rate of change of scalar function in space is known as gradient of the scalar field.

It is a vector quantity and its direction is normal to the scalar surface, that is the direction in which the rate of change is maximum.

If we take $\hat{n}$ as unit vector normal to the equiscalar surface, then

$$\text{Gradient } \phi = \left(\frac{d\phi}{dn}\right) \hat{n} = \left(\frac{d\phi}{dr}\right)_{max} \hat{n}$$



Gradient of Scalar Field

- The gradient of a scalar field at any point is the maximum rate of change of the field at that point.
The **gradient** of a scalar field V is a vector that represents both the magnitude and the direction of the maximum space rate of increase of V .
- It is a vector quantity, and its direction is normal to the equiscalar surface i.e. the direction in which rate of change is maximum.

Example: Electric field is related to the electric potential V by

$$\vec{E} = -\nabla V$$

If $\phi(x, y, z)$ is a scalar field, then gradient of ϕ is defined as

$$\text{grad } V = \nabla V = \left(\hat{i} \frac{\partial V}{\partial x} + \hat{j} \frac{\partial V}{\partial y} + \hat{k} \frac{\partial V}{\partial z} \right)$$

where
$$\nabla = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right)$$

Gradient of Scalar Field contd..

A mathematical expression for the gradient can be obtained by evaluating the difference in the field dV between points P_1 and P_2 of Figure, where V_1 , V_2 , and V_3 are contours on which V is constant. From calculus,

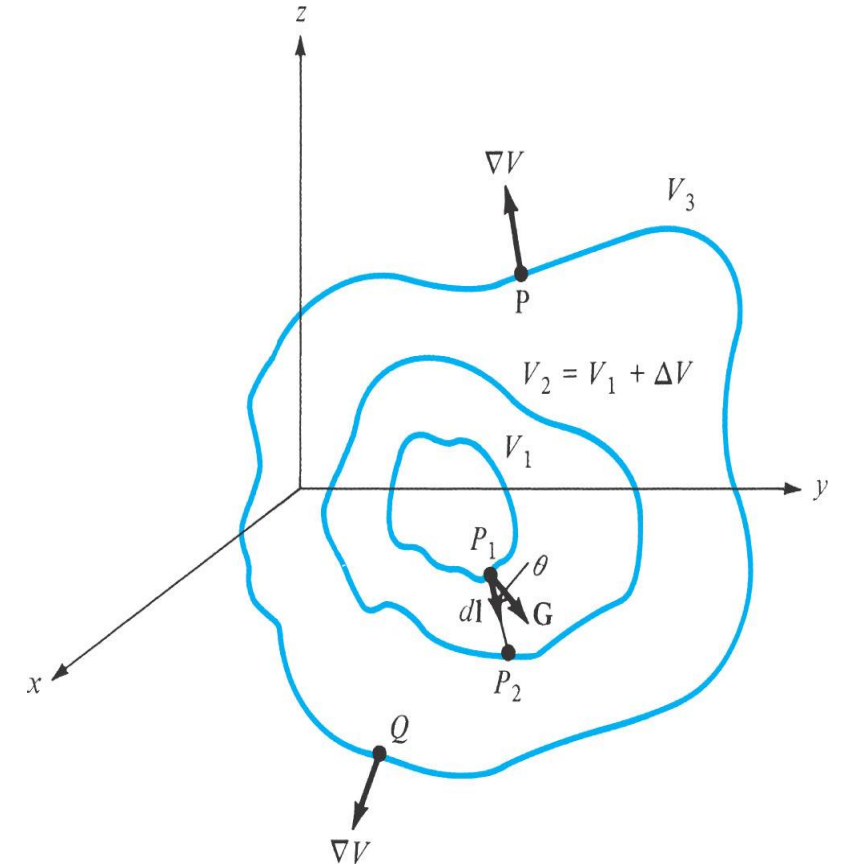
$$\begin{aligned} dV &= \frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy + \frac{\partial V}{\partial z} dz \\ &= \left(\frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z \right) \cdot (dx \mathbf{a}_x + dy \mathbf{a}_y + dz \mathbf{a}_z) \end{aligned} \quad \dots(1)$$

For convenience, let

$$\bullet \quad \mathbf{G} = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z \quad \dots(2)$$

Then

$$dV = \mathbf{G} \cdot d\mathbf{l} = G \cos \theta \, dl$$



Gradient of Scalar Field contd..

Or

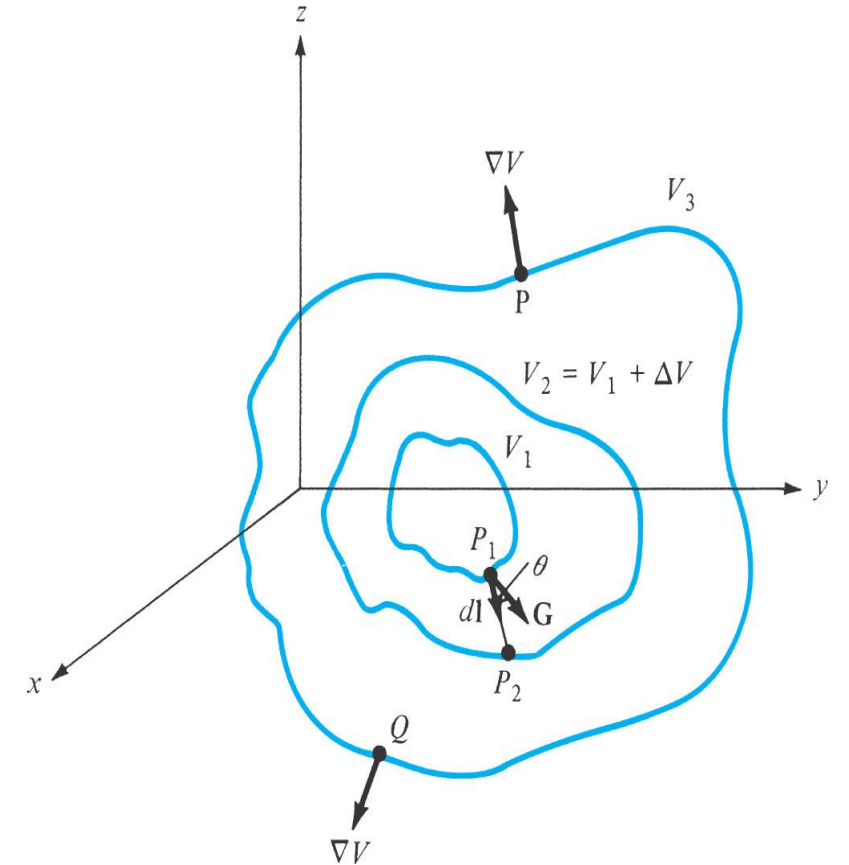
$$\frac{dV}{dl} = G \cos \theta \quad \dots(3)$$

where $d\mathbf{l}$ is the differential displacement from P_1 to P_2 and θ is the angle between $\mathbf{G}$ and $d\mathbf{l}$. From eq. (3), we notice that dV/dl is a maximum when $\theta=0$, that is, when $d\mathbf{l}$ is in the direction of $\mathbf{G}$. Hence,

$$\left. \frac{dV}{dl} \right|_{\max} = \frac{dV}{dn} = G \quad \dots(4)$$

where dV/dn is the normal derivative. Thus G has its magnitude and direction as those of the maximum rate of change of V . By definition, $\mathbf{G}$ is the gradient of V . Therefore:

$$\text{grad } V = \nabla V = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z \quad \dots(5)$$



Gradient of Scalar Field contd..

The gradient of V can be expressed in Cartesian coordinates as

$$\nabla V = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z$$

for cylindrical coordinates,

- $$\nabla V = \frac{\partial V}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V}{\partial z} \mathbf{a}_z \quad \dots(6)$$

- and for spherical coordinates,

- $$\nabla V = \frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi \quad \dots(7)$$

The following computation formulas on gradient, which are easily proved, should be noted:

$$(i) \quad \nabla(V + U) = \nabla V + \nabla U \quad \dots(8 a)$$

$$(ii) \quad \nabla(VU) = V\nabla U + U\nabla V \quad \dots(8 b)$$

$$(iii) \quad \nabla\left[\frac{V}{U}\right] = \frac{U\nabla V - V\nabla U}{U^2} \quad \dots(8 c)$$

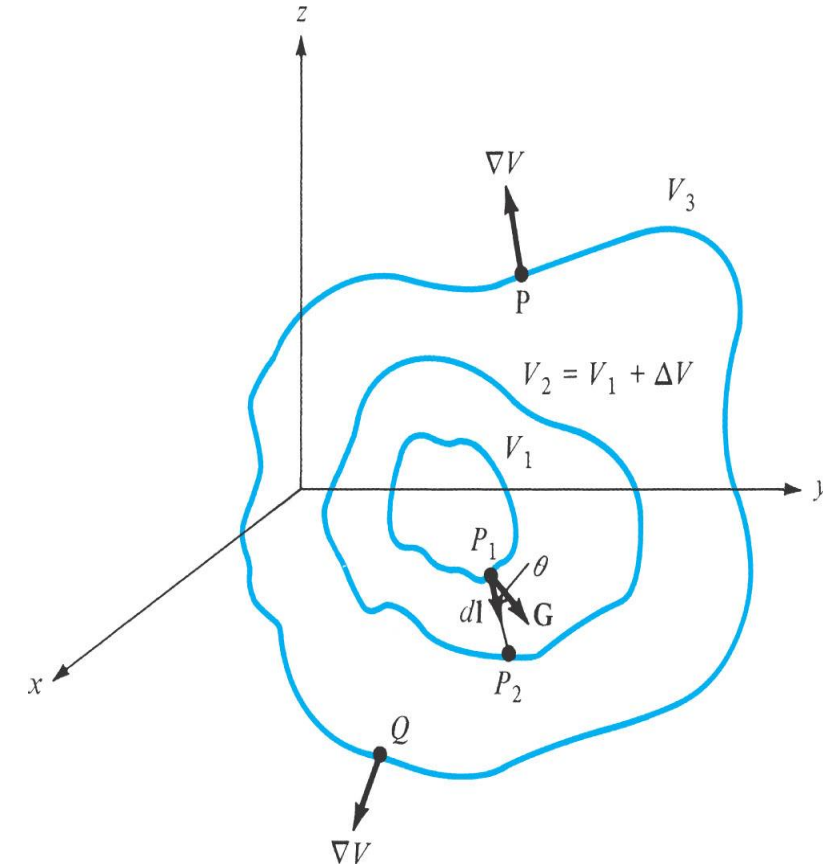
$$(iv) \quad \nabla V^n = nV^{n-1} \nabla V \quad \dots(8 d)$$

where U and V are scalars and n is an integer

Gradient of Scalar Field contd..

Also take note of the following fundamental properties of the gradient of a scalar field V :

- The magnitude of ∇V equals the maximum rate of change in V per unit distance.
- ∇V points in the direction of the maximum rate of change in V .
- ∇V at any point is perpendicular to the constant V surface that passes through that point (see points P and Q in figure).
- The projection (or component) of ∇V in the direction of a unit vector $\mathbf{a}$ is $\nabla V \cdot \mathbf{a}$ and is called the directional derivative of V along $\mathbf{a}$. This is the rate of change of V in the direction of $\mathbf{a}$. For example, dV/dl in eq. (3) is the directional derivative of V along P_1P_2 in Figure. Thus the gradient of a scalar function V provides us with both the direction in which V changes most rapidly and the magnitude of the maximum directional derivative of V .
- If $\mathbf{A} = \nabla V$, V is said to be the scalar potential of $\mathbf{A}$



Gradient of Scalar Field contd..

Example:

Find the gradient of the following scalar fields:

(a) $V = e^{-z} \sin 2x \cosh y$

(b) $U = \rho^2 z \cos 2\phi$

(c) $W = 10r \sin^2 \theta \cos \phi$

Solution:

$$\begin{aligned} \text{(a) } \nabla V &= \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z \\ &= 2e^{-z} \cos 2x \cosh y \mathbf{a}_x + e^{-z} \sin 2x \sinh y \mathbf{a}_y - e^{-z} \sin 2x \cosh y \mathbf{a}_z \end{aligned}$$

$$\begin{aligned} \text{(b) } \nabla U &= \frac{\partial U}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial U}{\partial \phi} \mathbf{a}_\phi + \frac{\partial U}{\partial z} \mathbf{a}_z \\ &= 2\rho z \cos 2\phi \mathbf{a}_\rho - 2\rho z \sin 2\phi \mathbf{a}_\phi + \rho^2 \cos 2\phi \mathbf{a}_z \end{aligned}$$

$$\begin{aligned} \text{(c) } \nabla W &= \frac{\partial W}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial W}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial W}{\partial \phi} \mathbf{a}_\phi \\ &= 10 \sin^2 \theta \cos \phi \mathbf{a}_r + 10 \sin 2\theta \cos \phi \mathbf{a}_\theta - 10 \sin \theta \sin \phi \mathbf{a}_\phi \end{aligned}$$

Flux of a Field

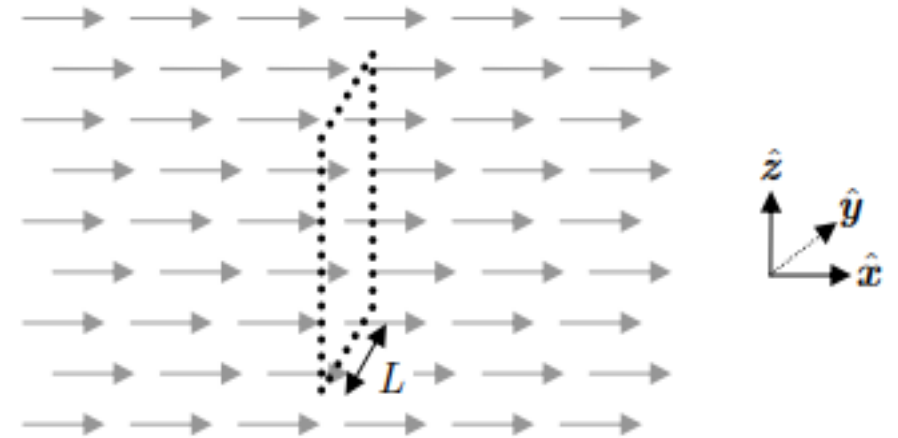
- The term flux has been used to study the flowing field.
- Flux is the amount of vector field passing through a surface per second.
- The **total flux** depends on strength of the field, the size of the surface it passes through, and their orientation.

- Consider a region of space in which there is a constant vector field,

$$\vec{E}(x, y, z) = a\hat{x}$$

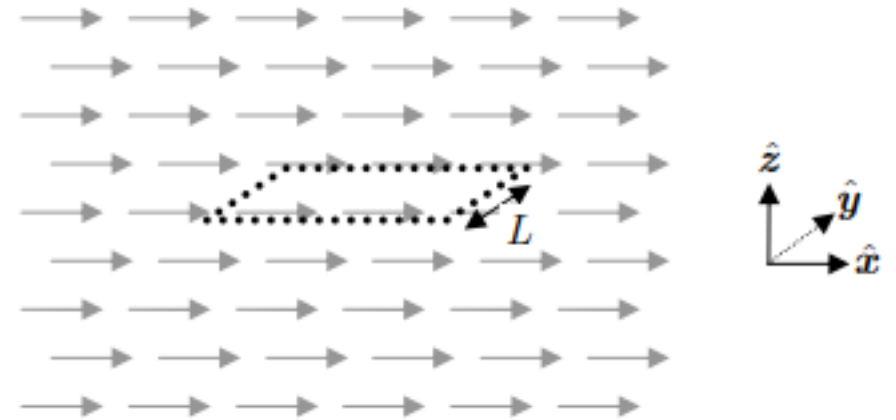
- Imagine that the vector field is a fluid flowing through space.
- Then, the flux of the field through an area is the amount of “fluid” flowing through that area. Now, in this case, the area through which fluid is flowing is L^2 , and the field strength is “a”, and so

$$\text{Flux} = aL^2$$



Flux of a Field....

- Now, consider a slightly different example.
- In this case, it should be pretty clear that none of the fluid is passing through the square, because the plane of the square is parallel to the fluid flow.



$$\boxed{\text{Flux} = 0}$$

Flux of a Field....

- If the area of the square is inclined at some angle to the fluid flow.
- The field vectors can be broken down into two components.
- Now, let's have a look at the flux of each of these components through the square:
- (i) The $a \sin \theta$ component is parallel to the plane, and therefore, as in the second example above, it will produce 0 flux.
- (ii) The $a \cos \theta$ component is perpendicular to the plane, and therefore, as in the first example above, it will produce a flux $aL^2 \cos \theta$
- Adding these two fluxes together

$$\text{Flux} = aL^2 \cos \theta$$

The general expression for flux is

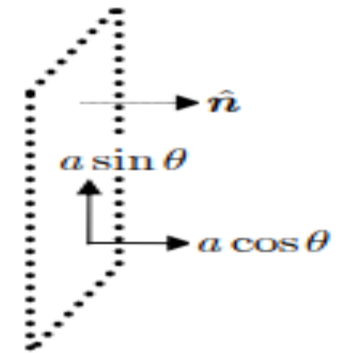
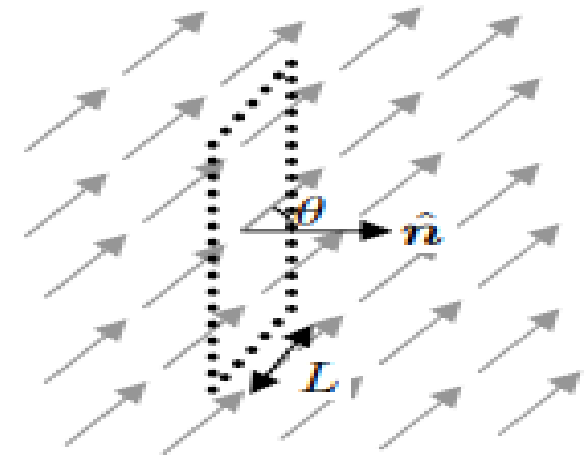
$$\text{Flux } \phi = (\mathbf{A} \cdot \mathbf{n}') \times \text{Area}$$

$$\phi = (\mathbf{A} \cdot \mathbf{n}) \, ds$$

Where $\hat{n}$ is a unit vector that points in a direction perpendicular to the area ds .

$$\text{Flux} = (\mathbf{E} \cdot \hat{n}) \times \text{Area}$$

$$\Phi = \mathbf{E} \cdot \hat{n} A$$



Divergence of Vector Field

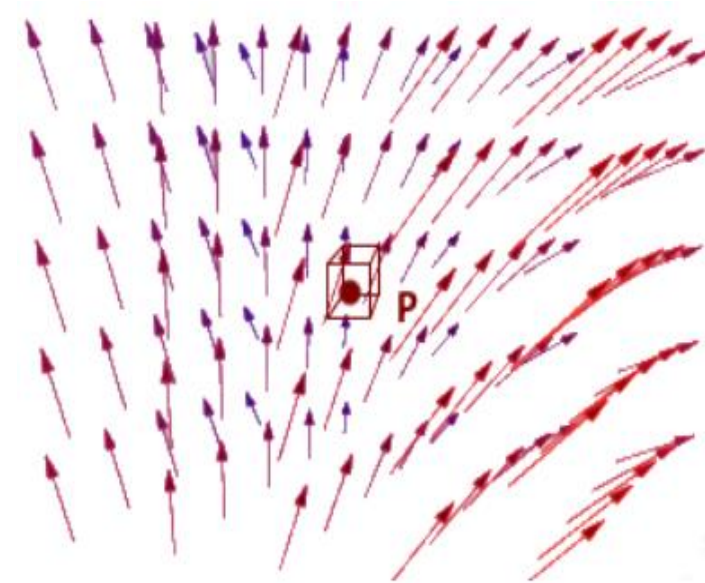
- The net outflow of the flux of a vector field $\mathbf{A}$ from a closed surface S is obtained from the integral $\oint_S \vec{A} \cdot \vec{dS}$. We now define the divergence of $\mathbf{A}$ as the net outward flow of flux per unit volume over a closed incremental surface.
- The **divergence** of $\mathbf{A}$ at a given point P is the *outward* flux per unit volume as the volume shrinks about P .

Hence,

$$\text{div } \mathbf{A} = \nabla \cdot \mathbf{A} = \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta v} \quad (\dots 1)$$

where Δv is the volume enclosed by the closed surface S in which P is located.

- Physically, we may regard the divergence of the vector field $\mathbf{A}$ at a given point as a measure of how much the field diverges or emanates or originates from that point.



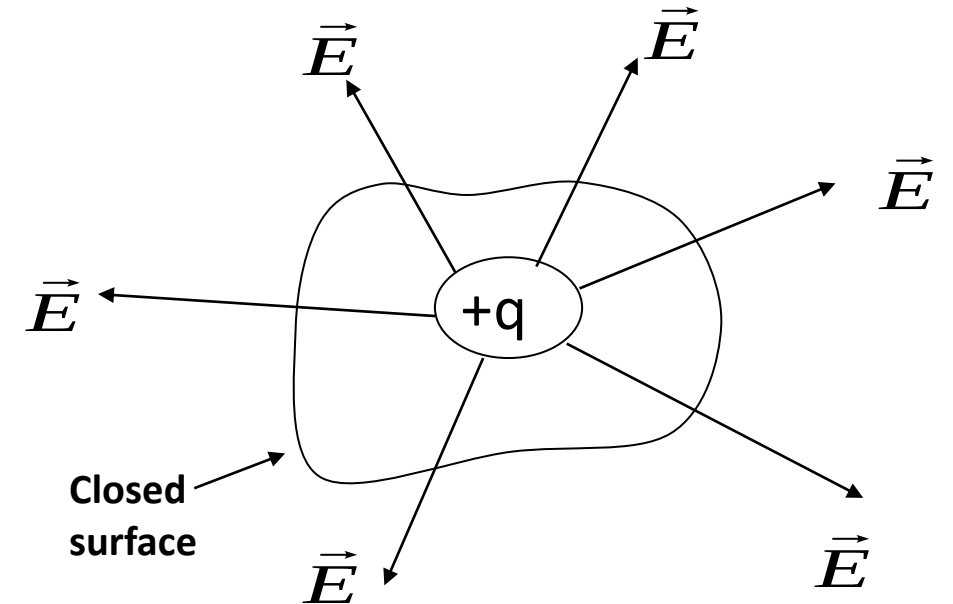
Divergence of Vector Field.....

Physical Significance:

1. If +q charge is placed inside a volume.
 - The net flux is emerging out of the closed surface.

$$\text{Div}\vec{E} = +ve$$

- Source of electric field lines placed inside a given volume.



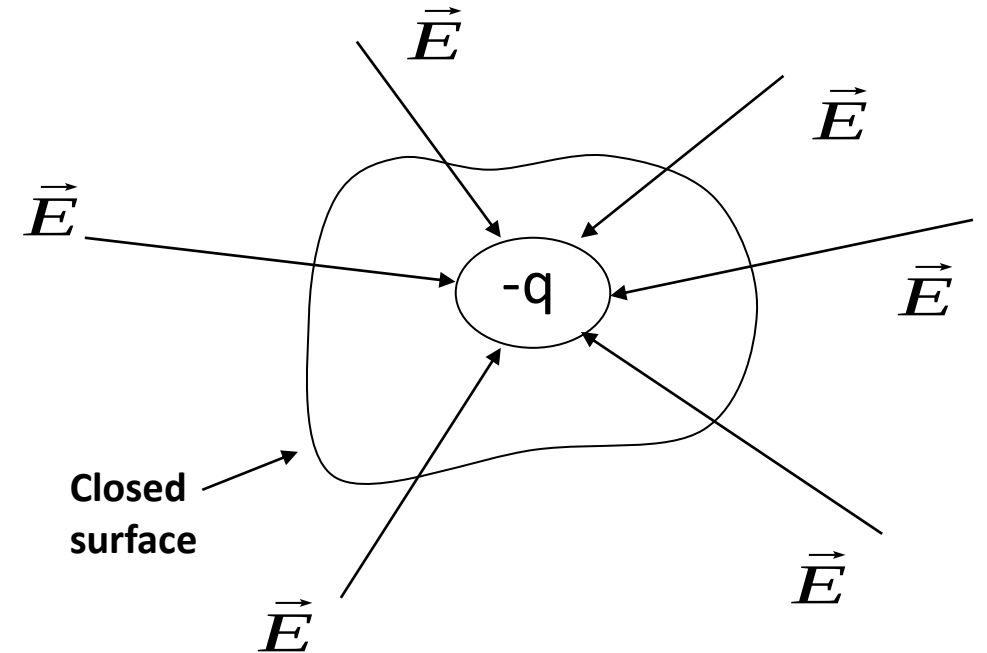
Divergence of Vector Field.....

2. If $-q$ charge placed inside a volume.

- The Electric field lines converge toward $-q$ charge the closed volume.

$$\text{Div}\vec{E} = -ve$$

- Sink of electric field lines placed inside a given volume.



Divergence of Vector Field.....

3. If +q charge is placed outside from closed surface net flux passing is zero i.e. net incoming flux will be equal to net emerging flux.

$$\text{Div} \vec{E} = 0$$

- No source or sink is present inside the volume.

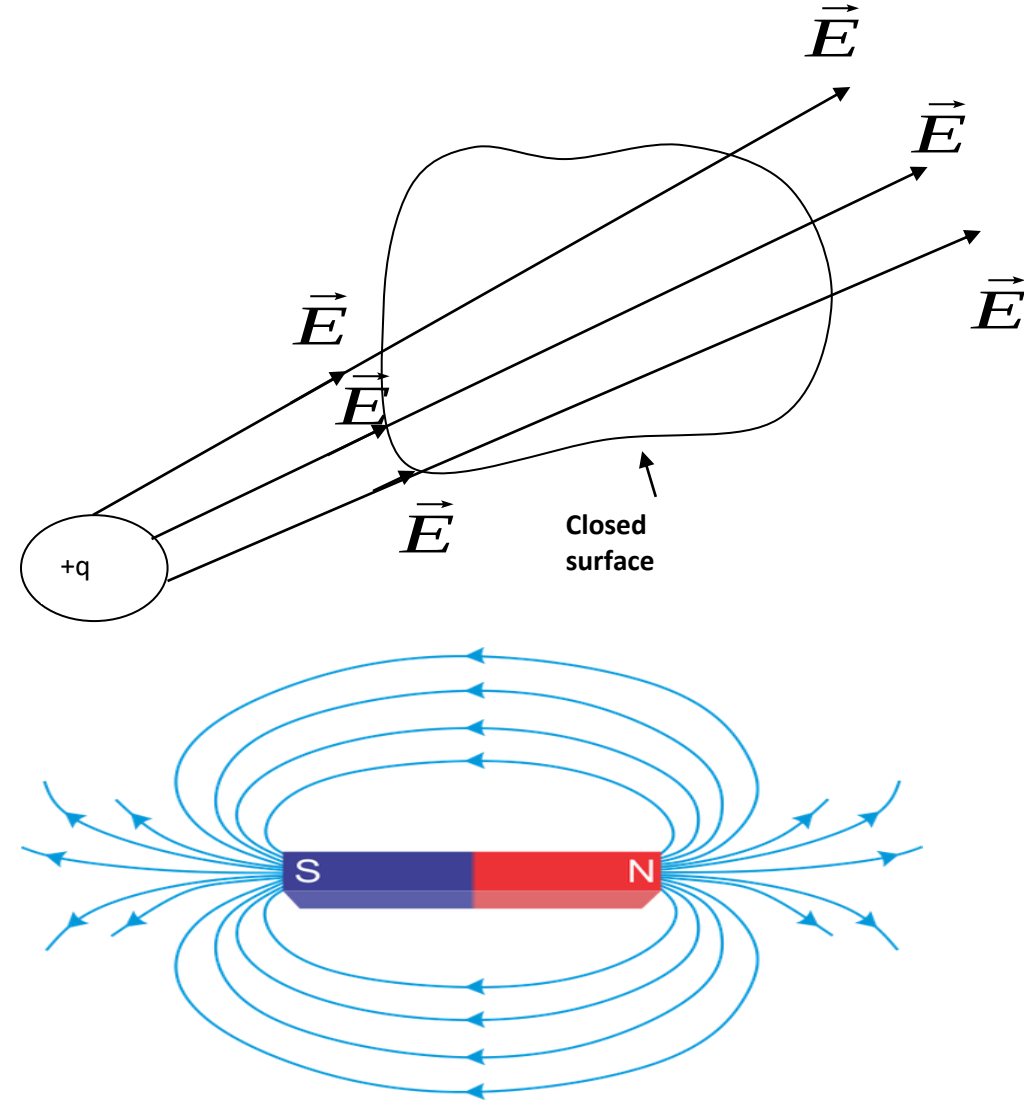
Example: Magnetic lines of force always form close loop i.e. no. of magnetic field lines entering into a given volume is equal to no. of field lines emerging.

So

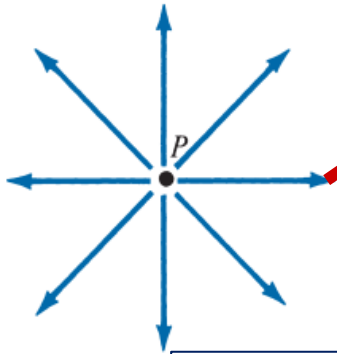
$$\text{Div} \vec{B} = 0$$

Vector field whose divergence is zero is called Solenoidal vector field.

i.e. if $\text{Div} \vec{A} = 0$, the $\vec{A}$ is a Solenoidal vector.

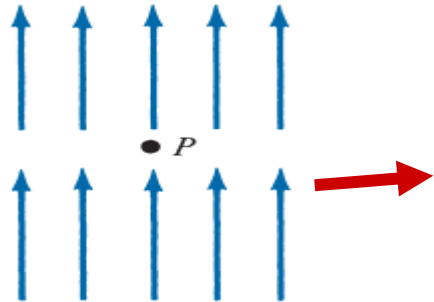
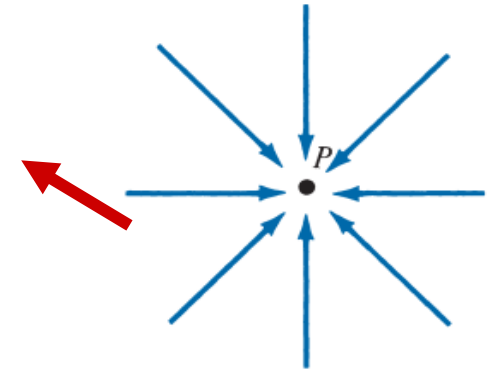


Divergence of Vector Field



This figure shows that the divergence of a vector field at point P is positive because the vector diverges (or spreads out) at P .

This figure shows that the divergence of a vector field at point P is negative because the vector converges at P .

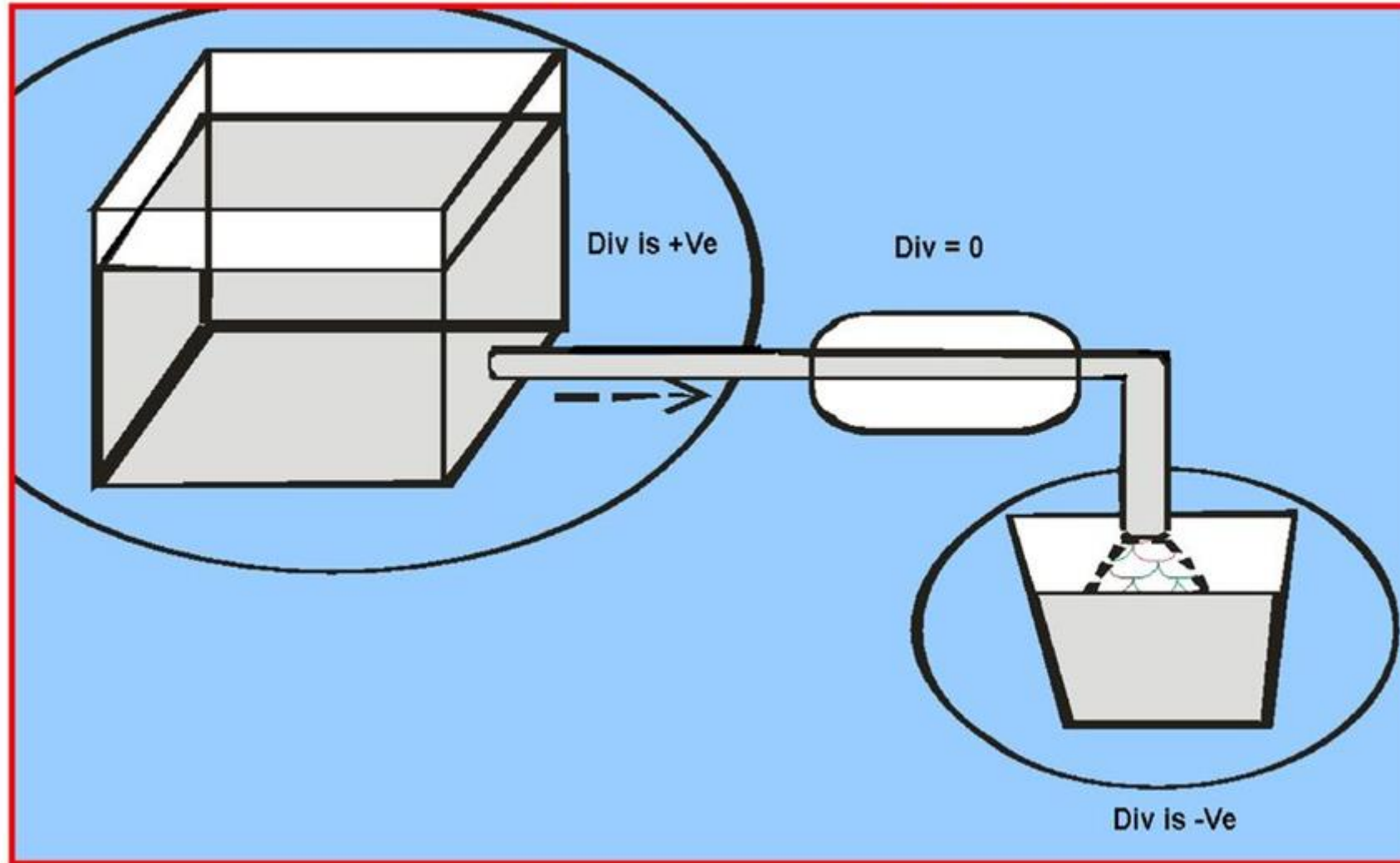


This figure shows that the divergence of a vector field at point P is zero at P .

The divergence of a vector field can also be viewed as simply the limit of the field's source strength per unit volume (or source density); it is positive at a source point in the field, and negative at a sink point, or zero where there is neither sink nor source.

$$\text{div } \mathbf{A} = \nabla \cdot \mathbf{A} = \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta v}$$

Example for Divergence



Divergence of Vector Field in Cartesian Coordinates

- Let the length, width and height of a rectangular parallelopiped of volume dV be dx , dy and dz respectively and the coordinates of its centre P be (x,y,z) . Let a vector field at the central point P be $\vec{A}$ and its components be A_x , A_y and A_z , then

$$\vec{A} = \hat{i}A_x + \hat{j}A_y + \hat{k}A_z$$

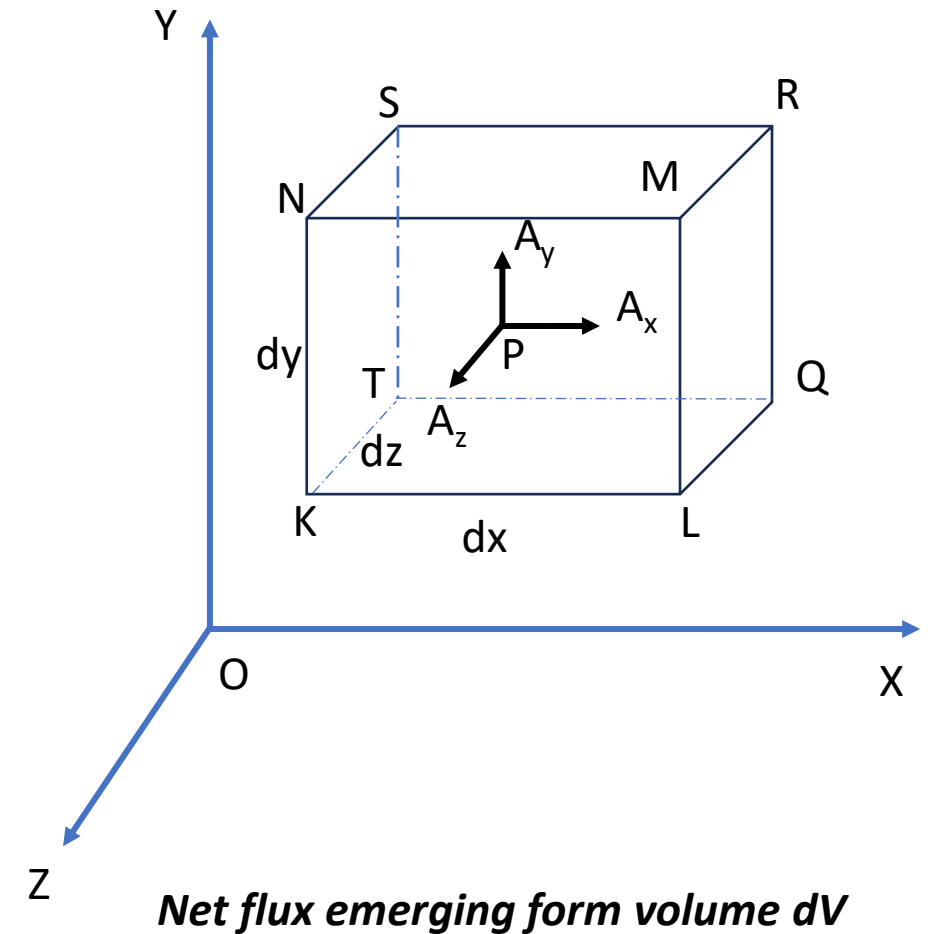
If the rate of change of the x-component A_x of the field with x is $\frac{\partial A_x}{\partial x}$, then the x-component of the field at a distance of $(dx/2)$ from P , i.e. at the surface $LQRM$ will be,

$$\left[A_x + \left(\frac{\partial A_x}{\partial x} \right) \frac{dx}{2} \right]$$

Similarly x component of the vector field at the surface $KTSN$ situated at a distance of $(-dx/2)$ from P will be equal to

$$\left[A_x - \left(\frac{\partial A_x}{\partial x} \right) \frac{dx}{2} \right]$$

$\therefore$ Flux emerging out of the face $LQRM$



Divergence of Vector Field in Cartesian Coordinates

∴ Flux emerging out of the face LQRM

$$= \left[A_x + \left(\frac{\partial A_x}{\partial x} \right) \frac{dx}{2} \right] dy \, dz$$

And the flux emerging out of the face KTSN

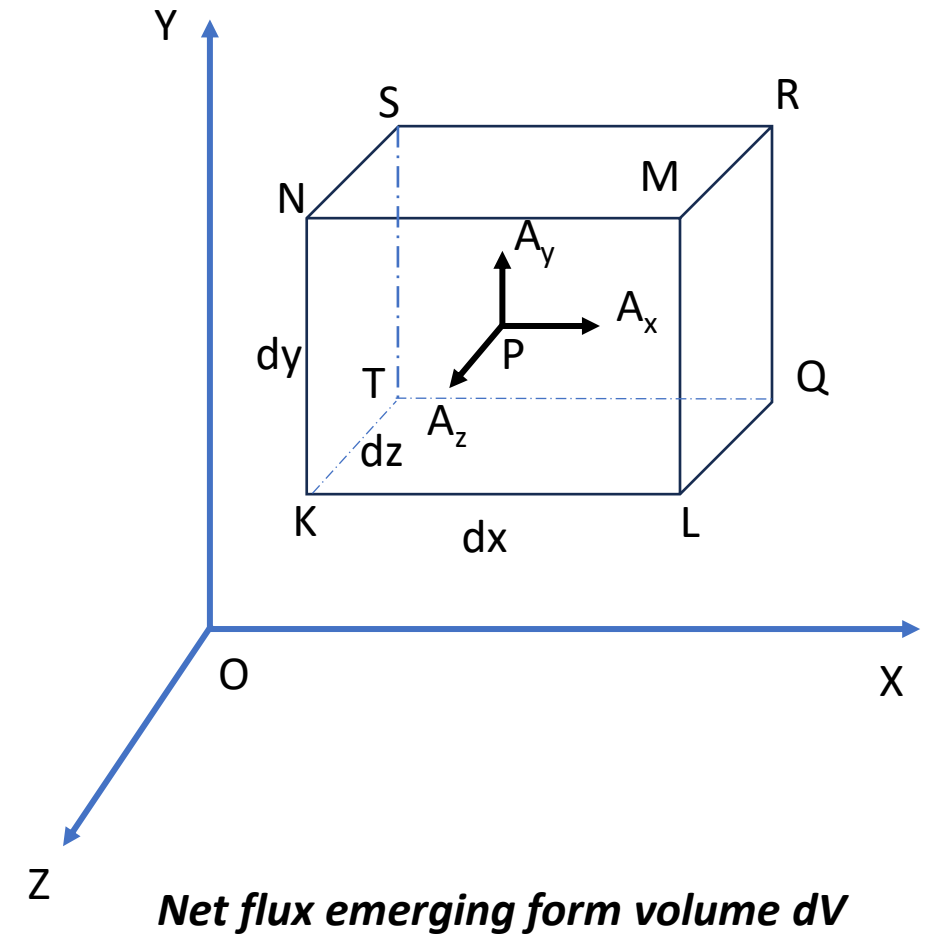
$$= \left[A_x - \left(\frac{\partial A_x}{\partial x} \right) \frac{dx}{2} \right] dy \, dz$$

∴ The flux emerging from the volume element along the x-direction

$$\begin{aligned} &= \left[A_x + \left(\frac{\partial A_x}{\partial x} \right) \frac{dx}{2} \right] dy \, dz - \left[A_x - \left(\frac{\partial A_x}{\partial x} \right) \frac{dx}{2} \right] dy \, dz \\ &= \left(\frac{\partial A_x}{\partial x} \right) dx \, dy \, dz = \left(\frac{\partial A_x}{\partial x} \right) dV \quad (\dots 1) \end{aligned}$$

Here $dV = dx \, dy \, dz$ = volume of the element

Here the flux emerging out of the element through a face is considered positive and the flux entering the element is considered negative.



Divergence of Vector Field in Cartesian Coordinates

Similarly the flux emerging out of the parallelopiped along the y-axes and z-axes will be $\left(\frac{\partial A_y}{\partial y}\right)dV$ and $\left(\frac{\partial A_z}{\partial z}\right)dV$ respectively.

Therefore total flux emerging out of the rectangular parallelopiped,

$$\begin{aligned}\oint_S \vec{A} \cdot d\vec{a} &= \left(\frac{\partial A_x}{\partial x}\right)dV + \left(\frac{\partial A_y}{\partial y}\right)dV + \left(\frac{\partial A_z}{\partial z}\right)dV \\ &= \left[\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}\right]dV\end{aligned}\quad \dots(2)$$

According to the definition of the del operator,

$$\begin{aligned}\nabla \cdot \vec{A} &= \left[\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}\right] \cdot [\hat{i}A_x + \hat{j}A_y + \hat{k}A_z] \\ &= \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}\end{aligned}\quad \dots(3)$$

Substituting equation (3) in (2), the total flux emerging from the faces of the volume element,

$$\oint_S \vec{A} \cdot d\vec{a} = (\nabla \cdot \vec{A})dV \quad \dots(4)$$

Divergence of Vector Field in Cartesian Coordinates

According to the definition of divergence

$$\text{div } \vec{A} = \lim_{dV \rightarrow 0} \left[\frac{1}{dV} \oint_S \vec{A} \cdot d\vec{a} \right] \quad \dots(5)$$

Comparing equation (4) and (5),

$$\text{div } \vec{A} = \nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

Therefore, the divergence of a vector function is equal to the scalar product of the del operator and the vector function.

Gauss Divergence Theorem

- To prove the divergence theorem, subdivide volume V enclosed by a closed surface S in a vector field $\vec{A}$ into a large number of infinitesimally small rectangular parallelepipeds. If the k th cell has volume dV_k and is bounded by surface S_k

$$\oint_S \vec{A} \cdot d\vec{a} = \sum_k \oint_{S_k} \vec{A}_k \cdot d\vec{a}_k \quad \dots(6)$$

But the total flux emerging from the faces of the infinitesimally small volume elements is (from equation (4))

$$\oint_{S_k} \vec{A}_k \cdot d\vec{a}_k = (\nabla \cdot \vec{A}_k) dV_k \quad \dots(7)$$

Substituting equation (7) in (6)

$$\oint_S \vec{A} \cdot d\vec{a} = \sum_k (\nabla \cdot \vec{A}_k) dV_k$$

Gauss Divergence Theorem

- If $dV_k \rightarrow 0$ then
$$\oint_S \vec{A} \cdot d\vec{a} = \int_v (\nabla \cdot \vec{A}) dV \quad \dots(8)$$

This theorem transforms the volume integral into the surface integral.

The **divergence theorem** states that the total outward flux of a vector field **A** through the *closed* surface *S* is the same as the volume integral of the divergence of **A**.

$$\oint_S \mathbf{A} \cdot d\mathbf{S} = \int_v \nabla \cdot \mathbf{A} dv$$

Line Integral of a Vector Field

Consider two points P and Q in a vector field. The path between P and Q is divided into displacement elements of length $dr_1, dr_2, dr_3, \dots$ etc. respectively. These displacements being small can be considered as straight lines as shown the figure. If the values of vector field at these small segments be $\vec{A}_1, \vec{A}_2, \vec{A}_3, \dots$ etc. respectively, then the sum of the dot products of all displacement and the vector fields at their positions is

$$w = \vec{A}_1 \cdot \vec{dr}_1 + \vec{A}_2 \cdot \vec{dr}_2 + \vec{A}_3 \cdot \vec{dr}_3 + \dots$$

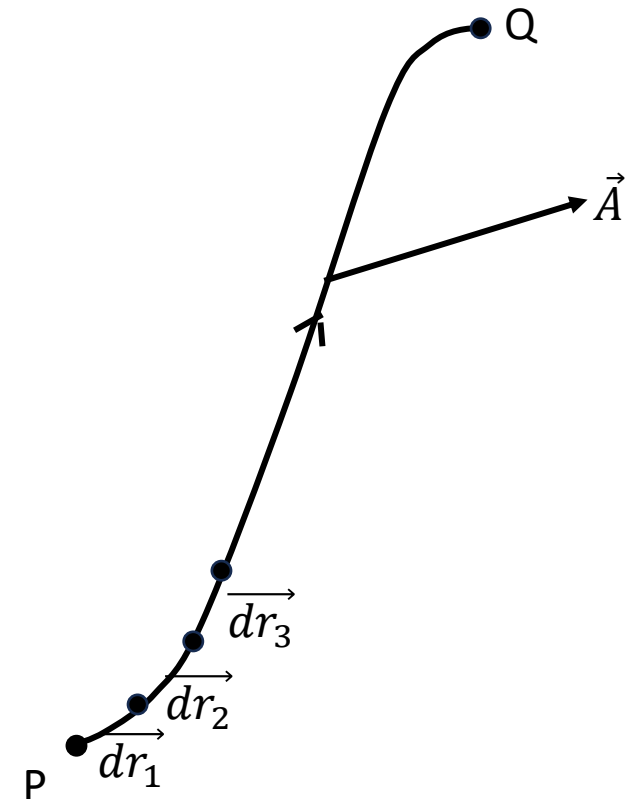
$$w = \sum_P^Q \vec{A} \cdot \vec{dr}$$

Or

If the length of the segment is infinitesimal i.e. $dr \rightarrow 0$, then this summation can be replaced by integration,

$$w = \lim_{dr \rightarrow 0} \sum_P^Q \vec{A} \cdot \vec{dr} \quad \text{or} \quad w = \int_P^Q \vec{A} \cdot \vec{dr}$$

This integral is known as the line integral of the vector field from point P to Q.



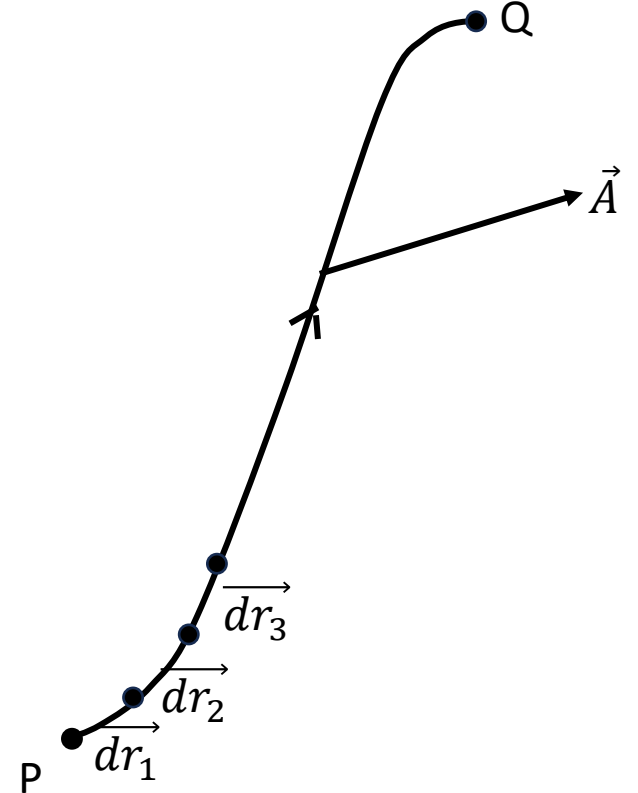
Physical Importance of Line Integral

- If in a vector field, the line integral of the field between two points does not depend on the path but depends only on the positions of the end points, then such a field is known as a conservative field.
- Electric field and Gravitational field are examples of conservative fields.
- If a vector $\vec{A}$ represents a force acting during a given displacement $\vec{dr}$, then $\vec{A} \cdot \vec{dr}$ will represent the work done by the force for the displacement $\vec{dr}$. Hence the total work done by the force field for the displacement from the point P to point Q is $\int_P^Q \vec{A} \cdot \vec{dr}$.
- If vector $\vec{A}$ represents electric field $\vec{E}$ Then $\int_P^Q \vec{E} \cdot \vec{dr}$ will represent the work done in taking unit positive charge from the point P to Q because electric field intensity is equal to the force acting on unit positive charge. This work done against electric field is called potential difference.

$$\Delta V = V_Q - V_P = - \int_P^Q \vec{E} \cdot \vec{dr}$$

If in any circuit a source (cell or battery) is used then line integral on complete path $\oint \vec{E} \cdot \vec{dl}$ is equal to the electromotive force of the source

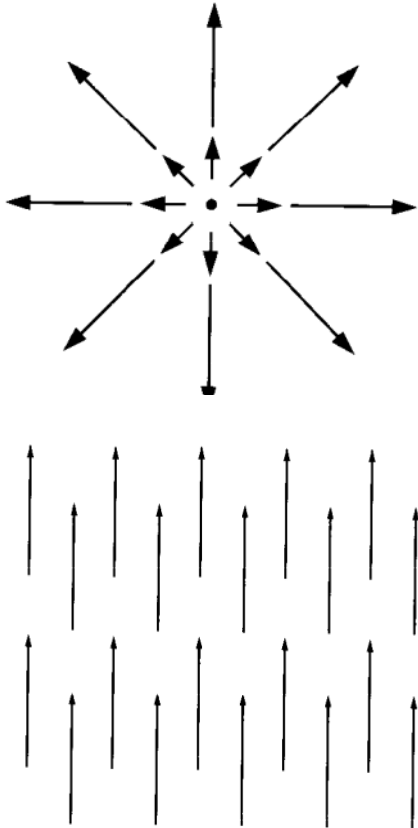
i.e. Electromotive Force (E.M.F) = $\oint \vec{E} \cdot \vec{dl}$



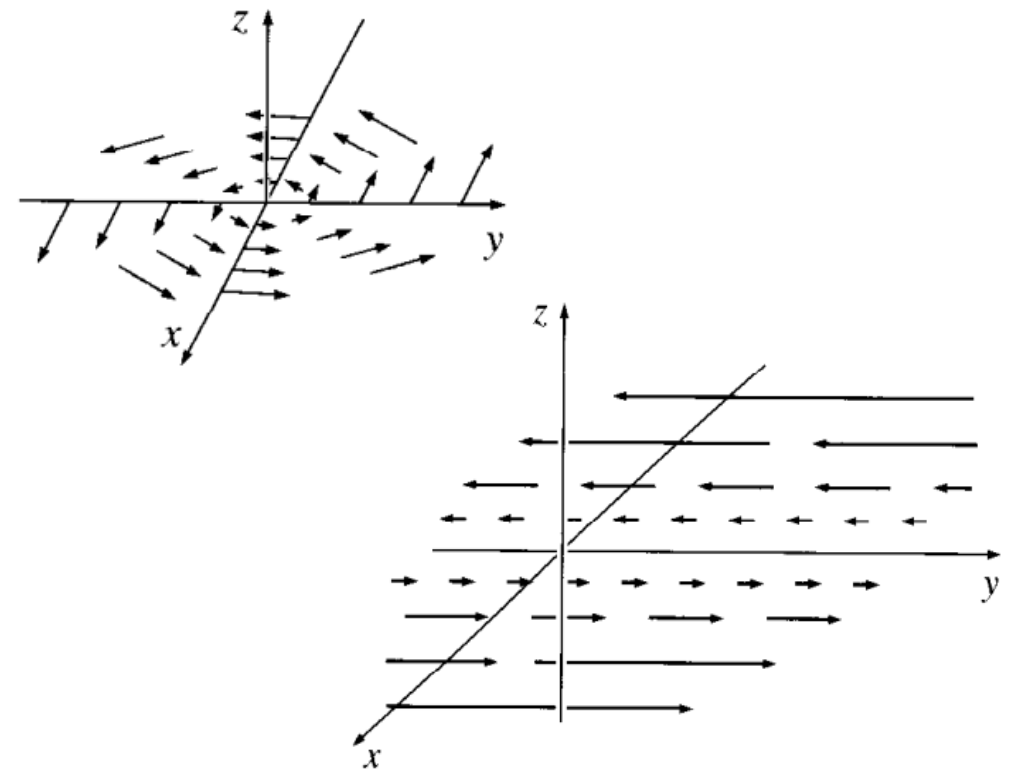
Curl of a Vector Field

The curl of a vector field is a measure of how much the vector curls around the point of observation.

Curl of vector field is zero



Curl of vector field is not zero



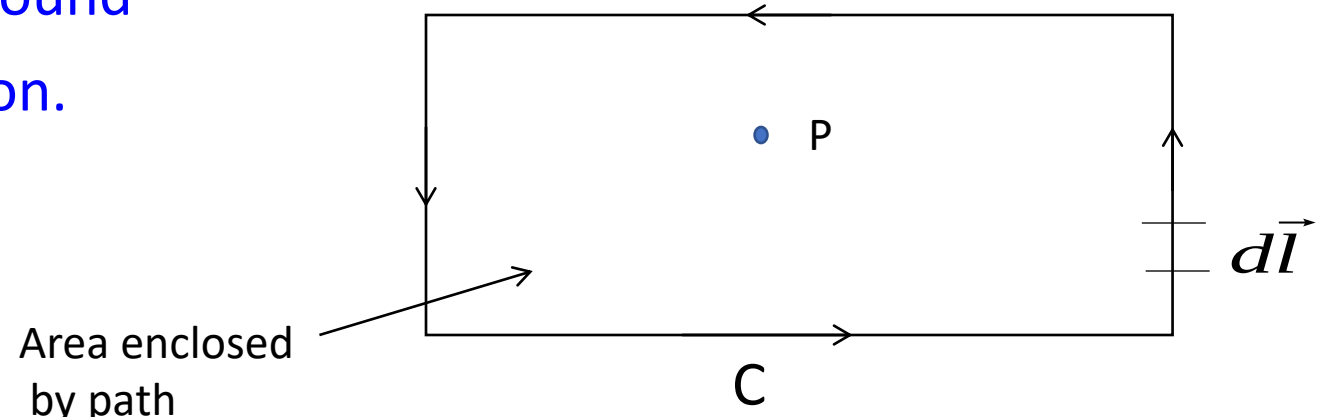
Curl of a Vector Field

- Curl of any vector field is related to line integral of that vector.
- Curl of a vector field is a vector quantity.

Consider a point P in a vector field $\vec{A}$ and path C enclose that point. $d\vec{l}$ is a small length element on this path. If area enclosed by the closed path is small so that vector field $\vec{A}$ same at each point of path then

$\oint_C \vec{A} \cdot d\vec{l}$ = line integral of vector field over a closed path.

- The line integral of a vector function around a closed path is also known as circulation.

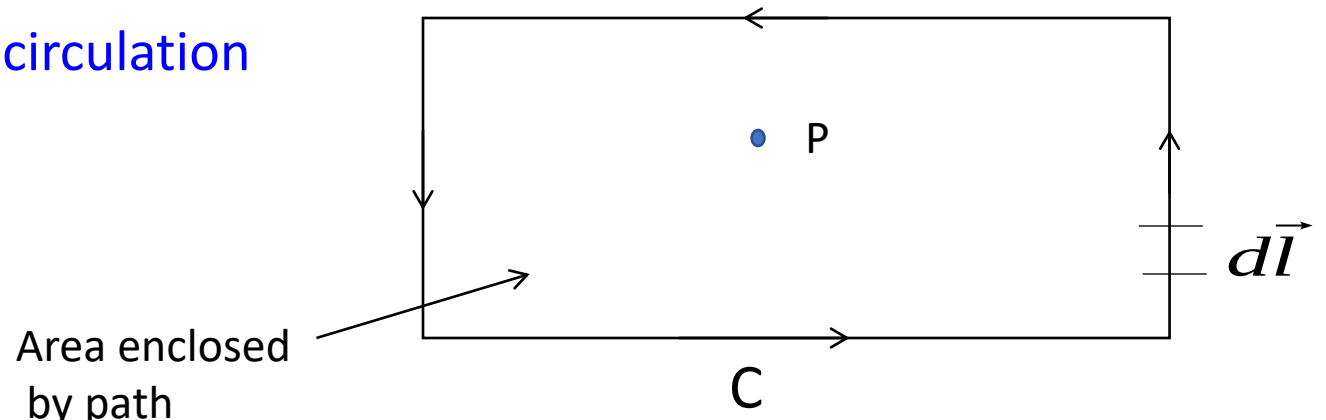


Curl of a Vector Field

- The line integral depends on the orientation of the surface enclosed by the closed path.
- If the surface enclosed by the closed path is kept in such a way that the line integral or circulation becomes maximum, then in this orientation the ratio of the line integral to the area enclosed by the path with the area tending to zero, is equal to the magnitude of the curl of the vector field at the position of the area element.

$$\text{curl} \vec{A} \cdot \hat{n} = \lim_{da \rightarrow 0} \left[\frac{\oint_C \vec{A} \cdot d\vec{l}}{da} \right]$$

- Its direction is perpendicular to the plane which contains the area element with maximum circulation around its boundary.



Curl of a Vector Field....

or

$$\text{curl} \vec{A} = \nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

Physical significance of curl:

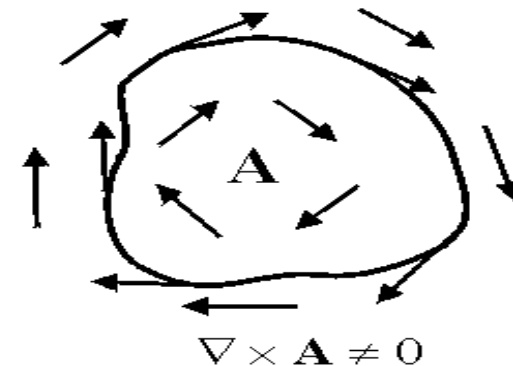
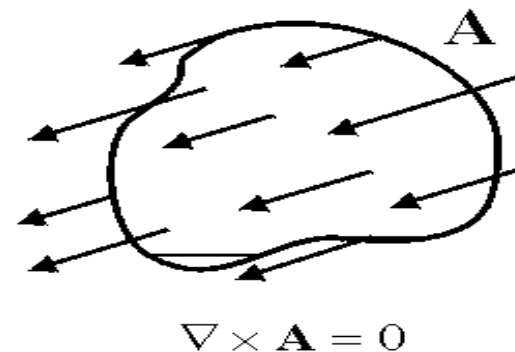
Curl of a vector field measure how much the vector $\vec{A}$ “curl around “ the point of observation. It comes from its rotational motion.

Example: For static field

$$\nabla \times \vec{E} = 0 \quad (\text{Maxwell III equation})$$

$$\nabla \times \vec{B} = \mu_o J \quad (\text{Maxwell IV equation})$$

If curl of a vector field is zero, these vectors are called irrotational vector field.



Curl of a Vector field

The **curl** of a vector field **A** is an axial (or rotational) vector whose magnitude is the maximum circulation of **A** per unit area as the area tends to zero and whose direction is the normal direction of the area when the area is oriented to make the circulation maximum.

That is

$$\text{curl } \mathbf{A} = \nabla \times \mathbf{A} = \left(\lim_{\Delta S \rightarrow 0} \frac{\oint_L \mathbf{A} \cdot d\mathbf{l}}{\Delta S} \right)_{\max} \mathbf{a}_n \quad (...1)$$

where the area ΔS is bounded by the curve L and $\mathbf{a}_n$ is the unit vector normal to the surface ΔS and is determined by using the right-hand rule.

In Cartesian coordinates the curl of **A** is easily found using

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

or

$$\nabla \times \mathbf{A} = \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] \mathbf{a}_x + \left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right] \mathbf{a}_y + \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right] \mathbf{a}_z \quad (...2)$$

Curl of a Vector field

In Cylindrical coordinates the curl of $\mathbf{A}$ is

$$\nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \mathbf{a}_\rho & \rho \mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

or

$$\nabla \times \mathbf{A} = \left[\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right] \mathbf{a}_\rho + \left[\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right] \mathbf{a}_\phi + \frac{1}{\rho} \left[\frac{\partial(\rho A_\phi)}{\partial \rho} - \frac{\partial A_\rho}{\partial \phi} \right] \mathbf{a}_z \quad (...3)$$

In Spherical coordinates the curl of $\mathbf{A}$ is

$$\nabla \times \mathbf{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{a}_r & r \mathbf{a}_\theta & r \sin \theta \mathbf{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix}$$

or

$$\nabla \times \mathbf{A} = \frac{1}{r \sin \theta} \left[\frac{\partial(A_\phi \sin \theta)}{\partial \theta} - \frac{\partial A_\theta}{\partial \phi} \right] \mathbf{a}_r + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial(r A_\phi)}{\partial r} \right] \mathbf{a}_\theta + \frac{1}{r} \left[\frac{\partial(r A_\theta)}{\partial r} - \frac{\partial A_r}{\partial \theta} \right] \mathbf{a}_\phi$$

(...4)

Curl of a Vector field

Note the following properties of the curl:

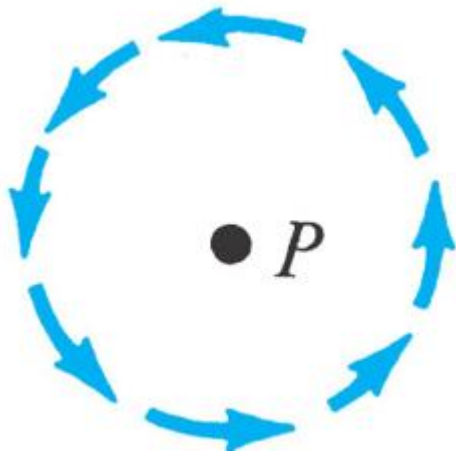
1. The curl of a vector field is another vector field.
2. $\nabla \times (\mathbf{A} + \mathbf{B}) = \nabla \times \mathbf{A} + \nabla \times \mathbf{B}$
3. $\nabla \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}) + (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B}$
4. $\nabla \times (V\mathbf{A}) = V\nabla \times \mathbf{A} + \nabla V \times \mathbf{A}$
5. The divergence of the curl of a vector field vanishes; that is, $\nabla \cdot (\nabla \times \mathbf{A}) = 0$.
6. The curl of the gradient of a scalar field vanishes; that is, $\nabla \times \nabla V = \mathbf{0}$ or $\nabla \times \nabla = \mathbf{0}$.

Curl of a Vector field

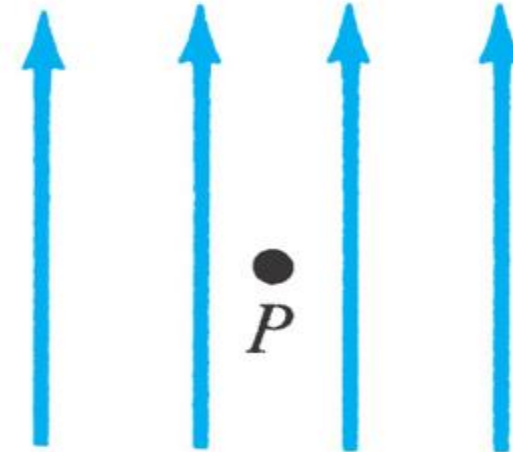
Physical Significance:

The curl provides the maximum value of the circulation of the field per unit area (or circulation density) and indicates the direction along which this maximum value occurs. The curl of a vector field $\mathbf{A}$ at a point P may be regarded as a measure of the circulation or how much the field curls around P . If the curl of a vector field is zero, these vectors are called an irrotational vector field.

For example.



This figure shows that the curl of a vector field around P is non-zero and directed out of the page.



This figure shows a vector field with zero curl.

Curl in Cartesian Coordinates

Let the value of a vector be $\vec{A}$ at any point O in the vector field and its components along the X,Y and Z axis be A_x , A_y and A_z respectively. Suppose a rectangular path KLMN of area element da is situated in the XY plain around the central point O. The length of the rectangular path along the X axis is dx and its breadth along the Y axis is dy . It is shown in the figure.

The vector field along the path KL

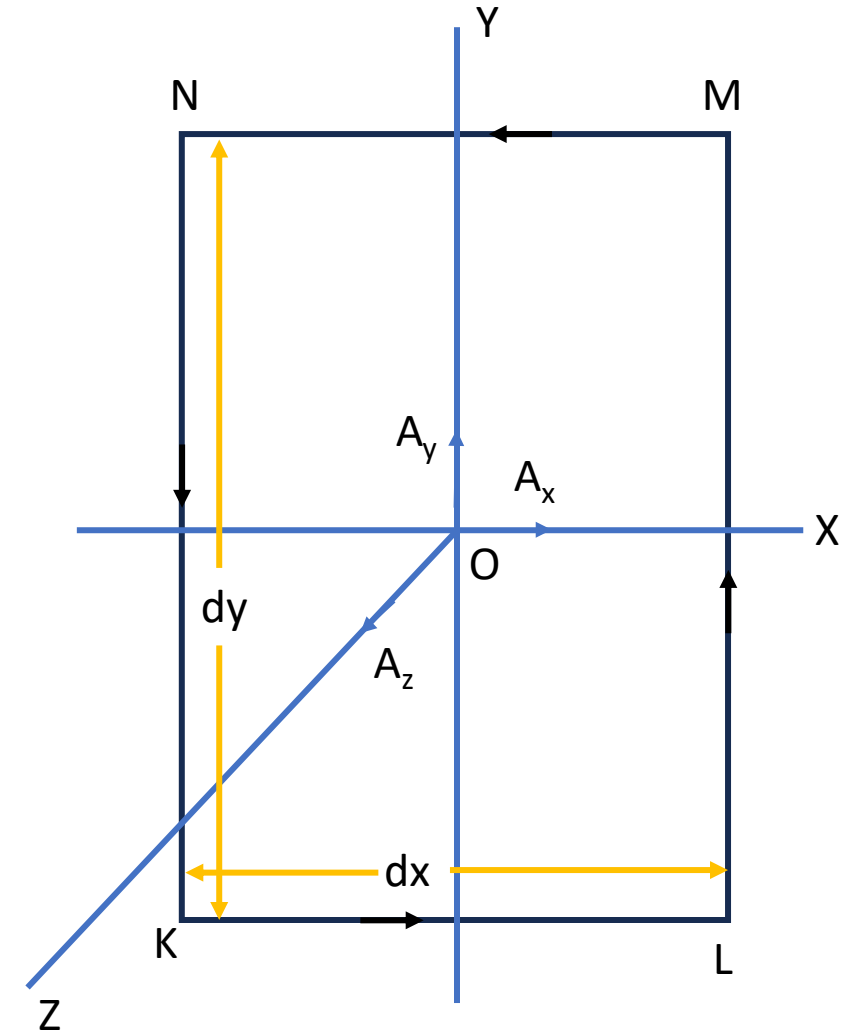
Component of the vector field along the X axis = A_x

$\therefore$ The rate of change of A_x in Y direction = $\frac{\partial A_x}{\partial y}$

$\therefore$ Decrease in the magnitude of the component A_x in traversing a distance $\left(-\frac{dy}{2}\right)$ from the central point O
$$= -\left(\frac{\partial A_x}{\partial y}\right) \frac{dy}{2}$$

$\therefore$ The component of the vector field along the path KL

$$= \left[A_x - \left(\frac{\partial A_x}{\partial y}\right) \frac{dy}{2} \right] \quad \dots(1)$$



Curl in Cartesian Coordinates

Similarly, the component of the vector field along the path MN

$$= - \left[A_x + \left(\frac{\partial A_x}{\partial y} \right) \frac{dy}{2} \right] \quad \dots(2)$$

The component of the vector field along the path LM

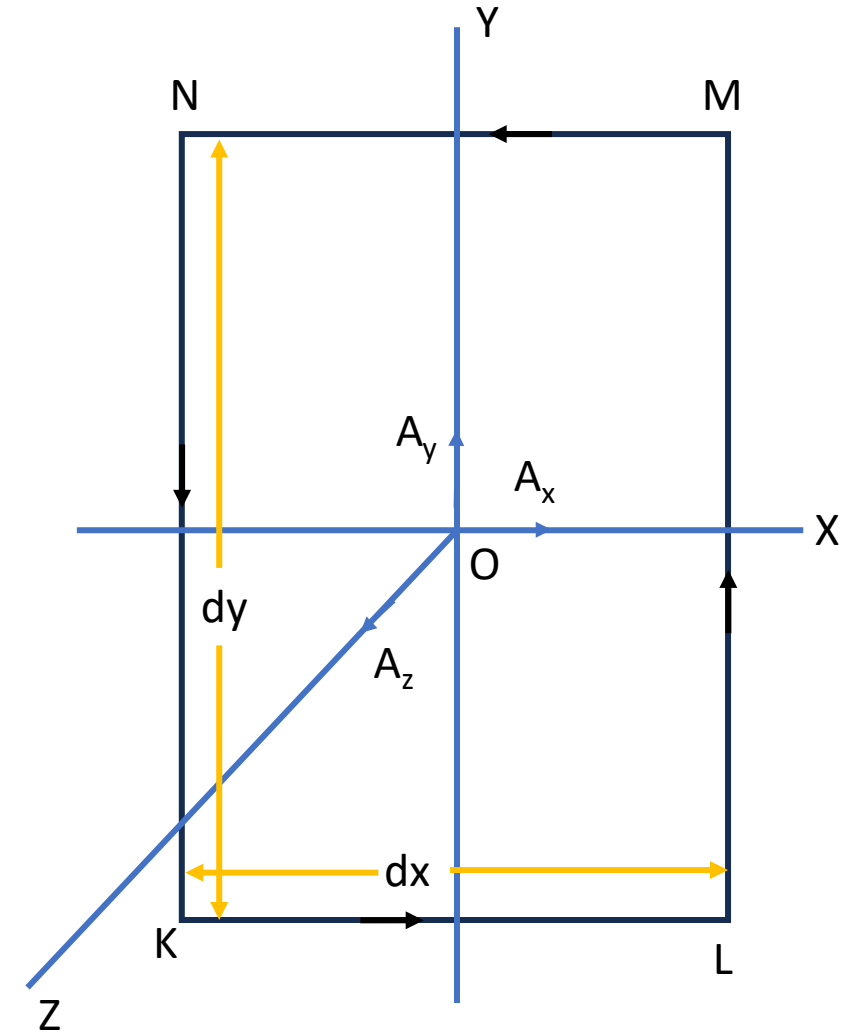
$$= \left[A_y + \left(\frac{\partial A_y}{\partial x} \right) \frac{dx}{2} \right] \quad \dots(3)$$

And the component of the vector field along the path NK

$$= - \left[A_y - \left(\frac{\partial A_y}{\partial x} \right) \frac{dx}{2} \right] \quad \dots(4)$$

We know that the line integral of the vector field $\vec{A}$ along the closed path KLMNK is equal to the sum of the line integrals of the vector field along their sides i.e.,

$$\oint_{KLMNK} \vec{A} \cdot d\vec{l} = \int_K^L \vec{A} \cdot d\vec{l} + \int_L^M \vec{A} \cdot d\vec{l} + \int_M^N \vec{A} \cdot d\vec{l} + \int_N^K \vec{A} \cdot d\vec{l}$$



Curl in Cartesian Coordinates

Substituting the vector components along the path from eq. (1) to (4)

$$\oint_{KLMNK} \vec{A} \cdot \vec{dl} = \left[A_x - \left(\frac{\partial A_x}{\partial y} \right) \frac{dy}{2} \right] dx + \left[A_y + \left(\frac{\partial A_y}{\partial x} \right) \frac{dx}{2} \right] dy \\ - \left[A_x + \left(\frac{\partial A_x}{\partial y} \right) \frac{dy}{2} \right] dx - \left[A_y - \left(\frac{\partial A_y}{\partial x} \right) \frac{dx}{2} \right] dy$$

$$\text{Or} \quad \oint_{KLMNK} \vec{A} \cdot \vec{dl} = \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) dx dy \quad \dots(5)$$

According to the definition of curl, the line integral of the vector field along a closed path having any surface element or the circulation per unit area is equal to the component of the curl of the vector field in a direction perpendicular to the plane of circulation (i.e. in the Z direction perpendicular to the XY plane)

$$(Curl \vec{A})_z = (Curl \vec{A}) \cdot \hat{k} = \lim_{dx dy \rightarrow 0} \left[\frac{1}{dx dy} \oint_{KLMNK} \vec{A} \cdot \vec{dl} \right]$$

Here $\hat{k}$ is the unit vector along the Z direction

Curl in Cartesian Coordinates

$$\text{Or} \quad \oint_{KLMNK} \vec{A} \cdot d\vec{l} = \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) dx dy \quad \dots(5)$$

According to the definition of curl, the line integral of the vector field along a closed path having any surface element or the circulation per unit area is equal to the component of the curl of the vector field in a direction perpendicular to the plane of circulation (i.e. in the Z direction perpendicular to the XY plane)

$$(curl \vec{A})_z = (curl \vec{A}) \cdot \hat{k} = \lim_{dxdy \rightarrow 0} \left[\frac{1}{dxdy} \oint_{KLMNK} \vec{A} \cdot d\vec{l} \right] \dots(6)$$

Here $\hat{k}$ is the unit vector along the Z direction

$$\text{Therefore, } (curl \vec{A})_z = (curl \vec{A}) \cdot \hat{k} = \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \dots(7)$$

Similarly if the rectangular path is situated in YZ and ZX plane, then the x- and y- components of curl $\vec{A}$ can be determined

$$(curl \vec{A})_x = (curl \vec{A}) \cdot \hat{i} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \dots(8)$$

$$\text{and } (curl \vec{A})_y = (curl \vec{A}) \cdot \hat{j} = \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \dots(9)$$

Curl in Cartesian Coordinates

From equations (7), (8) and (9) for the components of $\text{curl } \vec{A}$, we can obtain $\text{curl } \vec{A}$. Therefore

$$\begin{aligned}\text{curl } \vec{A} &= \hat{i}(\text{curl } \vec{A})_x + \hat{j}(\text{curl } \vec{A})_y + \hat{k}(\text{curl } \vec{A})_z \\ \text{curl } \vec{A} &= \hat{i} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \hat{j} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \hat{k} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \quad \dots(10)\end{aligned}$$

Equation (10) can also be written in the following form

$$\text{curl } \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

The above determinant is equal to the vector product of del operator and the vector $\vec{A}$.

$$\text{i.e.} \quad \text{curl } \vec{A} = \nabla \times \vec{A} \quad \dots(11)$$

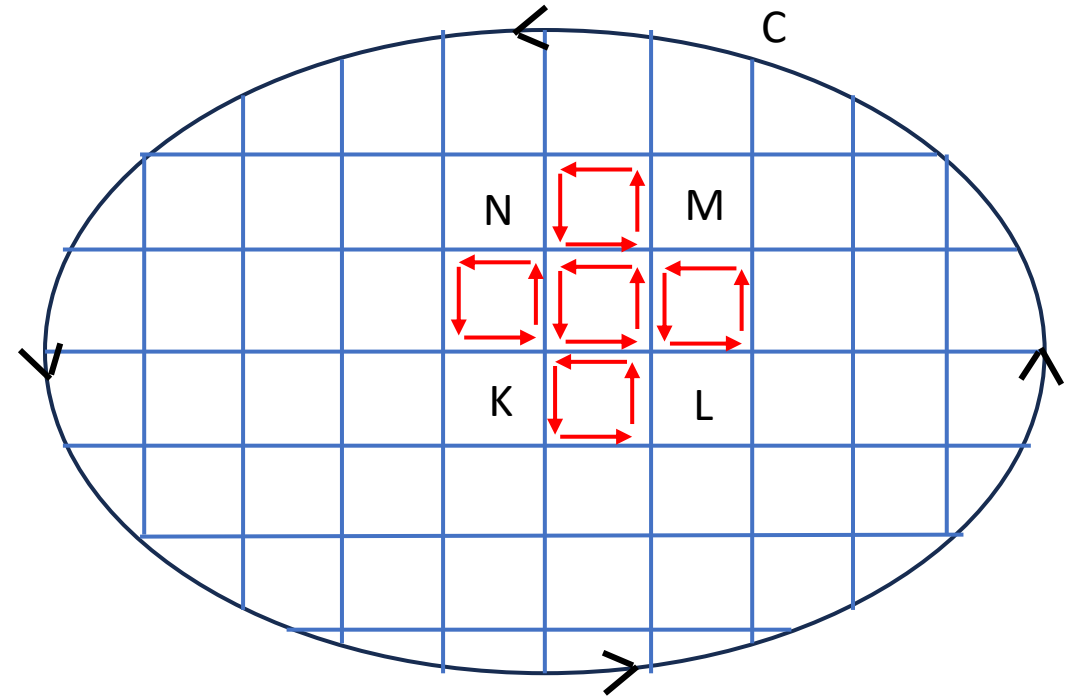
Stoke's Curl Theorem

- Suppose an area is enclosed by the closed path C in a vector field $\vec{A}$. This area can be considered to be made up of a large number of infinitesimal areas enclosed in infinitesimal closed paths (as shown in figure).
- Suppose in every area element, the circulation or curl of the vector field along the path is anticlockwise.
- Therefore the line integrals of the vector field at the common boundary of any two area elements will be mutually equal and opposite.
- For example

$$\int_K^L \vec{A} \cdot d\vec{l} = - \int_L^K \vec{A} \cdot d\vec{l}$$

$$\therefore \int_K^L \vec{A} \cdot d\vec{l} + \int_L^K \vec{A} \cdot d\vec{l} = 0$$

- Therefore, except for the outer boundary C, the sum of the line integrals along the common paths around all the area elements will be equal to zero. Hence the sum of line integrals along the boundaries of all the elements will be equal to the line integral of the vector field along the boundary C.



Stoke's Curl Theorem

$$\text{i.e.} \quad \oint_C \vec{A} \cdot d\vec{l} = \sum \oint_{dC} \vec{A} \cdot d\vec{l} \quad \dots(1)$$

According to the definition of curl,

$$\text{curl} \vec{A} \cdot \hat{n} = \lim_{da \rightarrow 0} \left[\frac{\oint_{dC} \vec{A} \cdot d\vec{l}}{da} \right]$$

$$\text{Or} \quad \oint_{dC} \vec{A} \cdot d\vec{l} = (\text{curl} \vec{A}) \cdot \hat{n} da \quad \dots(2)$$

Here $\hat{n}$ is the unit vector perpendicular to the $\vec{da}$ area element

$$\text{As } \hat{n} da = \vec{da}$$

Using equation (2) in equation (1)

$$\oint_C \vec{A} \cdot d\vec{l} = \sum (\text{curl} \vec{A}) \cdot \vec{da}$$

If the area $\vec{da}$ enclosed by the element dC tends to zero the summation be replaced by integration,

$$\oint_C \vec{A} \cdot d\vec{l} = \int_S (\text{curl} \vec{A}) \cdot \vec{da} \quad \dots(3)$$

Clearly, **the line integral of vector field along the limiting boundary of the area enclosed by a closed path is equal to the surface integral of the curl of the vector field. This is known as Stoke's Curl Theorem. This theorem is used in converting line integral into surface integral and vice versa.**

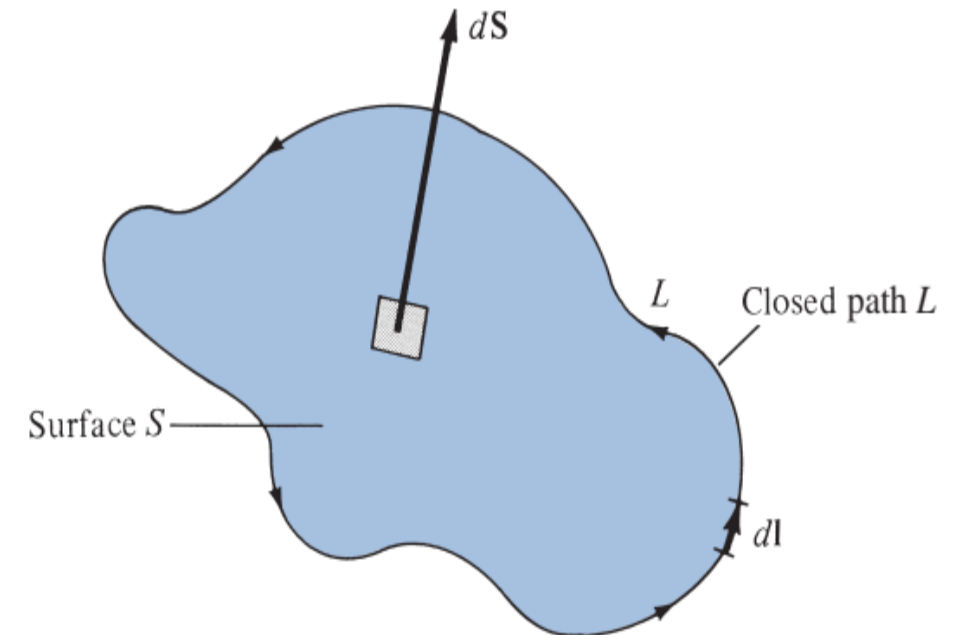
Stoke's Theorem

Also, from the definition of the curl of $\mathbf{A}$ in eq. (1), we may expect that

$$\oint_L \mathbf{A} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

This is called *Stokes's theorem*.

Stokes's theorem states that the circulation of a vector field $\mathbf{A}$ around a (closed) path L is equal to the surface integral of the curl of $\mathbf{A}$ over the open surface S bounded by), provided $\mathbf{A}$ and $\nabla \times \mathbf{A}$ are continuous on S .



Solid Angle

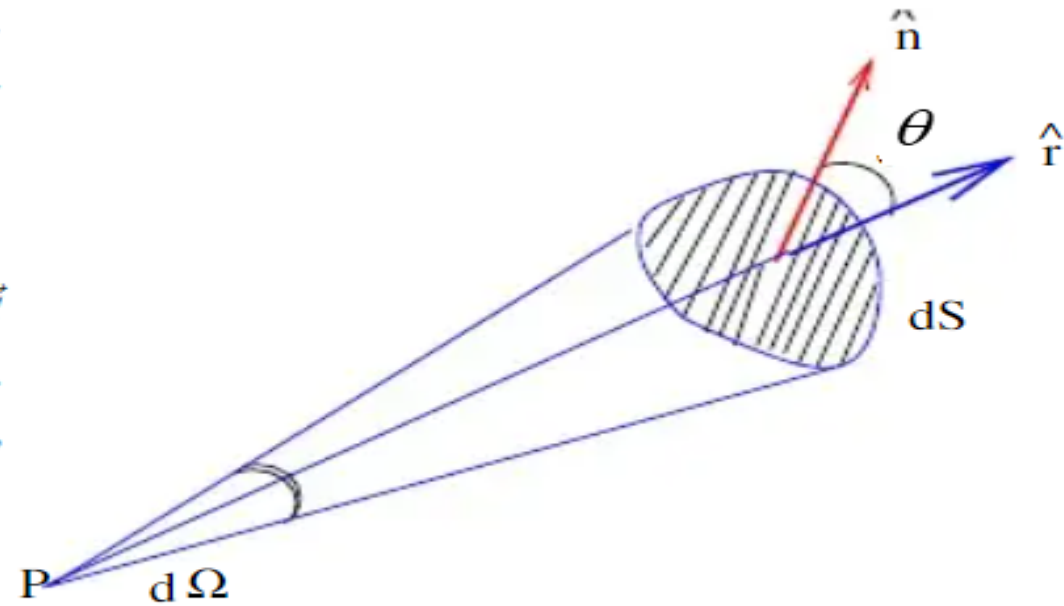
The concept of solid angle is a natural extension of a plane angle to three dimensions. Consider an area element dS at a distance r from a point P . Let $\hat{n}$ be the unit vector along the outward normal to dS .

The element of the solid angle subtended by the area element at P is defined as

$$d\Omega = \frac{dS_{\perp}}{r^2}$$

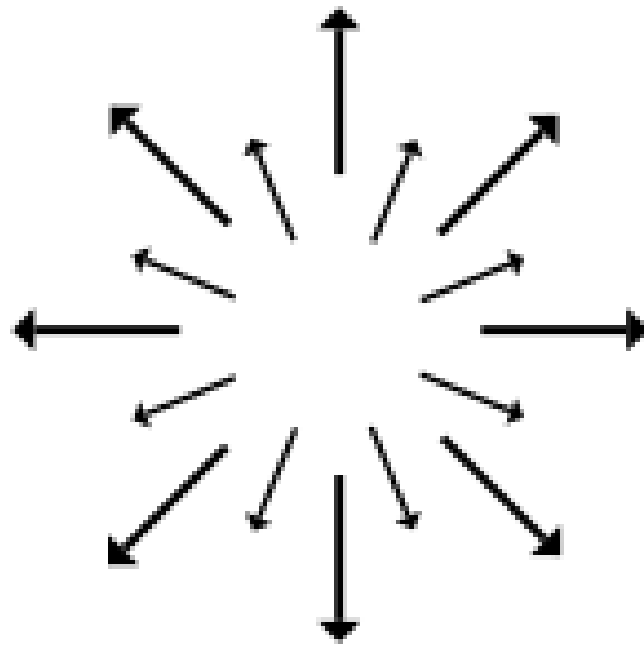
where $dS_{\perp}$ is the projection of $d\vec{S}$ along a direction perpendicular to $\vec{r}$. If θ is the angle between $\hat{r}$ and $\hat{n}$, then,

$$d\Omega = \frac{dS \cos \theta}{r^2}$$

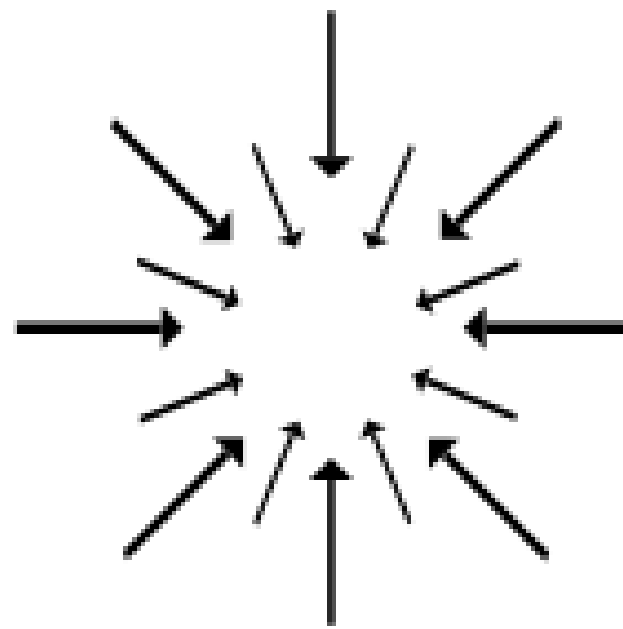


Solid angle is dimensionless. However, for practical reasons it is measured in terms of a unit called **steradian** (much like the way a planar angle is measured in terms of degrees).

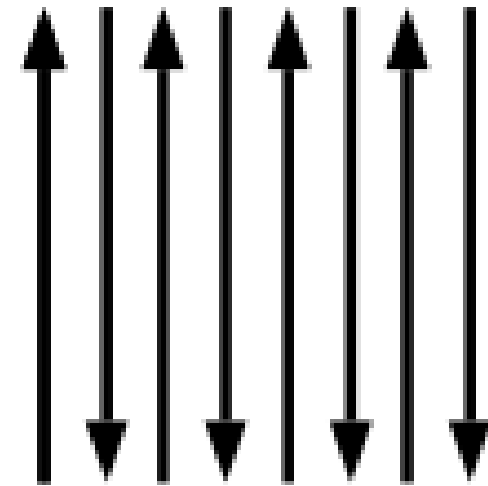
The maximum possible value of solid angle is 4π , which is the angle subtended by an area which encloses the point P completely.



$$\begin{aligned}\frac{\partial}{\partial x}(\mathbf{V}_x) &> 0 \\ \frac{\partial}{\partial y}(\mathbf{V}_y) &> 0 \\ \nabla \cdot (\mathbf{V}) &> 0\end{aligned}$$



$$\begin{aligned}\frac{\partial}{\partial x}(\mathbf{V}_x) &< 0 \\ \frac{\partial}{\partial y}(\mathbf{V}_y) &< 0 \\ \nabla \cdot (\mathbf{V}) &< 0\end{aligned}$$



$$\begin{aligned}\frac{\partial}{\partial x}(\mathbf{V}_x) &= 0 \\ \frac{\partial}{\partial y}(\mathbf{V}_y) &= 0 \\ \nabla \cdot (\mathbf{V}) &= 0\end{aligned}$$

Divergence of Vector Field

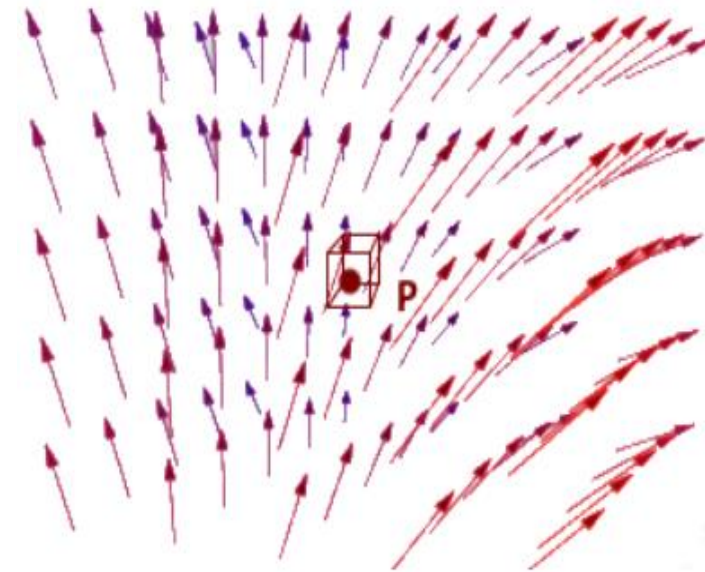
- The net outflow of the flux of a vector field $\mathbf{A}$ from a closed surface S is obtained from the integral $\oint_S \vec{A} \cdot \vec{dS}$. We now define the divergence of $\mathbf{A}$ as the net outward flow of flux per unit volume over a closed incremental surface.
- The **divergence** of $\mathbf{A}$ at a given point P is the *outward* flux per unit volume as the volume shrinks about P .

Hence,

$$\text{div } \mathbf{A} = \nabla \cdot \mathbf{A} = \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta v} \quad (\dots 1)$$

where Δv is the volume enclosed by the closed surface S in which P is located.

- Physically, we may regard the divergence of the vector field $\mathbf{A}$ at a given point as a measure of how much the field diverges or emanates or originates from that point.

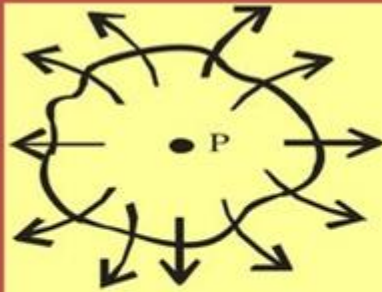


(i) Divergence

➤ The divergence of a vector $\mathbf{V}$ written as $\text{div } \mathbf{V}$ represents the

$$\text{div } \mathbf{V} = \nabla \cdot \mathbf{V} =$$

$$\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$



(a) positive divergence



(b) negative divergence



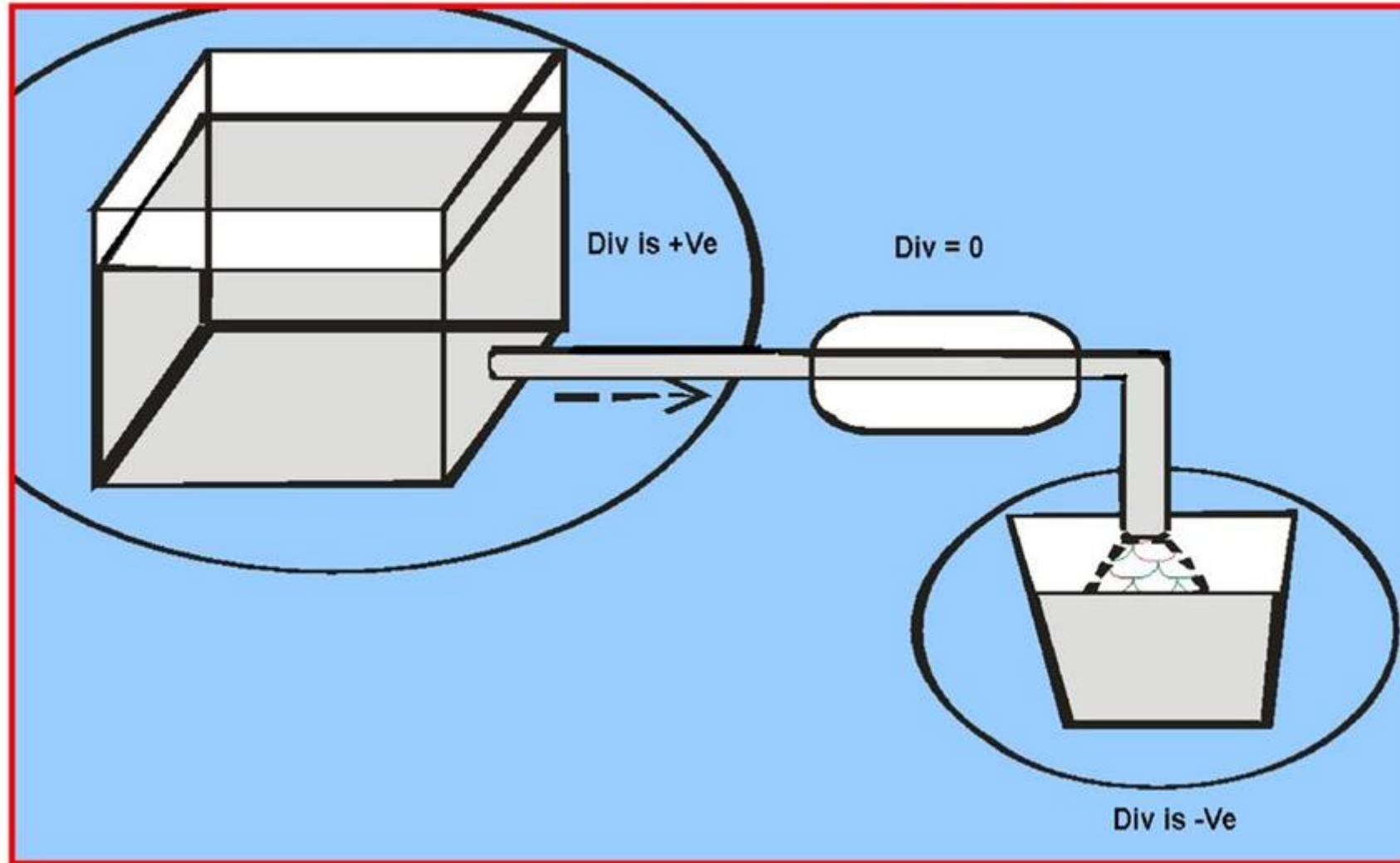
(c) zero divergence

Physical significance of divergence

- Physically the divergence of a vector quantity represents the rate of change of the field strength in the direction of the field.
- If the divergence of the vector field is positive at a point then something is diverging from a small volume surrounding with the point as a source.
- If it negative, then something is converging into the small volume surrounding that point is acting as sink.
- if the divergence at a point is zero then the rate at which something entering a small volume surrounding that point is equal to the rate at which it is leaving that volume.
- The vector field whose divergence is zero is called ***solenoidal***

$$\text{div } \mathbf{A} = \nabla \cdot \mathbf{A} = \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta v}$$

Example for Divergence

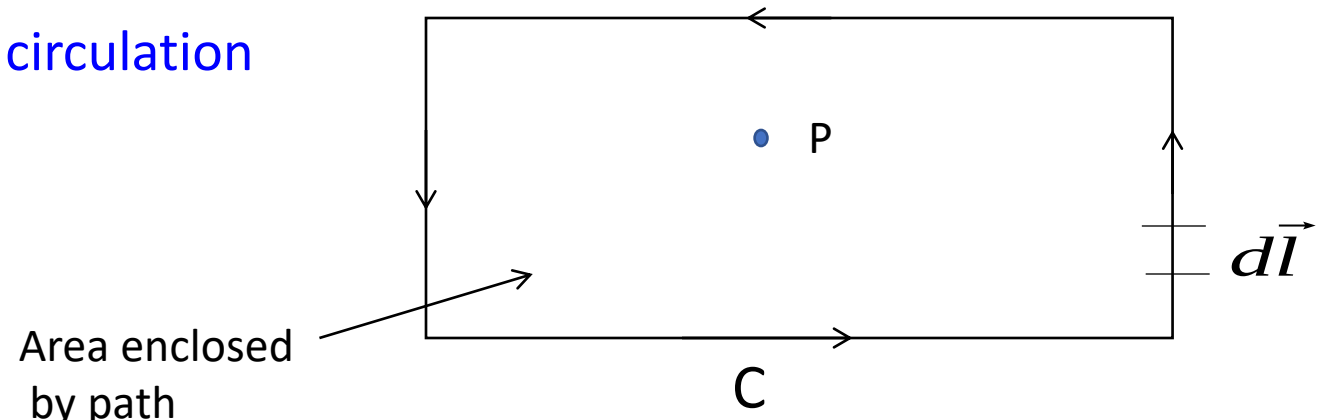


Curl of a Vector Field

- The line integral depends on the orientation of the surface enclosed by the closed path.
- If the surface enclosed by the closed path is kept in such a way that the line integral or circulation becomes maximum, then in this orientation the ratio of the line integral to the area enclosed by the path with the area tending to zero, is equal to the magnitude of the curl of the vector field at the position of the area element.

$$\text{curl} \vec{A} \cdot \hat{n} = \lim_{da \rightarrow 0} \left[\frac{\oint_C \vec{A} \cdot d\vec{l}}{da} \right]$$

- Its direction is perpendicular to the plane which contains the area element with maximum circulation around its boundary.



Curl of a Vector Field....

or

$$\text{curl} \vec{A} = \nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

Physical significance of curl:

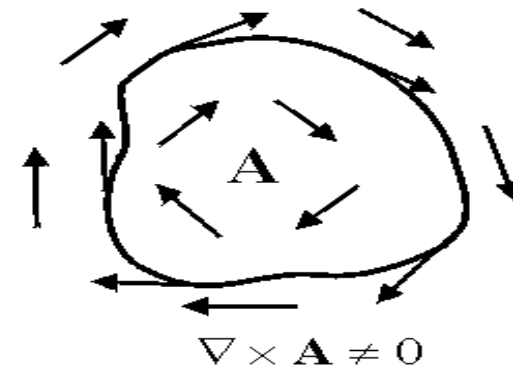
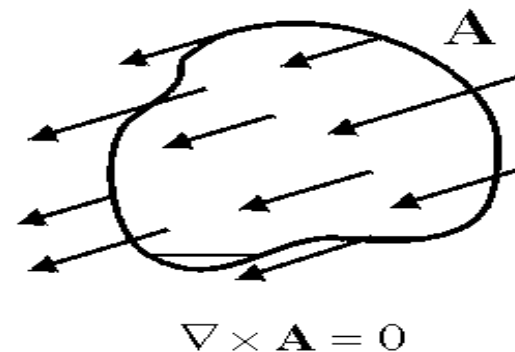
Curl of a vector field measure how much the vector $\vec{A}$ “curl around “ the point of observation. It comes from its rotational motion.

Example: For static field

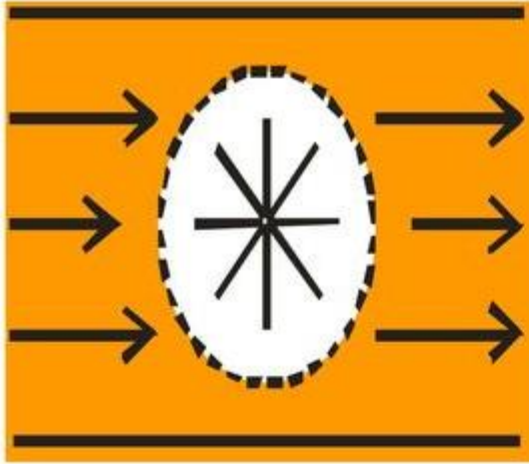
$$\nabla \times \vec{E} = 0 \quad (\text{Maxwell III equation})$$

$$\nabla \times \vec{B} = \mu_o J \quad (\text{Maxwell IV equation})$$

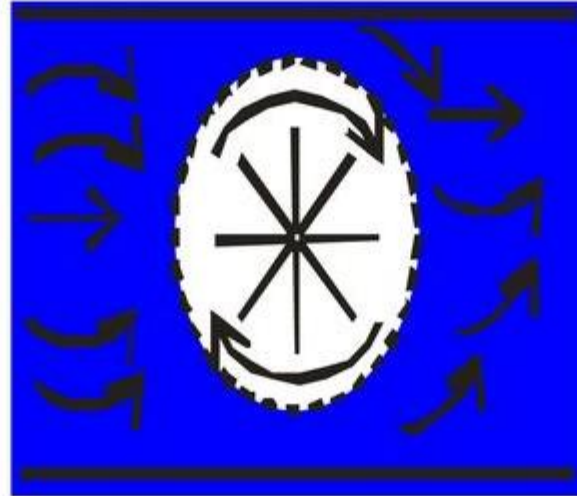
If curl of a vector field is zero, these vectors are called irrotational vector field.



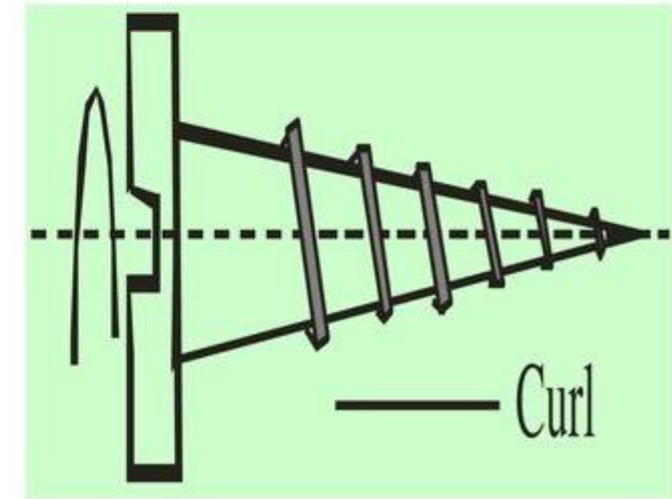
Representation of Curl



(I) No rotation of the paddle wheel represents zero curl



(II) Rotation of the paddle wheel showing the existence of curl



(III) direction of curl

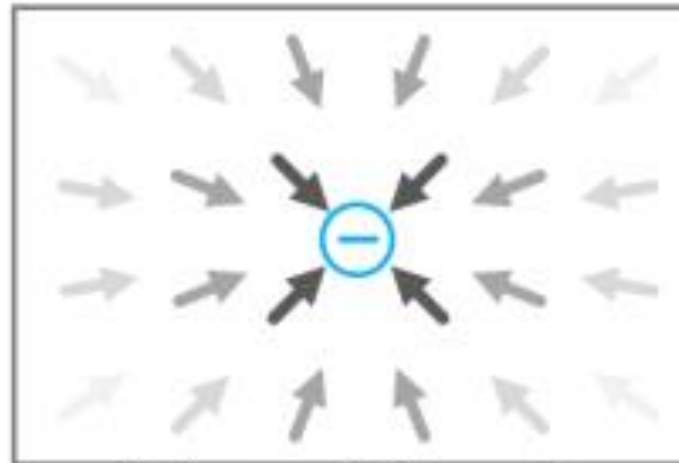
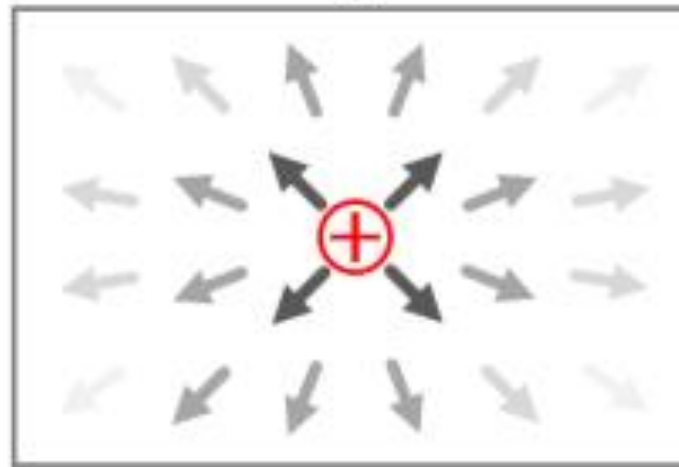
➤ For vector fields whose curl is zero there is no rotation of the paddle wheel when it is placed in the field, Such fields are called *irrotational*

Gradient



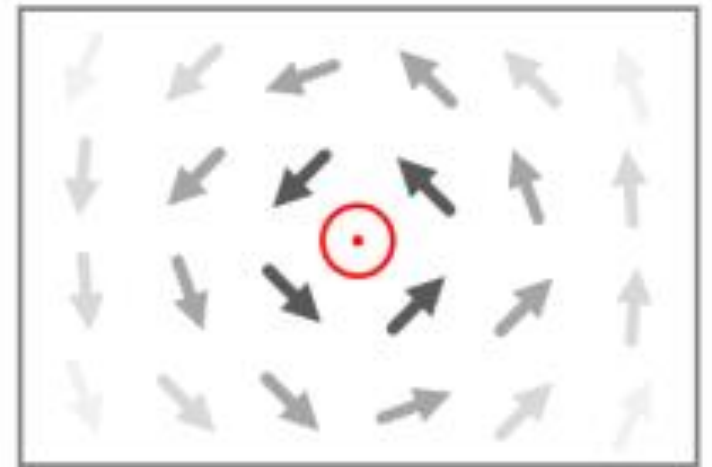
$\text{grad (scalar field)} = \text{vector}$

Divergence



$\text{div (vector field)} = \text{scalar}$

Curl



$\text{curl (vector field)} = \text{vector}$



Grad, Div, and the Curl

	Cartesian Coordinates	Cylindrical Coordinates	Spherical Coordinates
Conversion to Cartesian Coordinates		$x = \rho \cos \varphi \quad y = \rho \sin \varphi \quad z = z$	$x = r \cos \varphi \sin \theta \quad y = r \sin \varphi \sin \theta$ $z = r \cos \theta$
Vector A	$A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}$	$A_\rho \hat{\rho} + A_\varphi \hat{\varphi} + A_z \hat{z}$	$A_r \hat{r} + A_\theta \hat{\theta} + A_\varphi \hat{\varphi}$
Gradient $\nabla \phi$	$\frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k}$	$\frac{\partial \phi}{\partial \rho} \hat{\rho} + \frac{1}{\rho} \frac{\partial \phi}{\partial \varphi} \hat{\varphi} + \frac{\partial \phi}{\partial z} \hat{z}$	$\frac{\partial \phi}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial \phi}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial \phi}{\partial \varphi} \hat{\varphi}$
Divergence $\nabla \cdot A$	$\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$	$\frac{1}{\rho} \frac{\partial(\rho A_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial A_\varphi}{\partial \varphi} + \frac{\partial A_z}{\partial z}$	$\frac{1}{r^2} \frac{\partial(r^2 A_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial A_\theta \sin \theta}{\partial \theta}$ $+ \frac{1}{r \sin \theta} \frac{\partial A_\varphi}{\partial \varphi}$
Curl $\nabla \times A$	$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$	$\begin{vmatrix} \frac{1}{\rho} \hat{\rho} & \hat{\varphi} & \frac{1}{\rho} \hat{z} \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \varphi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\varphi & A_z \end{vmatrix}$	$\begin{vmatrix} \frac{1}{r^2 \sin \theta} \hat{r} & \frac{1}{r \sin \theta} \hat{\theta} & \frac{1}{r} \hat{\varphi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \varphi} \\ A_r & r A_\theta & r A_\varphi \sin \theta \end{vmatrix}$

Gauss Divergence Theorem

- It relates the volume integral of the divergence of a vector $\mathbf{V}$ to the surface integral of the vector itself.
- According to this theorem, if a closed S bounds a volume τ , then

$$\int_{\tau} \text{div } \mathbf{V} \, d\tau = \mathbf{V} \bullet d\mathbf{s}$$

(or)

$$\int_{\tau} (\nabla \bullet \mathbf{V}) d\tau = \int_S \mathbf{V} \bullet d\mathbf{s}$$

Stoke's Theorem

- It relates the surface integral of the curl of a vector to the line integral of the vector itself.
- According to this theorem, for a closed path C bounds a surface S ,

$$\int_S (\text{curl } \mathbf{V}) \bullet d\mathbf{s} = \oint_C \mathbf{V} \bullet d\mathbf{l}$$

Gauss's Law

Q.3(b) State and define Gauss law in integral and differential form.

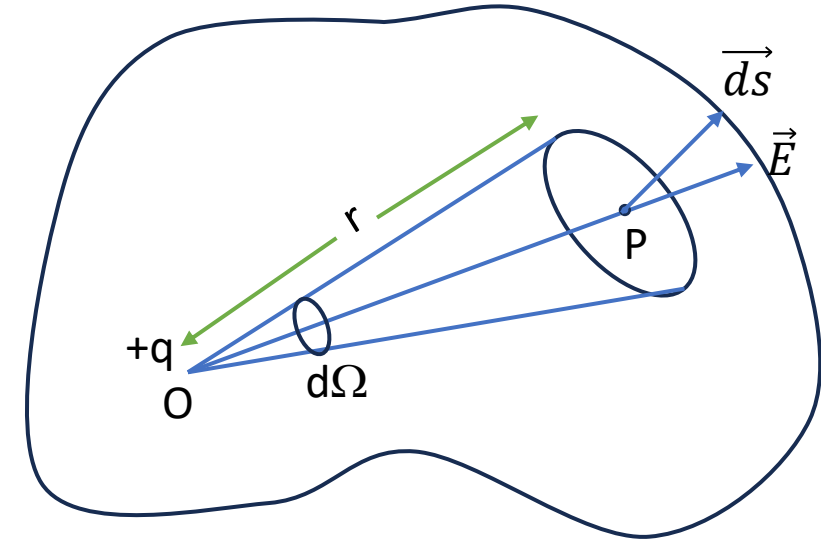
According to the Gauss's law for electrostatics, the total electric flux emerging from a closed surface situated in an electric field is equal to $\frac{1}{\epsilon}$ times the charge enclosed by it.

Suppose a point charge $+q$ placed at some point O is enclosed by an imaginary closed surface S (as shown in figure). The whole surface S can be divided into a number of small area elements. One of these area elements having area equal to $d\vec{s}$ is situated at P .

If the distance of the area element from the point charge $+q$ is r , then using the inverse square law, the intensity of electric field at the area element will be,

$$\vec{E} = \frac{1}{4\pi\epsilon} \frac{q}{r^2} \hat{r} \quad \dots(1)$$

Here $\hat{r}$ is the unit vector along OP and ϵ is the permittivity of the medium.



Gauss's Law

The electric flux emerging from the area element $\vec{ds}$ is given by

$$d\phi = \vec{E} \cdot \vec{ds}$$

or
$$d\phi = \frac{1}{4\pi\epsilon} \frac{q}{r^2} \hat{r} \cdot \vec{ds} \quad \dots(2)$$

If θ is the angle between $\hat{r}$ and $\vec{ds}$, then

$$\hat{r} \cdot \vec{ds} = |\hat{r}| ds \cos \theta = ds \cos \theta \quad (\text{from the concept of solid angle})$$

So that the electric flux
$$d\phi = \frac{1}{4\pi\epsilon} \frac{q}{r^2} ds \cos \theta \quad \dots(3)$$

From the definition of solid angle, the solid angle subtended at a point O by an area element ds at distance r is given by,

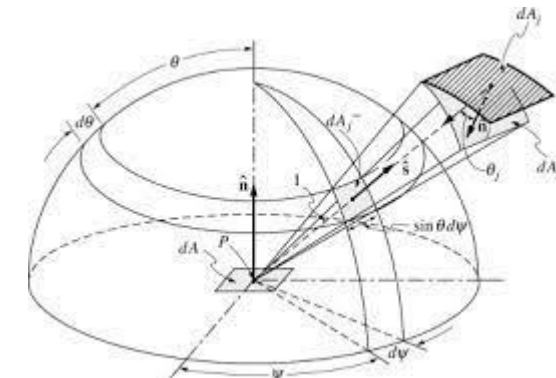
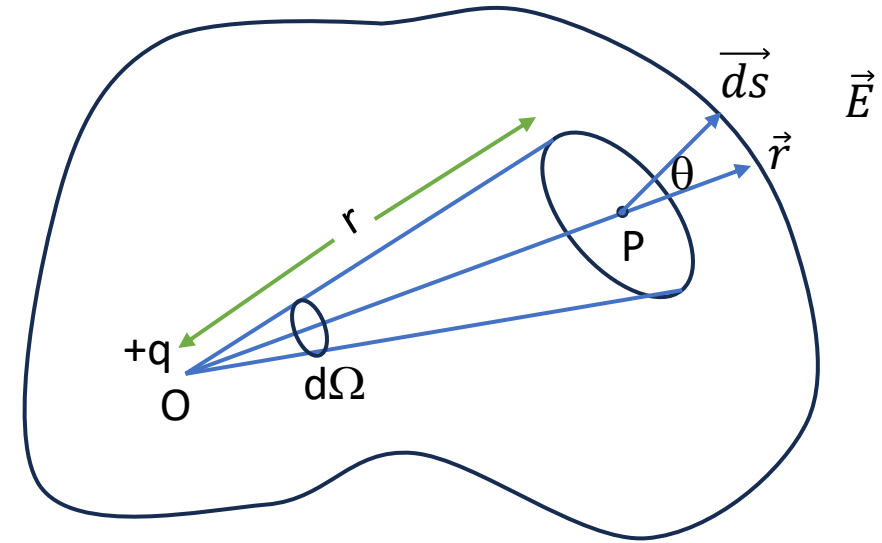
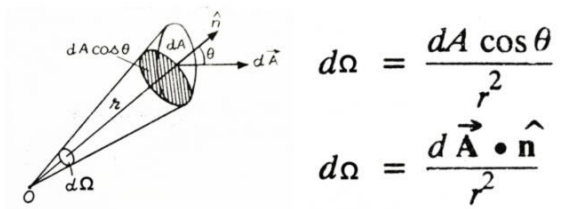
$$d\Omega = \frac{da}{r^2} = \frac{ds \cos \theta}{r^2}$$

$\therefore$ Electric flux emerging from the area element,

$$d\phi = \frac{q}{4\pi\epsilon} d\Omega \quad \dots(4)$$

Presented by Dr.Dheeraj Bhardwaj

Solid angle (steradian (sr))



Gauss's Law....

Since the Electric flux emerging from the area element,

$$d\phi = \frac{q}{4\pi\epsilon} d\Omega \quad \dots(4)$$

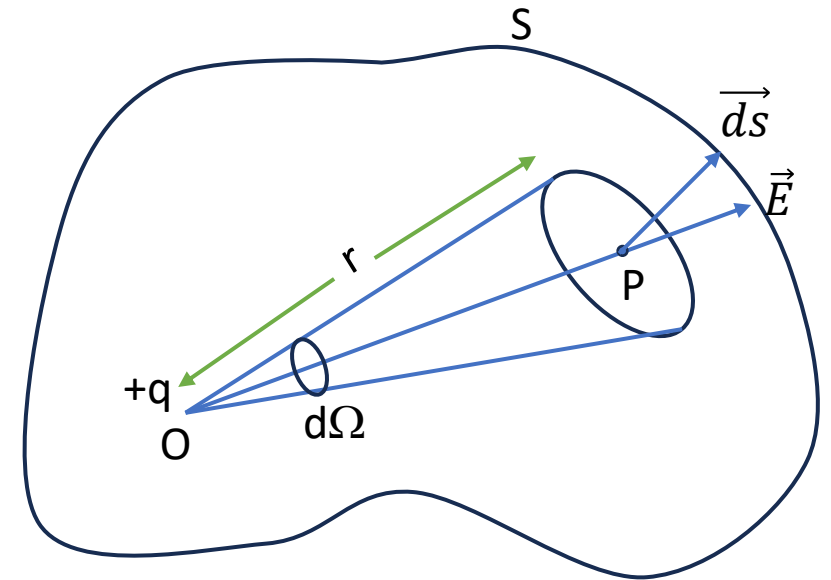
Electric flux is a scalar quantity, therefore the total flux emerging out from the closed surface will be equal to the sum of flux emerging from all the elements of the surface.

$\therefore$ Total flux emerging from the surface S,

$$\begin{aligned} \oint_S d\phi &= \oint_S \frac{q}{4\pi\epsilon} d\Omega \\ \phi &= \frac{q}{4\pi\epsilon} \oint_S d\Omega \quad \dots(5) \end{aligned}$$

But the solid angle ($d\Omega$) formed by a closed surface at any point situated inside it is equal to 4π

$$\begin{aligned} \therefore \phi &= \frac{q}{4\pi\epsilon} 4\pi \\ \phi &= \frac{q}{\epsilon} \quad \dots(6) \end{aligned}$$



Therefore, the total electric flux emerging from a closed surface situated in an electric field is equal to $\frac{1}{\epsilon}$ times the charge enclosed by it. This law is also applicable to charges distributed inside a closed surface.

Gauss's Law in Differential Form

Gauss's law expresses the relation between the electric flux emerging from an imaginary closed surface situated in an electric field and the charge existing inside this closed surface. **According to Gauss law, "the flux emerging out of a closed surface situated in an electric field is equal to the product of charge enclosed by the closed surface and $\frac{1}{\epsilon}$. Here ϵ is the permittivity of the medium".**

For vacuum, $\epsilon = \epsilon_0 = \frac{1}{36\pi} \times 10^{-9} \frac{C^2}{N \times m^2}$ or Farad/m

Suppose several charges $q_1, q_2, q_3, \dots$ etc. are enclosed by an imaginary closed surface S .

$\therefore$ Total enclosed charge $q = q_1 + q_2 + q_3 + \dots$

If the intensity of electric field at the closed surface due to these charges is $\vec{E}$ then the total electric flux emerging from the closed surface S will be,

$$\phi = \oint_S \vec{E} \cdot \vec{da} \quad \dots(1)$$

According to Gauss's law

$$\phi = \oint_S \vec{E} \cdot \vec{da} = \frac{q}{\epsilon} \quad \dots(2)$$

This is integral form of Gauss's Law

Gauss's Law in Differential Form

$$\phi = \oint_S \vec{E} \cdot \vec{da} = \frac{q}{\epsilon} \quad \dots(2)$$

If the charge enclosed by the closed surface is distributed in the enclosed volume and ρ is the charge density, then

$$\text{charge density} = \frac{\text{charge}}{\text{volume}}$$

or

$$\rho = \frac{dq}{dV} \quad \dots(3)$$

Here dq is the amount of charge in the volume element dV .

$$\therefore dq = \rho dV$$

$\therefore$ Total amount of charge in the volume enclosed by the complete surface S ,

$$q = \oint_S dq = \oint_V \rho dV \quad \dots(4)$$

Substituting equation (4) in equation (2),

$$\phi = \oint_S \vec{E} \cdot \vec{da} = \frac{1}{\epsilon} \oint_V \rho dV \quad \dots(5)$$

Gauss's Law in Differential Form

$$\phi = \oint_S \vec{E} \cdot \vec{da} = \frac{1}{\epsilon} \oint_V \rho dV \quad \dots(5)$$

Using **Gauss divergence theorem** for changing the surface integral of the electric field into volume integral,

$$\int_S \vec{E} \cdot \vec{da} = \int_V (\nabla \cdot \vec{E}) dV \quad \dots(6)$$

Comparing equations (5) and (6),

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon} \quad \dots(7)$$

The above equation is the differential form of Gauss's law.

Applications of Gauss's Law

Gauss law can be used to find electric field intensity at some distance due to different charge distribution.

Step to determine Electric field intensity:

- Draw a suitable Gaussian surface enclosing the point where we want to calculate electric field intensity.

Gaussian Surface:

An imaginary closed surface of any shape drawn in an electric field to solve problems concerning electric flux is called Gaussian surface. The shape of the Gaussian surface is chosen on the basis of symmetry of the problem, so that the expression for Gauss's law

$$\phi = \oint_S \vec{E} \cdot \vec{da} \quad \text{can be evaluated conveniently.}$$

- Calculate total electric flux using above formula.
- Calculate electric flux using equation $\phi = \frac{q}{\epsilon}$

- Compare the two equations to get electric field intensity

Applications of Gauss's Law

- Deduction of Coulomb's Law (Point Charge)
- Electric field due to a charged solid sphere (Uniformly Charged Sphere)
- Field due to line of charge
- Field due to uniformly charged plane (sheet of charge)
- Field due to infinite charged conducting plate
- Field due to two parallel charged plates.
- Field due to a spherically symmetric charge distribution (Volume distribution of charge)

Applications of Gauss's Law

Deduction of Coulomb's Law (Point Charge)

Let us consider electric field due to a single isolated positive charge of magnitude q . Field of a positive charge is in radially outward direction everywhere and magnitude of electric field intensity is same for all points at a distance r from the charge. We can assume Gaussian surface to be sphere of radius r enclosing the charge q . From Gauss's law

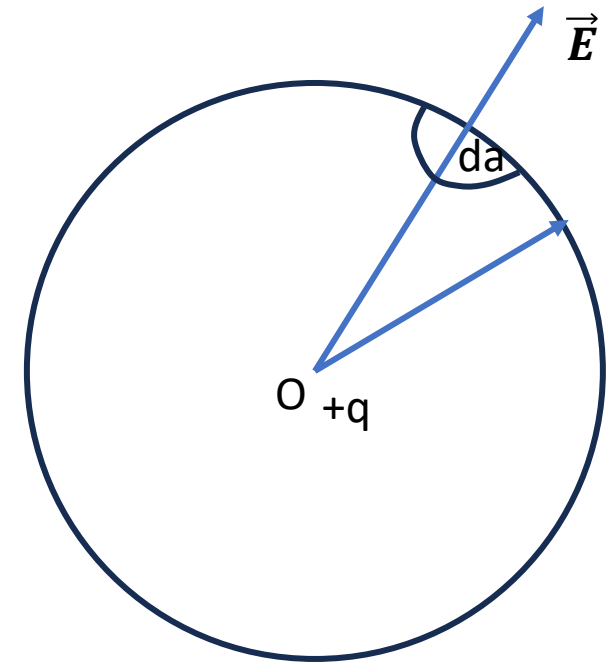
$$\oint \vec{E} \cdot \vec{da} = E \oint \vec{da} = \frac{q}{\epsilon_0}$$

Since E is constant at all points on the surface therefore,

$$EA = \frac{q}{\epsilon_0} \quad \text{or} \quad E = \frac{q}{\epsilon_0 A}$$

Since surface area of the sphere is $A = 4\pi r^2$, thus

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$



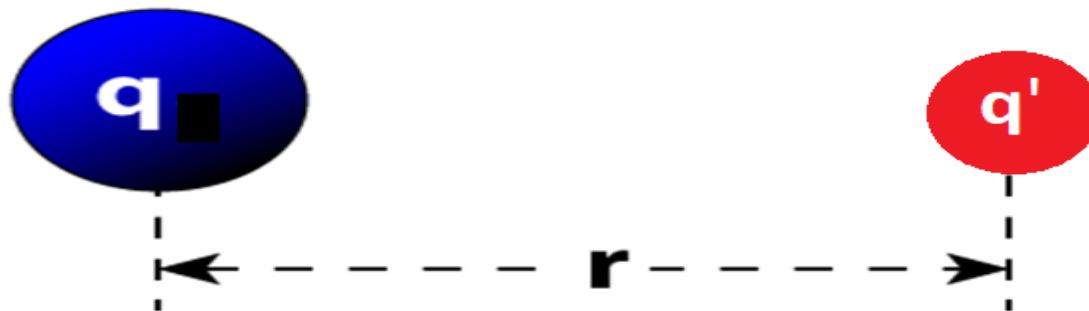
Applications of Gauss's Law

Now the force acting on a point charge q' at a distance r from point charge q is

$$F = q'E$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{qq'}{r^2}$$

This is nothing but the mathematical statement of Coulomb's law



Applications of Gauss's Law

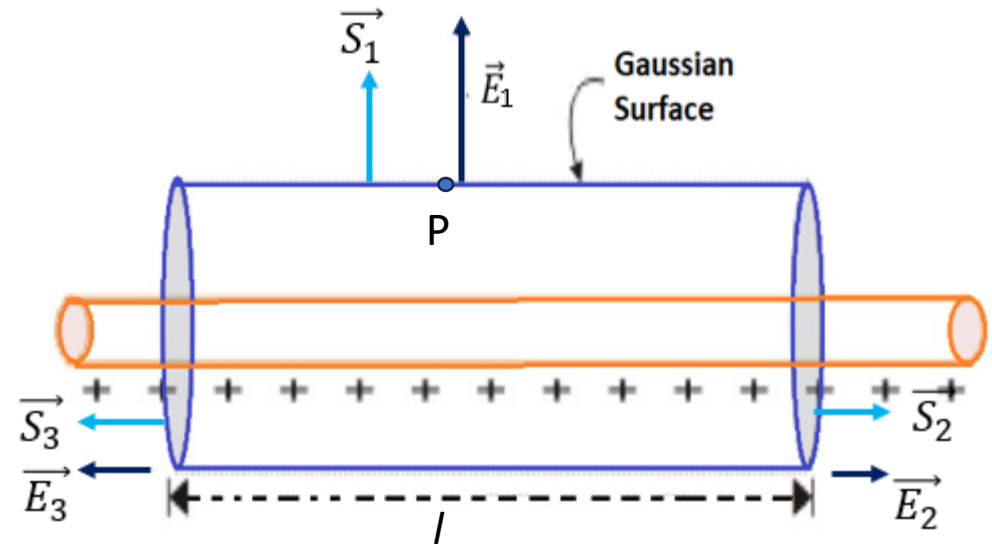
Electric Field due to line charge (Infinite Line Charge)

Let us consider a section of an infinite rod of charge with the linear charge density λ (charge per unit length) constant for all points on the line. Let the line charge is lying parallel to x- axis. Consider a point P at a distance r from the linear charge at which electric field strength has to be calculated. We choose a circular cylinder of radius r and length l as a Gaussian surface, closed at each end normal to the axis. The Gaussian surface can be divided into three sections S_1 where P is lying and S_2 and S_3 , the surface of the ends of Gaussian cylinder. The total charge enclosed by the Gaussian surface is $q = \lambda l$.

Now applying Gauss's law, we get

$$\oint \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$$

$$\text{or} \quad \left[\int_{S_1} \vec{E}_1 \cdot d\vec{s}_1 + \int_{S_2} \vec{E}_2 \cdot d\vec{s}_2 + \int_{S_3} \vec{E}_3 \cdot d\vec{s}_3 \right] = \frac{q}{\epsilon_0}$$



Cylindrical Gaussian Surface for calculation of electric field due to line charge

Applications of Gauss's Law

Electric Field due to line charge

Because of the symmetry $\vec{E}_2 = \vec{E}_3 = 0$

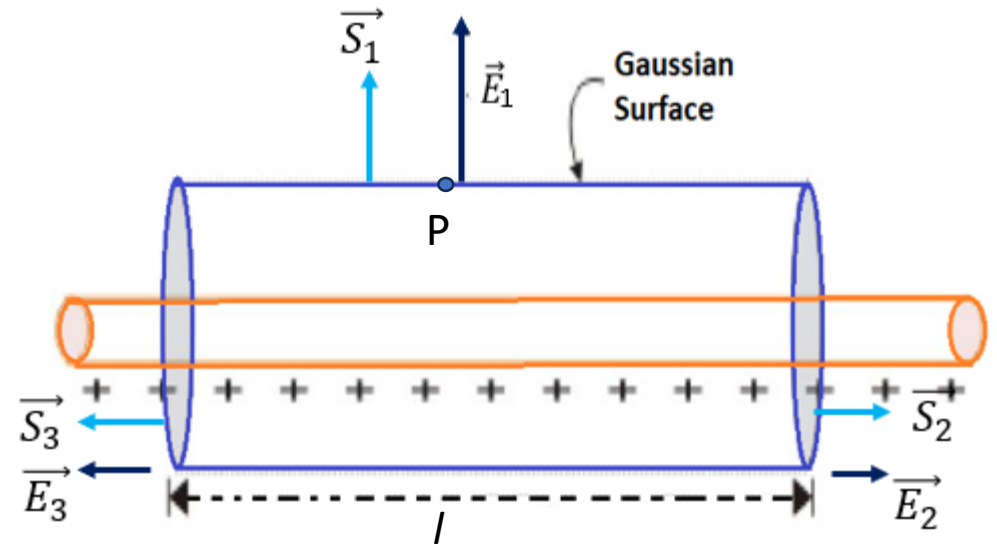
And let $\vec{E}_1 = \vec{E}$, then

$$\int_{S_1} \vec{E} \cdot d\vec{s}_1 = \frac{q}{\epsilon_0}$$

Or
$$\int_{S_1} \vec{E} \cdot d\vec{s}_1 = \frac{\lambda l}{\epsilon_0}$$

Since E is constant at the surface and $\vec{E}$ and S_1 are parallel everywhere, we have

$$\int_{S_1} \vec{E} \cdot d\vec{s}_1 = E \int_{S_1} d\vec{s}_1 = \frac{\lambda l}{\epsilon_0}$$



Cylindrical Gaussian Surface for calculation of electric field due to line charge

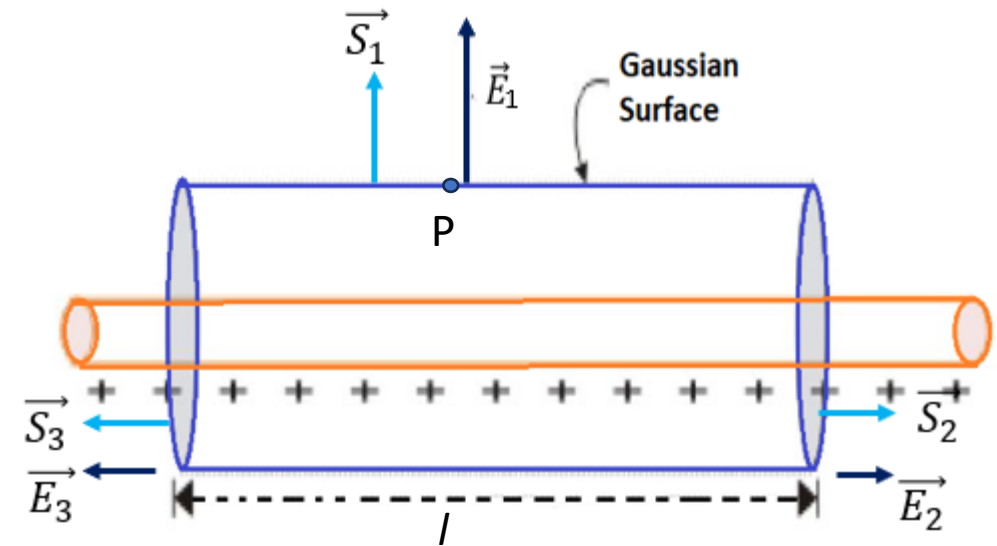
Applications of Gauss's Law

But $\int_{S_1} \vec{ds}_1 = 2\pi r l$

$\therefore E 2\pi r l = \frac{\lambda l}{\epsilon_0}$

Or $E = \frac{\lambda}{2\epsilon_0 \pi r}$

Which is radially directed along the cylinder.



Cylindrical Gaussian Surface for calculation of electric field due to line charge

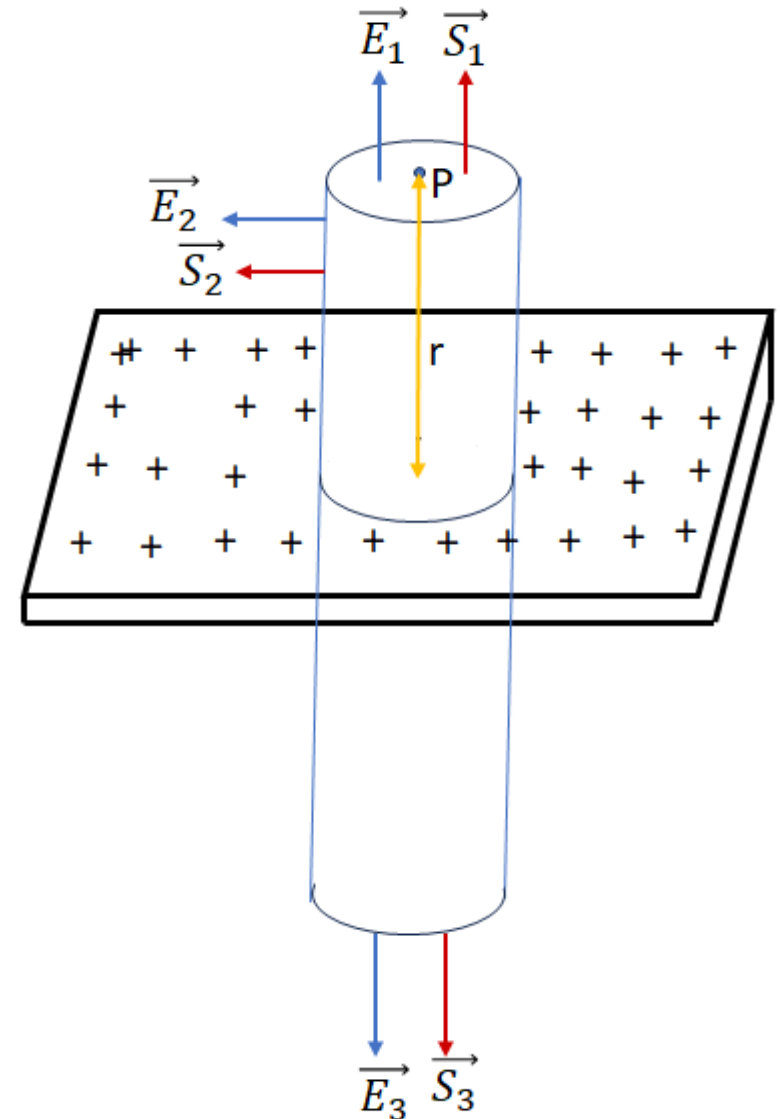
Applications of Gauss's Law

Electric Field due to a sheet of charge (Infinite Sheet of Charge)

Let us consider a portion of thin, non-conducting, infinite sheet of charge with surface charge density σ (charge per unit area) being constant. Consider a point P at a distance r from the surface of the sheet. Now draw a cylindrical surface having its upper end through P and lower end at a distance r from the sheet. Let S_1 , S_3 be the surface areas of the ends of the cylinder and S_2 the surface area of the cylindrical surface. From symmetry, E points at right angles to the ends of the cylinder and away from the plane. Since E does not pierce the cylindrical surface, there is no contribution to the flux from this surface. Thus, according to Gauss's law, we have the

$$\oint \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0} = \frac{\sigma S}{\epsilon_0}$$

$$\text{Or } \left[\int_{S_1} \vec{E}_1 \cdot d\vec{s}_1 + \int_{S_2} \vec{E}_2 \cdot d\vec{s}_2 + \int_{S_3} \vec{E}_3 \cdot d\vec{s}_3 \right] = \frac{\sigma S}{\epsilon_0}$$



Applications of Gauss's Law

Since $\vec{E}_2 = 0$, $\vec{E}_1 = -\vec{E}_3 = \vec{E}$ (say)

And $d\vec{s}_1 = -d\vec{s}_3 = d\vec{s}$ (say)

Also $\vec{E}_1 \parallel d\vec{s}_1$, $\vec{E}_3 \parallel d\vec{s}_3$

We have

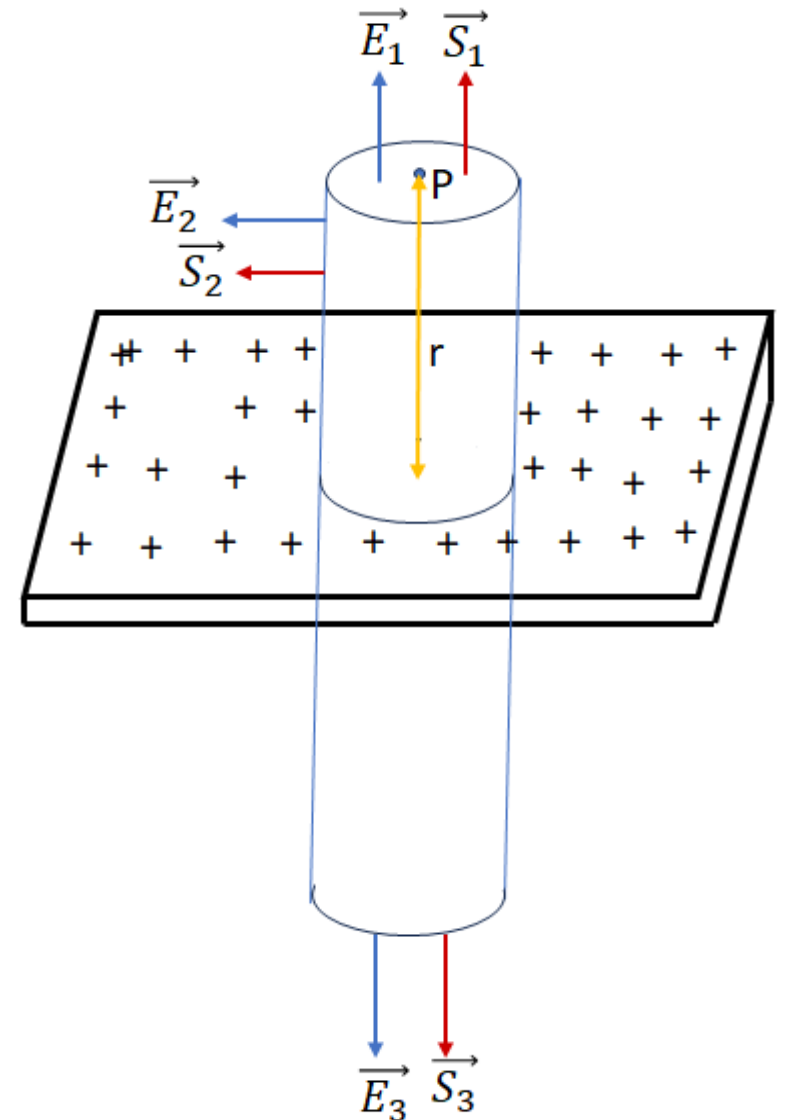
$$\int_{S_2} \vec{E}_2 \cdot d\vec{s}_2 = 0$$

$$\int_{S_1} \vec{E}_1 \cdot d\vec{s}_1 = \int \vec{E} \cdot d\vec{s} = E \int d\vec{s} = ES$$

$$\text{And } \int_{S_3} \vec{E}_3 \cdot d\vec{s}_3 = \int (-\vec{E}) \cdot (d\vec{s}) = E \int d\vec{s} = ES$$

$$\therefore ES = \frac{\sigma S}{2\epsilon_0} \quad \text{Or} \quad E = \frac{\sigma}{2\epsilon_0}$$

Thus, E is same for all points on each side of the plane.



Applications of Gauss's Law

Electric Field due to Infinite Parallel Sheets of Charge

Let P_1 and P_2 be two infinite plane parallel sheets carrying opposite uniform charge densities. Let a point P be lying in the region between two charged plates. Now in order to calculate the electric field at point P , draw a normal cylinder of cross-sectional area S through point P having one of its ends in the plate. Using Gauss's law, we have

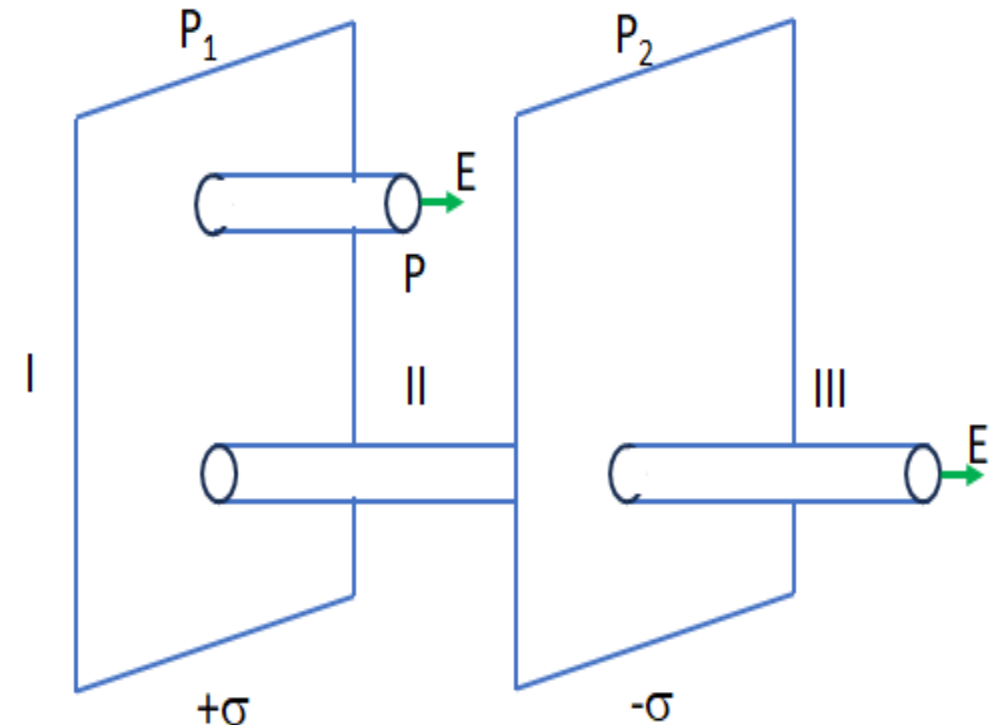
$$\oint_S \vec{E} \cdot \vec{ds} = \frac{q}{\epsilon_0}$$

Let σ be the surface charge density, then $q = \sigma S$

$\therefore$ Above equation becomes $ES = \frac{\sigma S}{\epsilon_0}$ [As $\vec{E}$ and $\vec{S}$ are parallel]

$$\text{or } E = \frac{\sigma}{\epsilon_0}$$

which implies the the field between two plates is uniform and independent of the separation between them.



Applications of Gauss's Law

Again consider a point Q lying extreme right of the two plates. Now draw a cylinder of cross-sectional area S through this point Q with one of its face lying on plate P_1 . Therefore from Gauss's law, we have

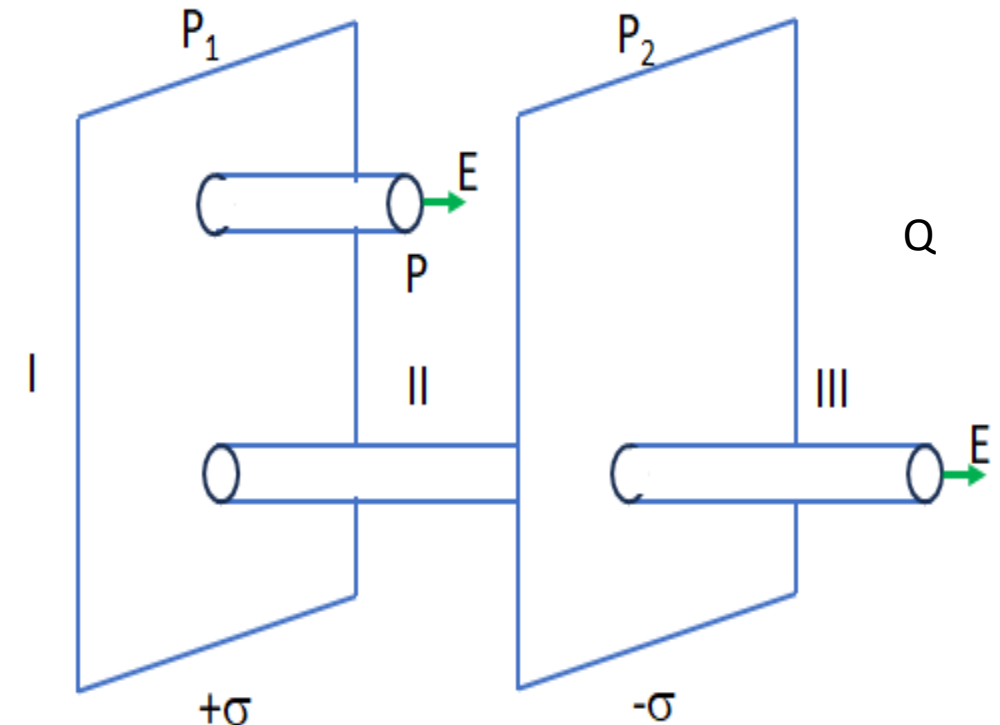
$$\oint_S \vec{E} \cdot \vec{ds} = \frac{q}{\epsilon_0}$$

Since the charge inside the Gaussian surface is zero i.e. $q = \sigma S - \sigma S$, therefore we have

$$\oint_S \vec{E} \cdot \vec{ds} = 0$$

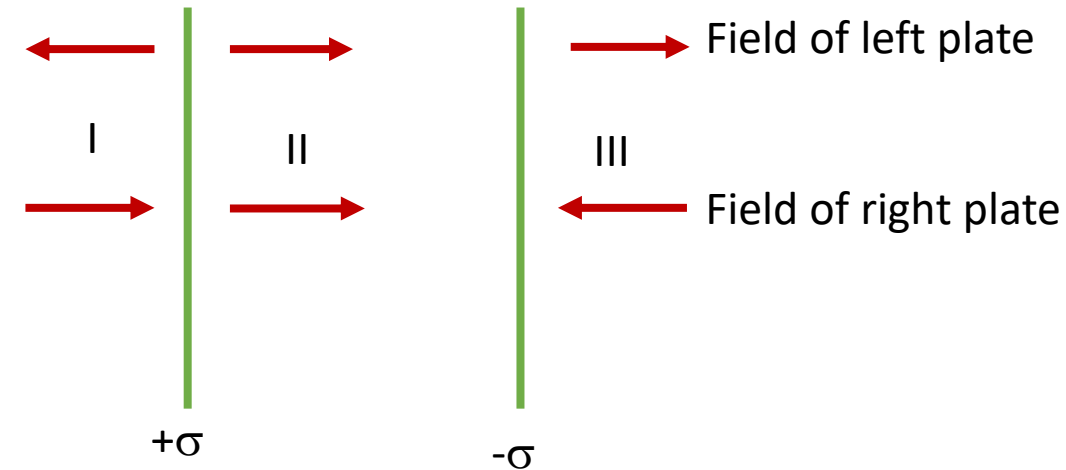
Or $ES=0 \Rightarrow E=0$

which clearly indicates that the electric field intensity due to two oppositely charged infinite plane sheets at a point outside the plates is zero.



Applications of Gauss's Law

We can also verify these results pictorially as follows:
We know that the electric field intensity at any point at a distance r due to an infinite plane sheet of charge having charge density σ is $\frac{\sigma}{2\epsilon_0}$. Therefore the left plate produces a field $\frac{\sigma}{2\epsilon_0}$ which point away from it (as shown in the figure) to the left in region I and to the right in region II and III. The right plate produces a field $\frac{\sigma}{2\epsilon_0}$ which points towards it to the right in regions I and II and to the left in region III. The two fields cancel in region I and II but they conspire in region III. Thus, between the plates the electric field intensity is $\frac{\sigma}{\epsilon_0}$ and outside is zero.



Applications of Gauss's Law

Q.4(a) Find the electric field at a point P outside a uniformly charged sphere.

Q.2(b) Apply Gauss law to find out the electric field due to a uniformly charged solid sphere at a point [5] outside and inside the sphere.

The electric field due to the Spherical Charge Distribution (Uniformly Charged Sphere)

Consider a sphere of charge Q having volume density ρ . Let r be the radius of sphere. Let P and P' be two observation points outside and inside the sphere respectively as shown in figure.

To calculate the electric field strength at P and P' , we consider Gaussian surfaces S_1 and S_2 respectively through points P and P' with O as their center. Let us consider $OP = r_o$ and $OP' = r_i$.

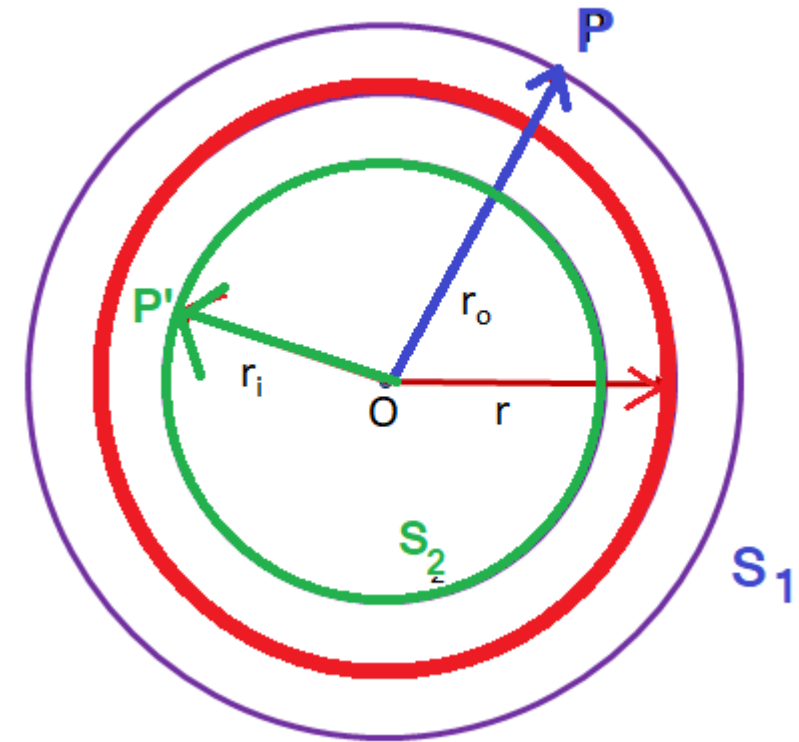
Now apply Gauss's law to the outer Gaussian surface (S_2) through P we have,

$$\oint_S \vec{E}_0 \cdot d\vec{s}_1 = \frac{Q}{\epsilon_0}$$

Where $\vec{E}_0$ is electric field at point P and charge $Q = \frac{4}{3}\pi r^3 \rho$

Or

$$\epsilon_0 \oint_S \vec{E}_0 \cdot d\vec{s}_1 = Q$$



A spherically symmetric charge distribution, showing two Gaussian surfaces

Applications of Gauss's Law

- Spherical Charge Distribution

$$\epsilon_0 E_0 \oint_S \vec{ds}_1 = Q$$

$$\epsilon_0 E_0 4\pi r_o^2 = Q$$

(The surface area of sphere is $4\pi r^2$)

or

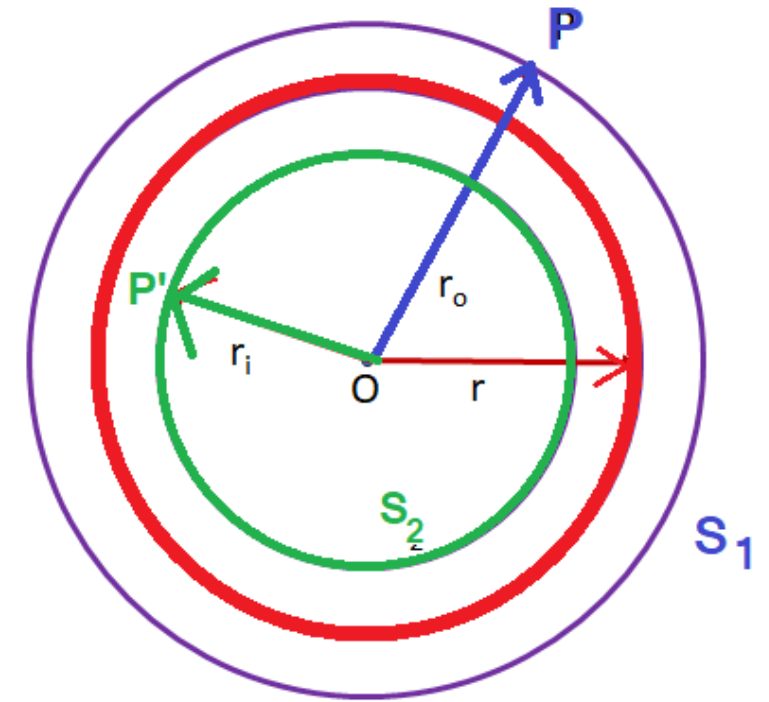
$$\vec{E}_0 = \frac{Q}{4\epsilon_0\pi r_o^2}$$

Thus, it is clear that the sphere of charge behaves as if the whole charge is concentrated at the center. Again apply Gauss's law to the inner Gaussian surface (S_2) through P' , we have

$$\oint_S \vec{E}_i \cdot \vec{ds}_2 = \frac{Q'}{\epsilon_0}$$

where $\vec{E}_i$ is the electric field at P' and Q' is the charge enclosed by the Gaussian surface S_2 . Therefore $Q' = \frac{4}{3}\pi r_i^3 \rho$

Presented by Dr.Dheeraj Bhardwaj



A spherically symmetric charge distribution, showing two Gaussian surfaces

Applications of Gauss's Law

- Spherical Charge Distribution

$$\text{Or } Q' = \frac{4}{3}\pi r_i^3 \rho \frac{Q}{\frac{4}{3}\pi r^3} = Q \frac{r_i^3}{r^3}$$

$$\therefore \oint_S \epsilon_0 \vec{E}_i \cdot \vec{ds}_2 = Q' = Q \frac{r_i^3}{r^3}$$

$$\epsilon_0 E_i \oint_S \vec{ds}_2 = Q \frac{r_i^3}{r^3}$$

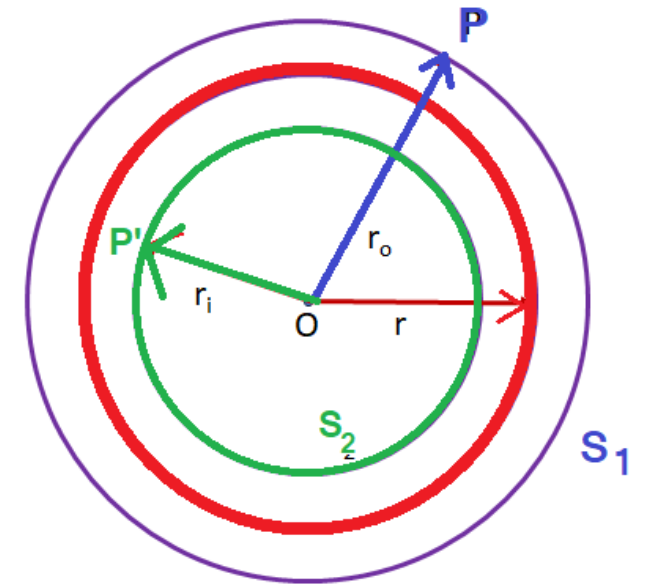
$$\epsilon_0 E_0 4\pi r_i^2 = Q \frac{r_i^3}{r^3}$$

$$E = \frac{Q}{4\pi\epsilon_0} \frac{r_i}{r^3}$$

The above relation shown that the intensity of the electric field inside the sphere of charge varies directly as a distance from the center.

Suppose $r_i = r$, then the electric field due to the sphere of the charge Q will be $\Rightarrow E = \frac{Q}{4\pi\epsilon_0} \frac{1}{r^2}$

which represents the electric field due to the sphere of radius r just on the surface of the sphere.



A spherically symmetric charge distribution, showing two Gaussian surfaces

Static Magnetic Fields or Magnetostatic fields

- Magnetostatics deals with the steady current or charges moving with constant velocity.
- In 1820, the scientist H.C. Oersted has discovered the relationship between electric and magnetic fields.
- He stated that current carrying conductor is always surrounded by a magnetic field.
- If a steady current flows through a conductor, a static magnetic field is produced around the conductor.
- The magnetic field is a representation of imaginary lines around the magnet which are also called as magnetic lines of force or magnetic flux lines.

Magnetostatics – Magnetic Flux Density

The *magnetic flux density*, $\mathbf{B}$, related to the magnetic field intensity in free space by

$$\mathbf{B} = \mu_0 \mathbf{H}$$

where μ_0 is the *free space permeability*, given in units of *henrys per meter*, or

$$\mu_0 = 4\pi \times 10^{-7} \text{ H / m}$$

The units of $\mathbf{B}$ are therefore (H)(A)/m², but it is more instructive to write *webers per meter squared*, or Wb/m², where Wb=(H)(A).

The amount of magnetic flux, ϕ , in webers, from magnetic field passing through a surface is found in a manner analogous to finding electric flux:

$$\phi = \int \vec{\mathbf{B}} \cdot \vec{ds}$$

Magnetostatics – Gauss's Law

A fundamental feature of magnetic fields that distinguishes them from electric fields is that the field lines form closed loops

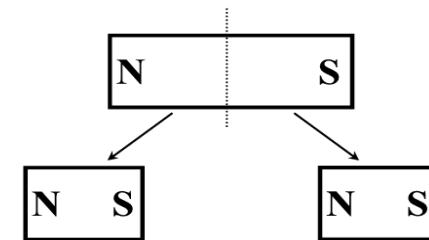
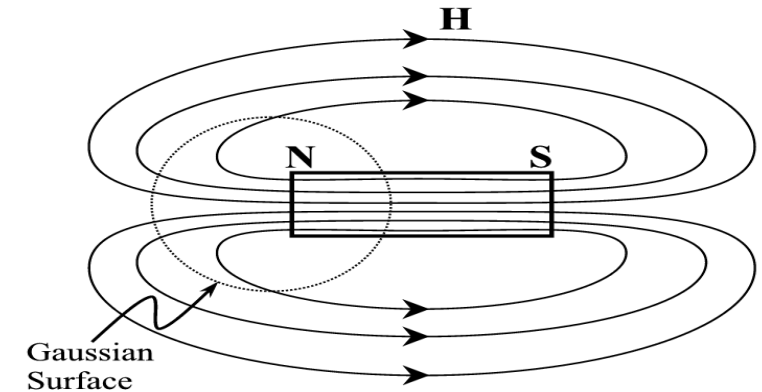
We cannot saw the magnet in half to isolate the north and the south poles; as Figure shows, if you saw a magnet in half you get two magnets.

Put another way, *you cannot isolate a magnetic pole.*

From this characteristic of magnetic fields, it is easy to see that the net magnetic flux passing through a Gaussian surface (a closed surface as shown in Figure) must be zero. What goes into the surface must come back out. Thus we have *Gauss's law for static magnetic fields*

$$\oint_S \vec{B} \cdot d\vec{s} = 0$$

This is also referred to as the law of conservation of magnetic flux.



Explain continuity equation. How it is used to modify the Ampere's law.

Ampere's Law

State Ampere circuital law. Translate its differential form from integral form. Find the electrostatic potential and no

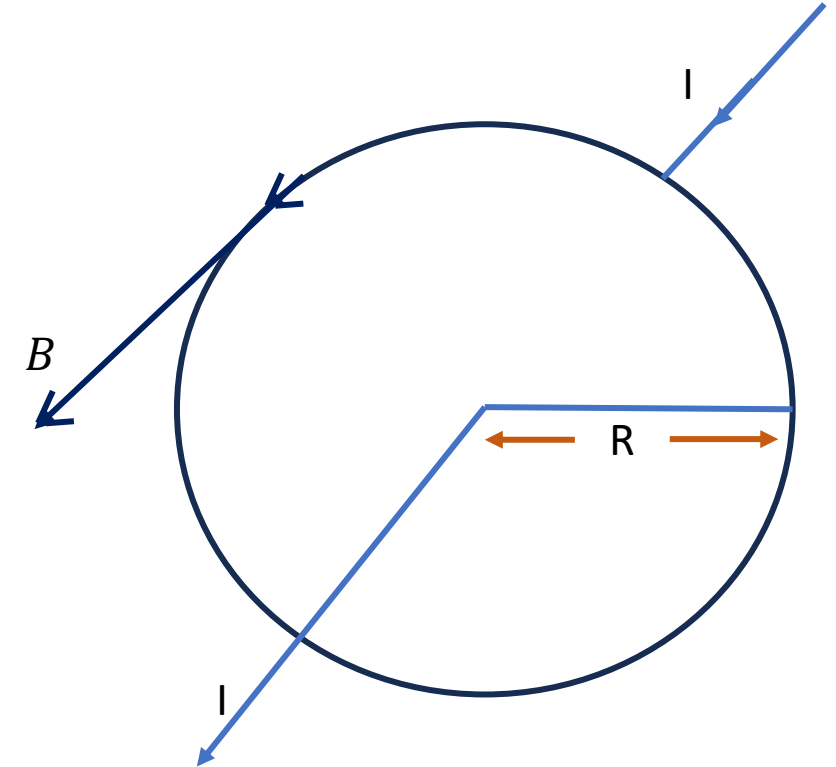
We know that moving charges produce a magnetic field, which in turn influences any other charge moving through it and it does not affect a stationary charge. Also, the magnetic effect is maximum when both the charges are moving parallel to each other and is minimum when the charges are moving perpendicular to each other.

It is well known that a current I produces a magnetic field $\vec{B}$ in a plane perpendicular to the direction of flow and its magnitude at a distance d is

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{2I}{d} \quad (\text{in SI}) \quad \dots(1)$$

Now consider a circular path or radius R in a plane perpendicular to the wire carrying I . Then magnetic field on every point of the circular path will be tangential to the path and is

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{2I}{R} \hat{dl} \quad (\text{in SI}) \quad \dots(2)$$



The Magnetic field on a circular path perpendicular to the wire carrying current I

Ampere's Law

Where $\hat{dl}$ is the unit vector along the path. The closed line integral of $\vec{B}$ on the circular path is

$$\oint \vec{B} \cdot \vec{dl} = \oint |\vec{B}| \hat{dl} \cdot \vec{dl} = \oint |\vec{B}| \hat{dl} \cdot |\vec{dl}| \hat{dl} = \oint B \cdot dl \quad (\text{As } \hat{dl} \cdot \hat{dl} = 1)$$

As B is constant everywhere on the circular path, we have

$$\oint B \cdot dl = B \oint dl = B (2\pi R) \Rightarrow \oint \vec{B} \cdot \vec{dl} = B (2\pi R) \quad \dots\dots(2b).. \text{ Since } \oint dl = 2\pi R$$

Also, because $d=R$, we have from eqⁿ- (1)

$$B = \frac{\mu_o}{4\pi} \frac{2I}{R}$$

Thus, the closed line integral of $\vec{B}$ on the circular path of radius R around the current carrying conductor is

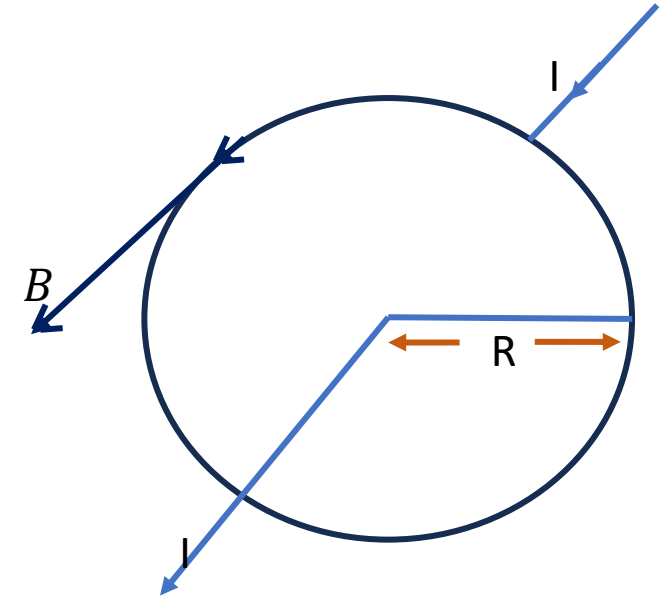
$$\oint \vec{B} \cdot \vec{dl} = \frac{\mu_o}{4\pi} \frac{2I}{R} (2\pi R) = \mu_o I \Rightarrow \oint \vec{B} \cdot \vec{dl} = \mu_o I \quad \dots(3) \text{ from eq}^n. 2(b)$$

The above equation is known as Ampere's law or Ampere's circuital law.

It is similar to the Gauss law in electrostatics. Also if the path goes round the current carrying conductor N times then

$$\oint \vec{B} \cdot \vec{dl} = N\mu_o I \quad \dots(4)$$

If $I = 0$; i.e. the current enclosed by the circular path is zero then $\oint \vec{B} \cdot \vec{dl} = 0$



Magnetic field on a circular path perpendicular to the wire carrying current I

Equation of Continuity

Q4 (a) Explain continuity equation.

From the principle of charge conservation, the time rate of decrease of charge within a given volume must be equal to the net outward current flow through the surface of the volume. Thus current I_{out} coming out of the closed surface is

$$I_{\text{out}} = \oint_S \mathbf{J} \cdot d\mathbf{S} = \frac{-dQ_{\text{in}}}{dt} \quad \dots(1)$$

where Q_{in} is the total charge enclosed by the closed surface. Invoking the divergence theorem, we write

$$\oint_S \mathbf{J} \cdot d\mathbf{S} = \int_v \nabla \cdot \mathbf{J} dv \quad \dots(2)$$

But

$$\frac{-dQ_{\text{in}}}{dt} = -\frac{d}{dt} \int_v \rho_v dv = -\int_v \frac{\partial \rho_v}{\partial t} dv \quad \dots(3)$$

Substituting equations (2) and (3) in (1)

$$\int_v \nabla \cdot \mathbf{J} dv = -\int_v \frac{\partial \rho_v}{\partial t} dv$$

Equation of Continuity

Q4 (a) Explain continuity equation.

Or

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

...(4)

which is called the *continuity of the current equation* or just *continuity equation*.

It must be kept in mind that the continuity equation is derived from the principle of conservation of charge and essentially states that there can be no accumulation of charge at any point.

For steady currents, $\frac{\partial \rho}{\partial t} = 0$, and hence $\nabla \cdot \mathbf{J} = 0$, showing that the total charge leaving a volume is the same as the total charge entering it.

the statements of the integral form of the continuity equation are given as:

- Whenever an additional q flows inward through the surface of any region, the amount of q in that region increases and decreases when the flow of q is outward.
- Whenever a new q is introduced in the region, the amount of q in that region increases and destroying q results in a decrease in the amount of q .
- To change the amount of q in the region, there is no other way than these two.

Module-2: Electromagnetic Theory: Gradient, Divergence and Curl, Statement of Gauss theorem & Stokes theorem, Gauss's law, Applications, Concept of electric potential, Relationship between E and V , Polarization of dielectrics, dielectric constant, Gauss's law in magnetostatics, Ampere's circuital law, Equation of continuity, Displacement current, Maxwell's equations.

Maxwell's Equations

In the year 1862, James Clerk Maxwell expressed the basic laws of electricity and magnetism in the form of four equations known as Maxwell's equations. These equations have been obtained from four basic laws of electricity and magnetism.

These laws are:

- ❖ Gauss's law for electric field,
- ❖ Gauss's law for magnetic field,
- ❖ Faraday's law of electromagnetic induction and
- ❖ Ampere's circuital law.

On the basis of mathematical analysis of these laws, Maxwell predicted that an accelerated charge or time varying current always emits electromagnetic waves. These waves travel with the speed of light in the vacuum. Light is also electromagnetic waves. The divergence and curl relations of electric and magnetic fields are called Maxwell equations.

Maxwell's I Equation/Divergence of Electrostatic Field

According to Gauss's law of electrostatic field the net outward flux of electric field $\vec{E}$ through any closed surface S is $\frac{1}{\epsilon_0}$ times of net charge inside the volume V enclosed by the surface.

$$\phi_E = \oint_S \vec{E} \cdot \vec{ds} = \frac{q}{\epsilon_0} \quad \dots(1)$$

where ϵ_0 is the permittivity of the free space

As
$$q = \int \rho dV \quad \dots(2)$$

where ρ = volume charge density

From equations (1) and (2)

$$\phi_E = \oint_S \vec{E} \cdot \vec{ds} = \frac{1}{\epsilon_0} \int_V \rho dV \quad \dots(3)$$

Maxwell's I Equation/Divergence of Electrostatic Field

Now applying Gauss Divergence theorem

$$\oint_s \vec{E} \cdot d\vec{s} = \int_V (\vec{\nabla} \cdot \vec{E}) dV \quad \dots(4)$$

From equations (3) and (4)

$$\begin{aligned} \int_V (\vec{\nabla} \cdot \vec{E}) dV &= \frac{1}{\epsilon_0} \int_V \rho dV \\ \Rightarrow \vec{\nabla} \cdot \vec{E} &= \frac{\rho}{\epsilon_0} \\ \Rightarrow \vec{\nabla} \cdot (\vec{D}) &= \rho \quad \dots(5) \end{aligned}$$

As $\vec{D} = \epsilon_0 \vec{E}$ = Electric flux density

Note:

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \text{ for points inside the charge distribution}$$

$$\vec{\nabla} \cdot \vec{E} = 0 \text{ for points outside the charge distribution}$$

Maxwell's II Equation/Divergence of Magnetostatic Field

Maxwell's II Equation:

According to Gauss's law of Magnetostatics the net outward flux of magnetic induction $\vec{B}$ through any closed surface is always zero.

$$\phi_B = \oint_S \vec{B} \cdot \vec{da} = 0 \quad \dots(6)$$

This is an integral form of Maxwell's II equation.

Applying Gauss- Divergence theorem

$$\oint_S \vec{B} \cdot \vec{da} = \int_V (\vec{\nabla} \cdot \vec{B}) dV \quad \dots(7)$$

From equations (6) and (7) we have

$$\begin{aligned} \int_V (\vec{\nabla} \cdot \vec{B}) dV &= 0 \\ \Rightarrow \vec{\nabla} \cdot \vec{B} &= 0 \quad \dots(8) \end{aligned}$$

This is differential form of Maxwell's II equation.

From equation we can conclude that magnetic monopoles do not exist.

Maxwell's III Equation/Curl of Electrostatic field

- According to Faraday's law of electromagnetic induction when the magnetic flux linked with a circuit changes, an e.m.f is induced in the circuit which is proportional to the rate of change of the flux linked with the circuit.

- i.e.
$$e = -\frac{\partial \phi_B}{\partial t} = -\frac{\partial}{\partial t} \int_S \vec{B} \cdot \vec{ds} \quad \dots(9)$$

- The e.m.f. can also be found by calculating the work done in carrying a unit charge completely around the loop. Thus

$$e = \oint_C \vec{E} \cdot \vec{dl} \quad \dots(10)$$

- Hence

$$\begin{aligned} e &= \oint_C \vec{E} \cdot \vec{dl} = -\frac{\partial}{\partial t} \int_S \vec{B} \cdot \vec{ds} \\ \Rightarrow \oint_C \vec{E} \cdot \vec{dl} &= -\int_S \frac{\partial \vec{B}}{\partial t} \cdot \vec{ds} \quad \dots(11) \end{aligned}$$

Maxwell's III Equation/Curl of Electrostatic field

According to Stoke's- curl theorem,

$$\oint_C \vec{E} \cdot d\vec{l} = \int_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{s} \quad \dots(12)$$

From equations (11) and (12)

$$\begin{aligned} \int_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{s} &= - \int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s} \\ \Rightarrow \vec{\nabla} \times \vec{E} &= - \frac{\partial \vec{B}}{\partial t} \quad \dots(13) \end{aligned}$$

For static field

$$\begin{aligned} \frac{\partial \vec{B}}{\partial t} &= 0 \\ \Rightarrow \vec{\nabla} \times \vec{E} &= 0 \end{aligned}$$

Maxwell's IV Equation/Divergence of Electrostatic Field

According to Ampere's circuital law the line integral of magnetic field around any closed loop is equal to the product of μ_0 times the total current passing through closed loop.

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 \sum I \quad \dots(14)$$

Total current can be written in terms of current density ($\vec{J}$)

$$\sum I = \int_S \vec{J} \cdot d\vec{a} \quad \dots(15)$$

From equations (14) and (15) we have

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 \int_S \vec{J} \cdot d\vec{a} \quad \dots(16)$$

This is an integral form of Maxwell's IV equation.

Applying Stoke's- Curl Theorem,

$$\oint_C \vec{B} \cdot d\vec{l} = \int_S (\vec{\nabla} \times \vec{B}) \cdot d\vec{a} \quad \dots(17)$$

Maxwell's IV Equation/Divergence of Electrostatic Field

From equations (16) and (17) we have

$$\begin{aligned}\int_s (\vec{\nabla} \times \vec{B}) \cdot \vec{da} &= \mu_o \int_s \vec{J} \cdot \vec{da} \\ \Rightarrow \int_s (\vec{\nabla} \times \vec{B}) \cdot \vec{da} &= \int_s (\mu_o \vec{J}) \cdot \vec{da} \\ \vec{\nabla} \times \vec{B} &= \mu_o \vec{J}\end{aligned}\quad \dots(18)$$

This is differential form of Maxwell's IV equation and called Ampere law. But it holds good for static field only.

Taking divergence of equation (18) on both sides

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{B}) = \vec{\nabla} \cdot (\mu_o \vec{J}) \quad \dots(19)$$

Using vector identity $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$

$$\Rightarrow \vec{\nabla} \cdot (\vec{\nabla} \times \vec{B}) = 0$$

So from equation (19) we have

$$\vec{\nabla} \cdot (\mu_o \vec{J}) = 0 \Rightarrow \vec{\nabla} \cdot \vec{J} = 0 \quad \dots(20)$$

Maxwell's IV Equation/Divergence of Electrostatic Field

From equation of continuity

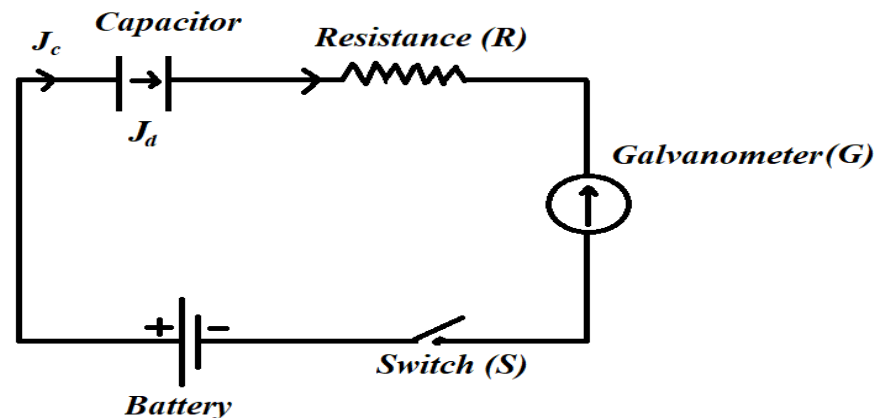
$$\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0 \quad \dots(21)$$

Where $\vec{J}$ = Current density and ρ = volume charge density

Comparing equations (20) and (21) we have

$$\frac{\partial \rho}{\partial t} = 0 \Rightarrow \rho = \text{constant} \quad \dots(22)$$

Above relation shows that equation (18) holds good only for static field which does not change with time and not valid for time varying field. This equation does not explain the current passing between two plates of capacitor.



Maxwell's IV Equation/Divergence of Electrostatic Field

At the instant switch S is on or off, the Galvanometer will show deflection. But as we know the current always passes through a closed circuit and there is no medium in between the two plates of a capacitor. So, a question arises that how the current passes through the capacitor. According to Maxwell the current passes through the capacitor due to time varying electric field. This current is known as displacement current.

As according to Faraday's law change in magnetic flux induces current similarly, change in electric flux also induces current (according to Maxwell). That is how current passes through the capacitor. So, Maxwell modified Ampere's law and introduced the concept of displacement current.

So, total current density

$$\vec{J} = \vec{J}_c + \vec{J}_d$$

Where $\vec{J}_c$ = conduction current density due to flow of free electron in conductor.

$\vec{J}_d$ = displacement current due to change in electric field between two plates of capacitor at the instant of On/Off switch.

Maxwell's IV Equation/Divergence of Electrostatic Field

So, from equation (20)

$$\vec{\nabla} \cdot \vec{J} = 0$$

$$\Rightarrow \vec{\nabla} \cdot (\vec{J}_c + \vec{J}_d) = 0$$

$$\Rightarrow \vec{\nabla} \cdot \vec{J}_c = -\vec{\nabla} \cdot \vec{J}_d \quad \dots(23)$$

From equation of Continuity

$$\vec{\nabla} \cdot \vec{J}_c + \frac{\partial \rho}{\partial t} = 0 \quad \dots(24)$$

From equations (23) and (24) we have

$$-\vec{\nabla} \cdot \vec{J}_d + \frac{\partial \rho}{\partial t} = 0 \Rightarrow \vec{\nabla} \cdot \vec{J}_d = \frac{\partial \rho}{\partial t} \quad \dots(25)$$

From Maxwell's I equation

$$\vec{\nabla} \cdot \vec{D} = \rho \quad \dots(26)$$

Maxwell's IV Equation/Divergence of Electrostatic Field

Using equations (25) and (26) we have

$$\vec{\nabla} \cdot \vec{J}_d = \frac{\partial}{\partial t} (\vec{\nabla} \cdot \vec{D})$$

$$\vec{\nabla} \cdot \vec{J}_d = \frac{\partial}{\partial t} (\vec{\nabla} \cdot \vec{D})$$

$$\Rightarrow \vec{\nabla} \cdot \vec{J}_d = \vec{\nabla} \cdot \left(\frac{\partial \vec{D}}{\partial t} \right)$$

$$\vec{J}_d = \frac{\partial \vec{D}}{\partial t} \quad \dots(27)$$

So Maxwell's IV equation can be modified as

$$\vec{\nabla} \times \vec{B} = \mu_o (\vec{J}_c + \vec{J}_d)$$

$$\vec{\nabla} \times \vec{B} = \mu_o \left(\vec{J}_c + \frac{\partial \vec{D}}{\partial t} \right)$$

$$\text{As } \vec{D} = \epsilon_o \vec{E}$$

$$\Rightarrow \vec{\nabla} \times \vec{B} = \mu_o \vec{J}_c + \mu_o \epsilon_o \frac{\partial \vec{E}}{\partial t} \quad \dots(28)$$

Maxwell's Equations in Some Particular Cases

Maxwell's equations in charge free space (where $\rho = 0$, $\vec{J} = 0$)

$$\vec{\nabla} \cdot \vec{E} = 0$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Maxwell's equations in isotropic medium

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{B} = \mu(\vec{J} + \epsilon \frac{\partial \vec{E}}{\partial t})$$

where ϵ is the permittivity of medium and μ is the permeability of the medium.

Maxwell's Equations

- Maxwell's equations combine the fundamental laws of electricity and magnetism .
- They are of profound importance in the analysis of most electromagnetic wave problems.
- These equations are the mathematical abstractions of certain experimentally observed facts and find their application to all sorts of problem in electromagnetism.
- Maxwell's equations are derived from Ampere's law, Faraday's law and Gauss law.

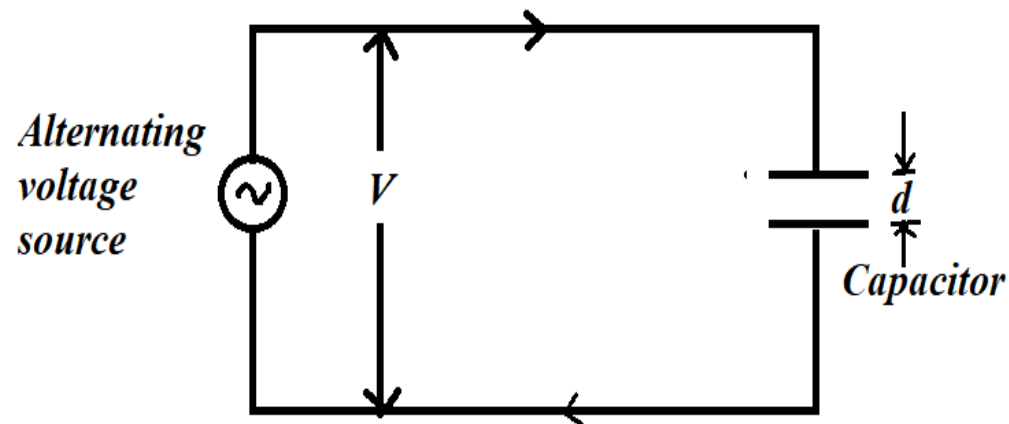
Maxwell's Equations Summary

Maxwell's Equations	Differential form	Integral form
1. Equation from electrostatics	$\nabla \cdot \vec{D} = \rho$	$\oint_s \vec{D} \cdot d\vec{s} = \int_v \rho dv$
2. Equation from magnetostatics	$\nabla \cdot \vec{B} = 0$	$\oint_s \vec{B} \cdot d\vec{s} = 0$
3. Equation from Faradays law	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\oint \vec{E} \cdot d\vec{l} = -\oint_s \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$
4. Equation from Ampere circuital law	$\nabla \times \vec{H} = \sigma \vec{E} + \frac{\partial \vec{D}}{\partial t}$	$\int_l \vec{H} \cdot d\vec{l} = \oint_s \left(\sigma \vec{E} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{s}$

Displacement Current

The nature of the current flow through a resistor is different from that through a capacitor. A constant voltage across a resistor produces a continuous flow of current of constant value. In capacitor there will be current only while voltage is changing. The current flowing through the capacitor is called displacement current. It passes through capacitor due to time varying electric field.

For capacitor connected to Alternating Voltage source as shown in figure.



Capacitor Connected to Alternating Voltage Source

Displacement Current

$$\text{Current } I_d = \frac{dQ}{dt} = \frac{CdV}{dt} \quad \dots(1)$$

$$\text{Where } C \text{ is capacitance} = \frac{\epsilon A}{d}$$

A = Area of plates

ϵ = permittivity of medium

d = separation between the plates

$$I_d = \frac{\epsilon A}{d} \frac{dV}{dt} \quad \dots(2)$$

Since $\vec{E} = \frac{V}{d}$, electric field intensity

$$I_d = \epsilon A \frac{d\vec{E}}{dt} \quad \dots(3)$$

Thus changing electric field within a capacitor is equivalent to a current through the capacitor. Maxwell called this current as the displacement current.

Displacement Current

From equation (3), we have

$$\vec{J}_d = \frac{I_d}{A} = \frac{d}{dt}(\epsilon \vec{E})$$

$$\vec{J}_d = \frac{d\vec{D}}{dt}$$

Since $\vec{D} = \epsilon \vec{E}$

Where $\vec{D}$ is electric displacement vector and $\vec{J}_d$ is displacement current density.

Curl of the Electric Field:

Let's take the simplest electric field:

Electric field due to a single point charge q is:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

We need to find the curl of it

$$\nabla \times \mathbf{E}(\mathbf{r}) = \frac{q}{4\pi\epsilon_0} \nabla \times \left(\frac{1}{r^2} \hat{\mathbf{r}} \right)$$

Take the area integral

$$\int_{surf} \nabla \times \mathbf{E}(\mathbf{r}) \cdot d\mathbf{a} = \frac{q}{4\pi\epsilon_0} \int_{surf} \nabla \times \left(\frac{1}{r^2} \hat{\mathbf{r}} \right) \cdot d\mathbf{a}$$

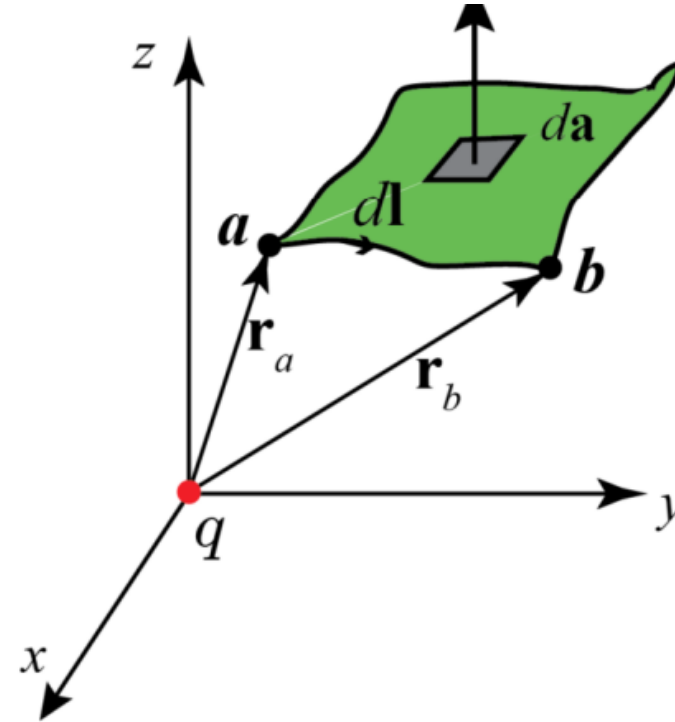
Use Stokes's theorem

$$\int_{surf} \nabla \times \mathbf{E}(\mathbf{r}) \cdot d\mathbf{a} = \frac{q}{4\pi\epsilon_0} \oint \left(\frac{1}{r^2} \hat{\mathbf{r}} \right) \cdot d\mathbf{l} = \frac{q}{4\pi\epsilon_0} \oint \frac{1}{r^2} dr$$

since

$$d\mathbf{l} = dr\hat{\mathbf{r}} + r d\theta\hat{\boldsymbol{\theta}} + r \sin\theta d\phi\hat{\boldsymbol{\phi}}$$

$$\int_{surf} \nabla \times \mathbf{E}(\mathbf{r}) \cdot d\mathbf{a} = \frac{q}{4\pi\epsilon_0} \times \frac{1}{r} \Big|_{r_a}^{r_a} = 0 \quad \text{implies} \quad \boxed{\nabla \times \mathbf{E} = 0}$$



Curl of an electric field ($\nabla \times \mathbf{E}$) is zero. We have shown this for the simplest field, which is the field of a point charge. But it can be shown to be true for any electric field, as long as the field is static.

if the field is dynamic, that is, what if the field changes as a function of time?

Ans:- $\nabla \times \mathbf{E} = - \frac{d\mathbf{B}}{dt}$

Gauss's Law from Coulomb's Law:

If Coulomb's Law and Gauss's law have the same information content, can we derive Gauss's law from Coulomb's law?

Coulomb's law gives the electric field due to a volume charge $\rho(\mathbf{r}')$

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\hat{\mathbf{r}}}{r^2} \rho(\mathbf{r}') d\tau$$

Take the divergence of both sides of the equation

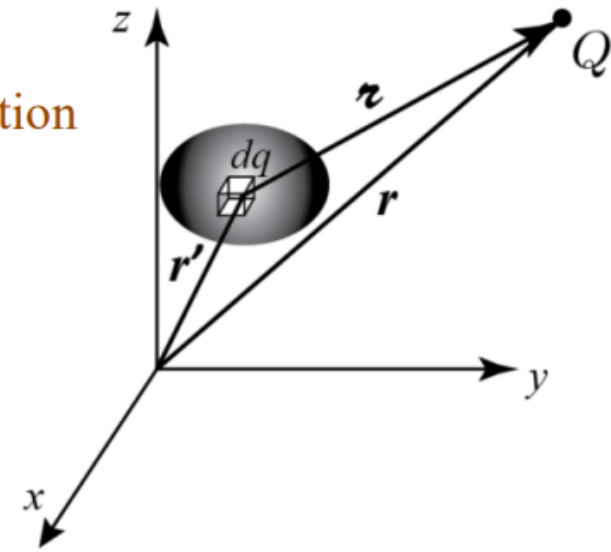
$$\nabla \cdot \mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) \rho(\mathbf{r}') d\tau$$

We have: $\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 4\pi \delta(\mathbf{r}) = 4\pi \delta(\mathbf{r} - \mathbf{r}')$

$$\text{Therefore, } \nabla \cdot \mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int 4\pi \delta(\mathbf{r} - \mathbf{r}') \rho(\mathbf{r}') d\tau = \frac{\rho(\mathbf{r})}{\epsilon_0}$$

$$\boxed{\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}}$$

The divergence of electric field is equal to the charge density divided by ϵ_0



BOUNDARY CONDITIONS FOR THE ELECTROMAGNETIC FIELDS VECTORS: B , E , D , AND H , (AT THE INTERFACE BETWEEN TWO MEDIA)

In order to discuss the behaviours of electromagnetic waves at the boundary.

We shall now investigate the boundary conditions which the time dependent electromagnetic field vectors $\vec{B}$, $\vec{E}$, $\vec{D}$ and $\vec{H}$, must satisfy at the interface between two different media. Boundary conditions are:

1. The normal component of magnetic induction $\vec{B}$ is **continuous** across boundary, i.e.,

$$B_{1n} = B_{2n}.$$

2. The tangential component of $\vec{E}$ is **continuous** across the interface, i.e.,

$$E_{1t} = E_{2t}.$$

3. The normal component of electric displacement $\vec{D}$ is **discontinuous** across the interface, i.e.,

$$D_{1n} - D_{2n} = \sigma.$$

4. The tangential component of magnetic induction $\vec{H}$ is **continuous** across the surface separating two dielectrics, i.e.,

$$H_{1t} = H_{2t}.$$

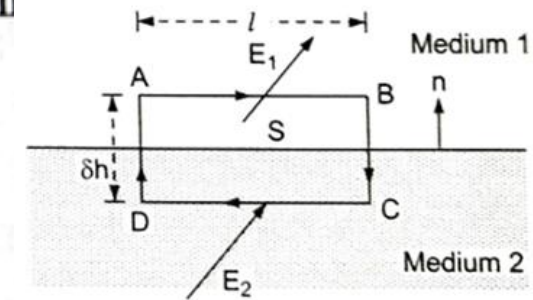


Fig. 12.5 Rectangular loop at the interface between two media

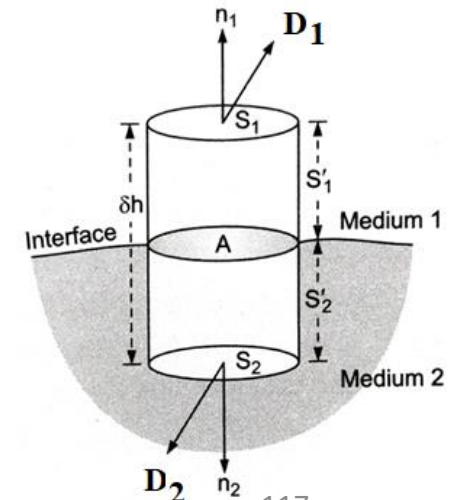


Fig. 12.4 Pill box shaped surface at the interface of two media.

Prove that The tangential component of $\vec{E}$ is *continuous* across the interface, i.e., $E_{1t} = E_{2t}$.

Q. Develop boundary conditions for $\vec{E}$ separated by two dielectric media of different permittivity

Boundary Condition for $\vec{E}$

The boundary condition, which the tangential component of electric field must satisfy, may be obtained from Maxwell's equation

$$\text{curl } \vec{E} = - \frac{\partial \vec{B}}{\partial t} \quad \dots(67)$$

At the interface, we construct a rectangular loop $ABCD$ bounding a surface S as shown in Fig. (12.5). Integrating eqn. (67) over the surface bound by rectangular loop $ABCD$, we have

$$\int_S \text{curl } \vec{E} \cdot \vec{n} dS = - \int_S \frac{\partial \vec{B}}{\partial t} \cdot \vec{n} dS \quad \dots(68)$$

Transforming the surface integral of the left hand side into a line integral over the path $ABCD$ with the help of Stoke's theorem, eqn. (68) yields.

$$\begin{aligned} \int_{ABCD} \vec{E} \cdot d\vec{l} &= \int_S \frac{\partial \vec{B}}{\partial t} \cdot \vec{n} dS \\ \text{or } \int_{AB} \vec{E}_1 d\vec{l} + \int_{CD} \vec{E}_2 d\vec{l} + \text{contribution from sides } BC \text{ and } DA &= - \int_S \frac{\partial \vec{B}}{\partial t} \cdot \vec{n} dS \quad \dots(69) \end{aligned}$$

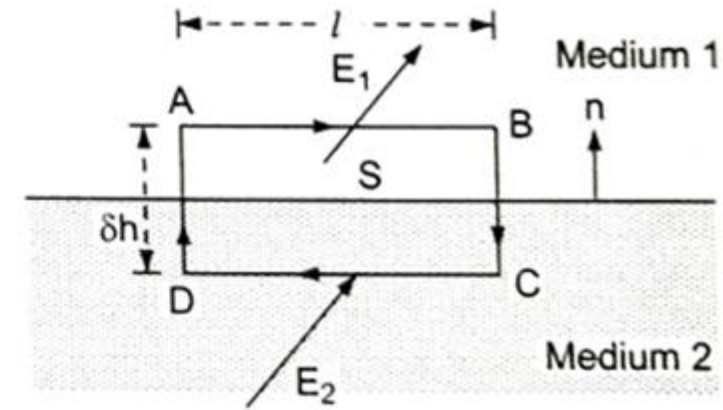


Fig. 12.5 Rectangular loop at the interface between two media

$$\text{or } \int_{AB} \vec{E}_1 d\vec{l} + \int_{CD} \vec{E}_2 d\vec{l} + \text{contribution from sides } BC \text{ and } DA = - \int_S \frac{\partial \vec{B}}{\partial t} \cdot \vec{n} dS \quad \dots(69)$$

If the loop is now shrunk (letting $\delta h \rightarrow 0$), the contribution from sides BC and DA will vanish. Provided $\partial \vec{B} / \partial t$ is everywhere finite, the right hand side of eqn. (69) also vanishes because the area of integrating also vanishes *i.e.*, no flux can be enclosed. Thus when $\delta h \rightarrow 0$.

$$\int_{AB} \vec{E}_1 d\vec{l} + \int_{CD} \vec{E}_2 d\vec{l} = 0$$

or

$$\vec{E}_1 \vec{AB} + \vec{E}_2 \vec{CD} = 0$$

or

$$\vec{E}_1 \vec{AB} - \vec{E}_2 \vec{AB} = 0$$

or

$$\vec{E}_{1t} = \vec{E}_{2t}$$

...(70)

where E_{1t} and E_{2t} are the tangential components of the electric field in the two media. Equation shows that the ***tangential components of E are continuous across the interface.***

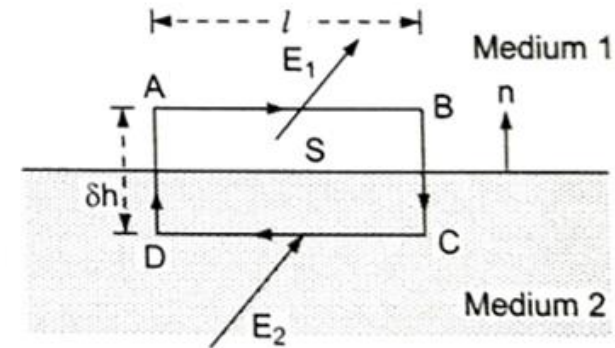


Fig. 12.5 Rectangular loop at the interface between two media

Prove that The normal component of electric displacement $\vec{D}$ is *discontinuous* across the interface, i.e., $D_{1n} - D_{2n} = \sigma$.

Q. Develop boundary conditions for $\vec{D}$ separated by two dielectric media of different permittivity

Boundary Condition of Electric Displacement $\vec{D}$

The boundary condition for the electric displacement $\vec{D}$ can be obtained from Maxwell's equation

$$\text{div } \vec{D} = \rho. \quad \dots(71)$$

Integrating eqn. (71) over the pill box shaped volume V of Fig. 12.4, we obtain

$$\int_V \text{div } \vec{D} dV = \int_V \rho dV \quad \dots(72)$$

Transforming the volume integral of left hand side of eqn. (72) in surface integral with the help of Gauss divergence theorem, we get

$$\oint_S \vec{D} \cdot \vec{n} dS = \int_V \rho dV$$

$$\text{or } \int_{S_1} \vec{D}_1 \cdot \vec{n}_1 dS + \int_{S_2} \vec{D}_2 \cdot \vec{n}_2 dS + \text{contribution from walls} = \int_V \rho dV$$

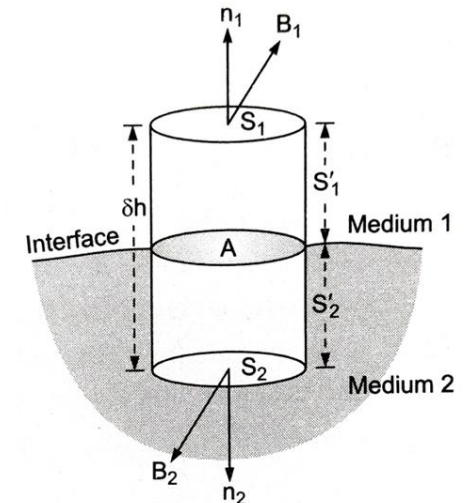


Fig. 12.4 Pill box shaped surface at the interface of two media.

Now let us suppose that the surfaces S_1 and S_2 approach each other towards the interface so that $S_1 = S_2 = A$, i.e., $\delta h \rightarrow 0$ the contribution from walls tends to zero. Now, instead of volume charge density ρ , the concept of surface charge density σ must be used, i.e.,

$$\lim_{\delta h \rightarrow 0} \int_V \rho dV = \int_V \rho dA = \sigma A$$

$$\int_{S_1} \vec{D}_1 \cdot \vec{n}_1 dS + \int_{S_2} \vec{D}_2 \cdot \vec{n}_2 dS = \sigma A$$

or
$$\vec{D}_1 \cdot \vec{n}_1 S_1 + \vec{D}_2 \cdot \vec{n}_2 S_2 = \sigma A$$

or
$$\vec{D}_1 \cdot \vec{n}_1 + \vec{D}_2 \cdot \vec{n}_2 = \sigma \text{ as } S_1 = S_2 = A$$

or
$$\vec{D}_1 \cdot \vec{n}_1 + \vec{D}_2 \cdot \vec{n}_2 = \sigma$$

or
$$\boxed{D_{1n} - D_{2n} = \sigma}$$

where D_{1n} and D_{2n} are components of the electric displacement in the two media normal to the interface in the direction of $\vec{n}_1$. Evidently, *the normal component of electric displacement is not continuous across the interface and changes by an amount equal to the free surface density of charge at the interface.*

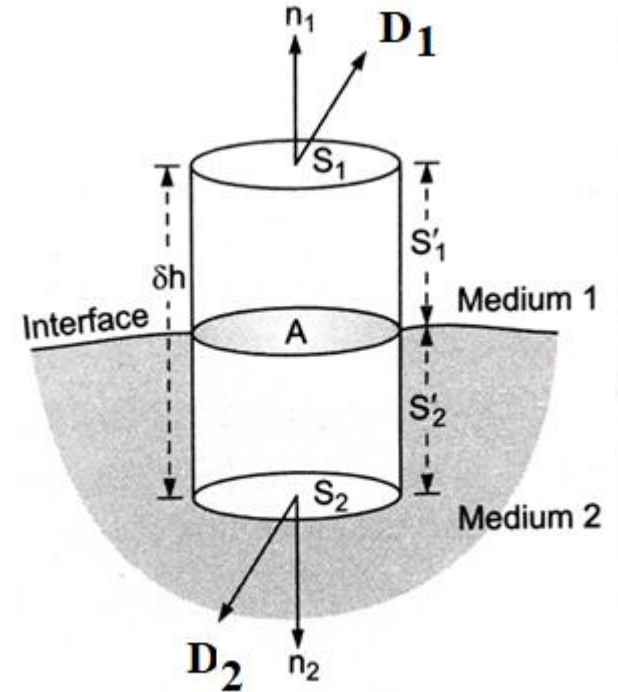


Fig. 12.4 Pill box shaped surface at the interface of two media.

DIELECTRIC POLARIZATION

Consider a dielectric bar containing polar or nonpolar molecules. Let us place the bar in an external electric field, say between the plates of a charged capacitor. The presence of the external electric field can cause the shifting of the centres of gravity of the positive and negative charges to convert the non-polar molecules into polar molecules. The result will be the alignment of dipoles along the electric field resulting in a net dipole moment of the dielectric bar Fig. 11.25. The *presence of* positive and negative charges, in the neighbourhood of each other, will cancel the effect of each other within the body of the dielectric. But on the surfaces of the bar at right angles to the external field $\vec{E}_0$, there will appear a net positive and negative charge as shown in Fig. 11.25. The appearance of the net positive and negative charges on the surfaces of the bar will set up an electric field $\vec{E}_p$ within the dielectric in a direction opposite to the applied field. And net electric field within the dielectric will be the vector sum of $\vec{E}_0$ and $\vec{E}_p$, i.e., $\vec{E} = \vec{E}_0 + \vec{E}_p$. As a result, the resultant electric intensity inside the dielectric is diminished, i.e., $E < E_0$ therefore, the actual field acting on the molecules or atoms of a dielectric is E instead of E_0 .

The phenomenon of appearance of the charges on the surfaces of the dielectric bar induced by an external electric field is called polarization. Since dipoles align along the electric field, thus, the process of dipole alignment in a dielectric is also known as polarization. Quantitatively, the polarization P is measured as net electric dipole moment per unit volume, i.e., $P = \frac{\Sigma p}{V}$, where p is the average dipole moment induced on each molecule of a dielectric having volume V . The charge on the plates of the capacitor is termed as free charge and the charge induced on the surfaces of the dielectric is called bound charge.

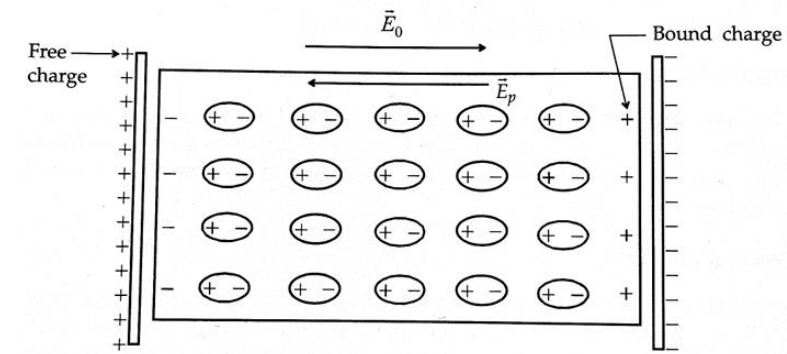


Fig. 11.25 Alignment of Atomic Dipoles when Dielectric Slab is Placed in an Electric Field $\vec{E}_0$

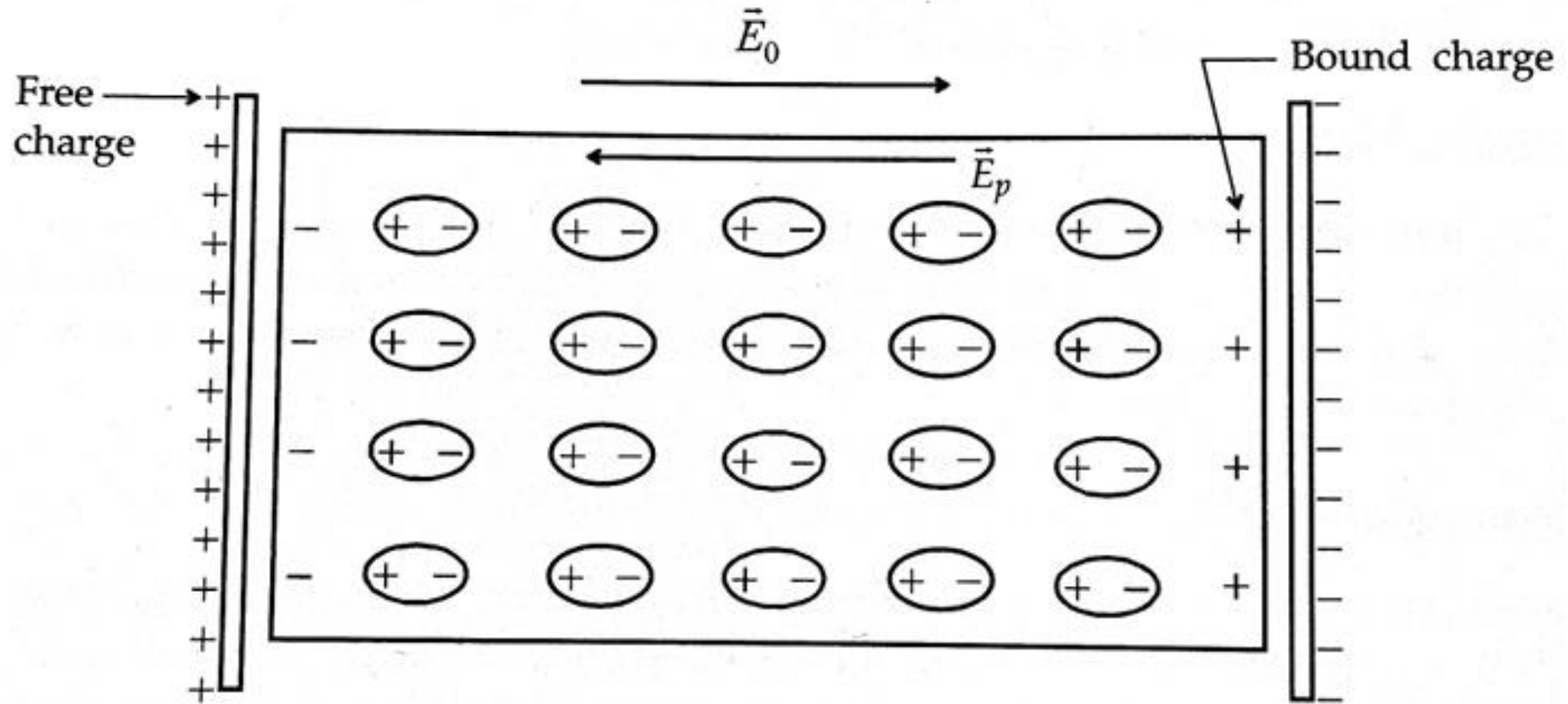


Fig. 11.25 Alignment of Atomic Dipoles when Dielectric Slab is Placed in an Electric Field $\vec{E}_0$

Q4 (b) Prove the relation $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$.

Establish the RELATION BETWEEN $\vec{D}$, $\vec{E}$ AND $\vec{P}$ (THREE ELECTRIC VECTORS)

Q. Deduce the relationship among the three vectors D , E , P in the metre-kilogram-second (mks) or SI system is: $D = \epsilon_0 E + P$ (ϵ_0 is a constant, the permittivity of a vacuum)

RELATION BETWEEN $\vec{D}$, $\vec{E}$ AND $\vec{P}$ (THREE ELECTRIC VECTORS)

Now we will derive the relationship among three electric vectors D , E and P . Let us consider a free charge q . According to Gauss's law the total electric flux due to this charge will be

$$\oiint \vec{E}_0 \cdot d\vec{A} = q/\epsilon_0 \quad \Rightarrow \quad \oiint \epsilon_0 \vec{E}_0 \cdot d\vec{A} = q$$

or $\epsilon_0 E_0 A = q$

$$\Rightarrow E_0 = \frac{q}{\epsilon_0 A} \quad \dots(49)$$

where E_0 is the electric field due to free charge q . When dielectric is present, induced charge on surface appears, say $-q'$, so that Gaussian surface encloses $q - q'$ charge. Gauss's law then becomes

$$\oiint \epsilon_0 \vec{E} \cdot d\vec{A} = q - q'$$

$$\oiint \epsilon_0 \vec{E} \cdot d\vec{A} = q - q'$$

or

$$\epsilon_0 EA = q - q'$$

$\Rightarrow$

$$E = \frac{q - q'}{\epsilon_0 A} = \frac{q}{\epsilon_0 A} - \frac{q'}{\epsilon_0 A}$$

or

$$\frac{q}{\epsilon_0 A} = E + \frac{q'}{\epsilon_0 A}$$

or

$$\frac{q}{A} = \epsilon_0 E + \frac{q'}{A} = \epsilon_0 E + \sigma_p \quad \dots(49a)$$

From eqn. (49), we have $\Rightarrow E_0 = \frac{q}{\epsilon_0 A} \quad \dots(49)$

$$\frac{q}{A} = \epsilon_0 E_0,$$

Also

$$\epsilon_r = \frac{E_0}{E}$$

$\therefore$

$$\frac{q}{A} = \epsilon_0 \epsilon_r E = \epsilon E = D \quad \dots(49b)$$

$$\epsilon_r = \frac{\text{Capacity with dielectric material}}{\text{Capacity with vacuum}} = \frac{\epsilon A/d}{\epsilon_0 A/d}$$

$$\Rightarrow \text{relative permittivity } \epsilon_r = \frac{\epsilon}{\epsilon_0}$$

Therefore, from eqn. (49a) and (49b), we have $D = \epsilon_0 E + \sigma_p$

$$D = \epsilon_0 E + P$$

$$(\because \sigma_p = P) \quad \dots(50)$$

The electric dipole moment will be $\sigma_p Al$, where Al represent the volume of the dielectric bar. If P be the σ_p . *dielectric polarization*, then electric dipole moment is PAI . Therefore, we have

$$PAI = \sigma_p \cdot Al \Rightarrow P = \sigma_p$$

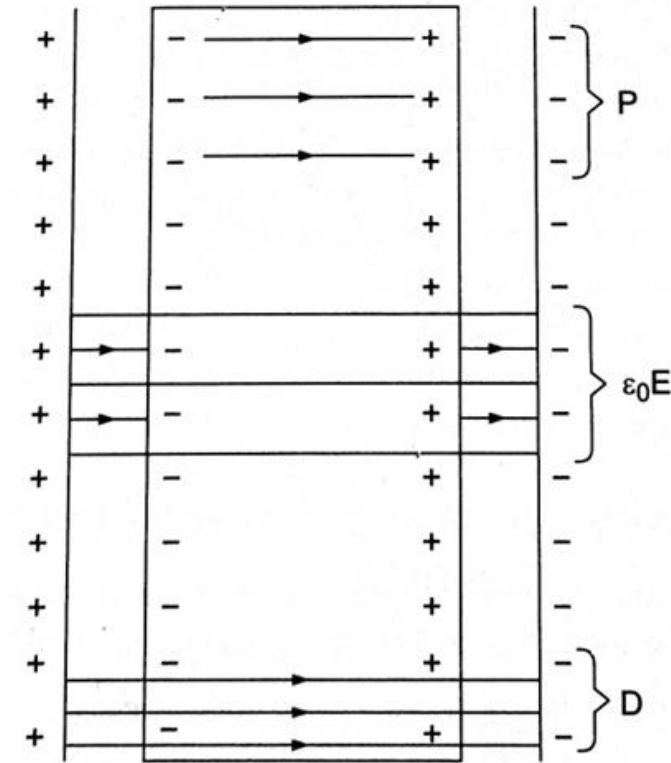


Fig. 11.30
Figures Shows Lines Associated with D (free Charges), $\epsilon_0 E$ (all Charges) and P (Polarization Charges).

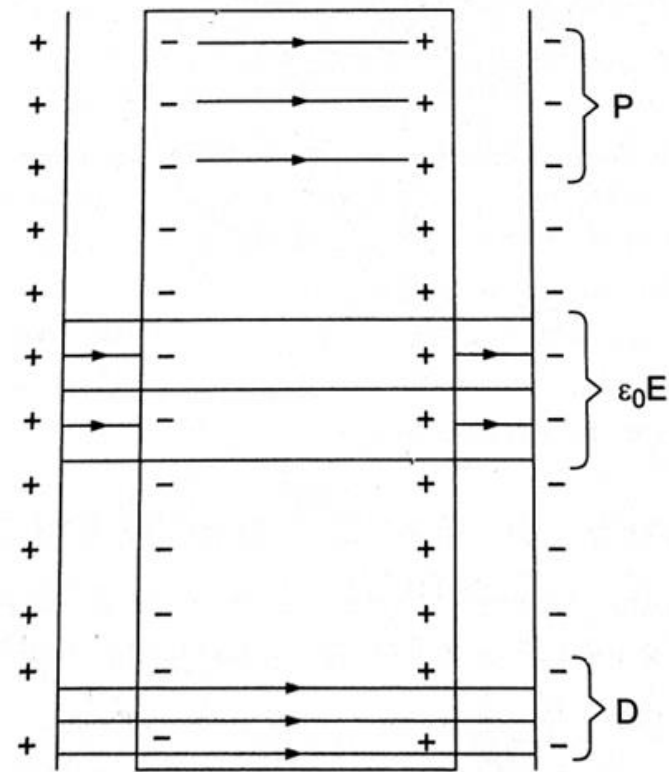


Fig. 11.30 Figures Shows Lines Associated with D (free Charges), $\epsilon_0 E$ (all Charges) and P (Polarization Charges).

It is noted here that D is connected with free charges only and P is connected with polarization charges only but E is connected with both free and polarization charges.

Figure shows lines of D , P and E where lines of D Begin and end on the free charges. lines of P begin and end on the polarization (bound) charges and lines of E reflect the presence of both types of charges.

Q. Prove that the Electric field at a point is defined as the gradient of the potential at that point.

OR

Q. Formulate the relation between E and V (or U) in differential form

RELATIONSHIP BETWEEN ELECTRIC FIELD (**E**) AND ELECTRIC POTENTIAL (**U or V**)

We have seen that the electric potential is given by

$$U(\vec{r}) = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r}$$

Now take gradient of above expression, we have

$$\vec{\nabla} U = \vec{\nabla} \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r} \right) = \frac{1}{4\pi\epsilon_0} q \vec{\nabla} \left(\frac{1}{r} \right)$$

Since, $\vec{r} = \hat{i}x + \hat{j}y + \hat{k}z$, and $r = (x^2 + y^2 + z^2)^{\frac{1}{2}}$

$$\text{we have } \vec{\nabla} \left(\frac{1}{r} \right) = \vec{\nabla} (x^2 + y^2 + z^2)^{-\frac{1}{2}} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (x^2 + y^2 + z^2)^{-\frac{1}{2}}$$

$$= -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} (\hat{i} 2x + \hat{j} 2y + \hat{k} 2z)$$

$$= -r^{-3} \vec{r} = -\frac{1}{r^2} \frac{\vec{r}}{r} = -\frac{\hat{r}}{r^2} \quad (\because \vec{r} = |\vec{r}| \hat{r})$$

$$\therefore \vec{\nabla} U = \frac{1}{4\pi\epsilon_0} q \left(\frac{-\hat{r}}{r^2} \right) = -\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

or

$$\vec{\nabla} U = -\vec{E}$$

$$\left(\because \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \right)$$

or

$$\vec{E} = -\vec{\nabla} U$$

...(3)

Thus, electric field at a point is defined as the gradient of the potential at that point.

Q.3(a) Show that electrostatic field is conservative in nature.

Suppose a charge Q is situated in an electric field at points A and B . The path from A to B is a closed path. The electric field does work to move a test charge from point A to B . To determine the work done, we sum up the work done for all the small segments that the path A to B is divided into. This kind of integral evaluated along a line is known as a line integral.

The line integral of the electric field along A to B

$$\int_A^B \vec{E} \cdot d\vec{l}$$

The line integral of the electric field along B to A

$$\int_B^A \vec{E} \cdot d\vec{l}$$

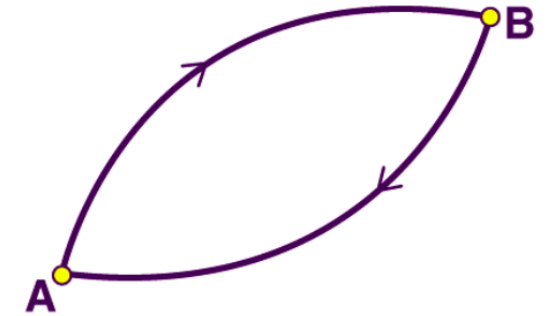
Hence, the line integral along the closed path as shown in the figure above is given by

$$\int_A^B \vec{E} \cdot d\vec{l} + \int_B^A \vec{E} \cdot d\vec{l}$$

But,

$$\int_B^A \vec{E} \cdot d\vec{l} = - \int_A^B \vec{E} \cdot d\vec{l}$$

$$\int_A^B \vec{E} \cdot d\vec{l} + \int_B^A \vec{E} \cdot d\vec{l} = \int_A^B \vec{E} \cdot d\vec{l} + - \int_A^B \vec{E} \cdot d\vec{l} \Rightarrow \int_A^B \vec{E} \cdot d\vec{l} + \int_B^A \vec{E} \cdot d\vec{l} = 0$$



Question:- Show that the electrostatic field E is conservative in nature

This demonstrates that the line integral of an electric field along a closed path is zero.

Therefore, the work done by the electric field is independent of the path, meaning it relies only on points A and B .

This confirms that the electric field is indeed conservative in nature.

**Thank
You**

